Preamble

- **Title:** Beyond Pattern Recognition: A New Paradigm Uniting Symbolicism and Connectionism
- **Author:** Chao Zheng | Independent Researcher
- **Abstract:** This study challenges the traditional view of neural networks as statistical pattern recognizers, proposing and validating a new paradigm capable of awakening their endogenous precise reasoning capabilities. By employing programmatically generated ideal data and a parallel, non-autoregressive solving framework, this paradigm transforms standard Transformers from probabilistic imitators into deterministic rule executors. Through successful application across a range of tasks including symbolic rules, algorithm fitting, visual reasoning, physical simulation, and interpretability, I have compellingly demonstrated the universality of this paradigm. Furthermore, I systematically evaluated various representative architectures (MLP, CNN, RNN, UNET, Diffusion) on the multi-step evolution task of one-dimensional cellular automata, confirming that multiple non-Transformer models also possess similar capabilities. This discovery strongly proves that the capability for precise algorithmic execution is an inherent potential of connectionist systems, rather than the patent of any specific architecture. Through the **Neural Sculpting Paradigm**, this work systematically demonstrates for the first time that standard neural networks possess the intrinsic potential to precisely execute arbitrary algorithms, opening a brand-new path for building highly reliable, explainable, and general-purpose artificial intelligence systems. **Code is available at:** https://github.com/ball-lightning6/neural-sculpting-paradigm.

Main Body

Act I: The Reasoning Gap in Modern AI

Deep neural networks have achieved immense success in pattern recognition tasks. However, in domains requiring precise, multi-step logical reasoning, their exhibition of "hallucinations" and logical inconsistencies remains a fundamental bottleneck. This limits the application of AI in highly reliable fields such as science and mathematics. This paper posits that the root of this dilemma is not an inherent defect of neural networks, but largely stems from a fundamental mismatch between the inductive bias of their architectures and the intrinsic structure of the tasks.

Based on this core insight, I propose and verify a novel, concise, and powerful solver-style reasoning paradigm—which I term the **Neural Sculpting Paradigm**. By abandoning autoregression, adopting single-step structured solution outputs, and utilizing programmatically generated ideal data, this paradigm systematically transforms standard neural networks (including but not limited to [2] Transformer, MLP, CNN, RNN, etc.) from statistical imitators into precise rule executors. Using the Transformer as the primary experimental platform, my extensive experiments across a series of tasks—spanning

symbolic rules, algorithm learning, visual reasoning, physical simulation, and interpretability —not only verify the universal effectiveness of this paradigm but also prove, through experiments like semantic shuffling, that the models learn abstract algebraic structures rather than superficial symbolic patterns. My work provides a concise, general, and scalable new path for constructing trustworthy, interpretable AI systems with high-order reasoning capabilities.

Act II: A Brief History of Neuro-Symbolic Computing

"Neuro-symbolic computing," which combines the powerful learning capabilities of neural networks with the rigorous reasoning capabilities of symbolic systems, has long been the **Holy Grail** of the artificial intelligence field. Previous explorations have mainly unfolded along two paths.

The first path is "Architectural Modification." Work in this category attempts to endow models with reasoning capabilities by designing special network architectures that mimic symbolic calculation processes. Pioneering explorations can be traced back to the [6] Neural Turing Machine, which introduced external memory modules, and the subsequent [7] Differentiable Neural Computer. This idea was further continued in later [11] Neural Module Networks and [10] Graph Neural Networks. The common characteristic of these works is the attempt to enhance neural network capabilities by "designing specialized hardware for reasoning." However, these specialized architectures are often difficult to transfer to new task domains and face the dilemma of complex design.

The second path is "Logic Embedding." Work in this category attempts to "embed" external symbolic logic rules into the learning process of neural networks in a differentiable form. Whether it is the [8] Neural Theorem Prover, which converts rules into loss function constraints via fuzzy logic, or [9] DeepProbLog, which treats neural networks as part of a probabilistic logic program, the core idea is to use an "external" logic system to "guide" or "constrain" an "internal" neural network.

However, the aforementioned paths share a common, underlying "hybrid" philosophy, viewing "learning" and "reasoning" as two heterogeneous capabilities that need to be "glued" together externally.

In fundamental contrast to this, my work discovers that reasoning is a "native, dormant" capability of neural network architectures; it does not need to be "mixed," but "awakened." I found that through a novel, concise training paradigm—namely, the combination of "solver-style output" and "ideal data"—a standard, general-purpose neural network (primarily validated using Transformers) can spontaneously and systematically learn and execute precise symbolic rules. My contribution lies not in designing a more complex "model," but in discovering a more correct "method" to unlock the immense potential already existing within the models.

Act III: The Neural Sculpting Paradigm

The core idea of the new paradigm I discovered is not to design a brand-new model architecture, but through a fundamental restructuring of the learning environment and task definition, systematically proving the powerful intrinsic reasoning capabilities of neural networks. The forms of these reasoning problems are not limited to symbolic rules, algorithmic problems, visual reasoning, physical simulation, or interpretability. To summarize in one sentence, it addresses a class of problems with clear transformation relationships: for a corresponding input, there is a mechanically executable symbolic process that transforms into a unique, precise output.

The main experiments of my research are based on the standard Transformer architecture and its variants in the image domain, such as [3] ViT or [4] Swin Transformer, which show excellent performance in handling complex long-range dependencies. However, supplementary experiments indicate that architectures like MLP, CNN, and RNN also possess such capabilities, and on some tasks, they may even converge faster. This proves that this capability is not unique to Transformers but is an inherent capability of neural networks.

The paradigm is built upon the following three mutually synergistic core principles.

3.1. Core Engine: Neural Architectures Biased Towards "Abstract Relations"

Although the Transformer architecture serves as the main validation platform in this study, experiments prove that various neural network architectures, including MLP, CNN, and RNN, possess similar capabilities for precise rule fitting. The reason likely lies in the fact that these architectures, through different mechanisms—whether self-attention, convolutional locality, or recurrent state transitions—ensure deterministic function approximation from input to output under ideal data conditions. In particular, the core self-attention mechanism of the Transformer architecture fundamentally breaks spatial constraints. It allows any element in a sequence to interact directly and in parallel with all other elements and calculate the "relational strength" between them. This architecture, naturally, possesses a profound "isomorphism" with the intrinsic structure of "reasoning" tasks. It is not "looking at" the neighborhood of pixels; it is "understanding" the "relational graph" of the entire symbol system. Experiments found that whether it is Transformer, ViT, Swin Transformer, etc., they all possess the same capabilities, such as the previously listed symbolic rules, algorithmic problems, visual reasoning, physical simulations, interpretability, etc. Therefore, I can conclude that the classification of these problems is merely superficial; there may be no essential difference between them.

3.2. Output Paradigm: From "Sequential Imitation" to "Parallel Solving"

Current large model reasoning capabilities mainly adopt the autoregressive method of predicting the next token, whereas I adopt a one-shot prediction method that completely abandons autoregression. On such reasoning problems, the reason why this method is more successful can be explained as follows: the method of abandoning autoregression makes the format of input data and output data more pure and consistent, while the method of autoregressively predicting the next token naturally cannot achieve this; it may still be more

suitable for Large Language Models. Moreover, this "prediction-by-word" mode has a fundamental defect: error accumulation. Any minor error early in the sequence can be continuously amplified in subsequent generation, leading to a complete collapse of the logical chain. My "one-step" format forces the model to perform global synergistic calculations and self-consistency constraints on all its internal "neurons" in a single forward pass, thereby transforming the reasoning process from a fragile "linear guessing chain" into a robust "parallel constraint solving" process.

Another reason is that autoregressive large models naturally perform multi-task learning because they receive inputs of different lengths. If the number of such tasks is excessive, it is highly likely to lose precision, resulting in hallucinations.

3.3. Data Paradigm: From "Realistic Noise" to "Ideal Laws"

Traditional machine learning paradigms are often forced to distill knowledge from data full of noise, accidental correlations, and uncertainty due to the authenticity of the data. Such data is often noisy at both input and output ends, and there is no clear, precise transformation rule. Therefore, in this kind of training data, neural networks can only manifest pattern recognition capabilities. However, for learning precise, deterministic rules, this "dirty data" is itself a powerful "source of interference."

My paradigm adopts a diametrically opposite "data philosophy." I use programmatic scripts to generate massive amounts of logically absolutely pure "ideal data." In these data, there exists only a unique, inevitable, and precise logical relationship between input and output, without any statistical "shortcuts" or "accidental patterns." This extremely high "signal-to-noise ratio" learning environment greatly reduces the difficulty of learning. It leaves the model with no choice but to learn the only invariant "signal" hidden behind all samples—that is, the underlying rule itself. My extensive experiments (see Chapter 4) prove that under this paradigm, the model only needs to cover a negligible fraction of the total input space to achieve perfect generalization over the entire problem space.

Additionally, the training data must share the same distribution as the input space (or at least, this seems the most convenient approach currently). I achieved this condition using random sampling in the training set generation script. This paradigm also cannot generalize outside the input space; or in other words, since the training data accounts for a very small proportion of the entire input space, the model allows for precise generalization across the vast remaining input space excluding the training data, which is analogous to generalization in past paradigms.

3.4. Paradigm Interpretation: Sculpting Precise Rules with Gradient Descent

In summary, my paradigm can be understood as a philosophical practice of using connectionist processes to infinitely approximate the ideals of symbolism. This is like a sculptor facing a rough block of marble (a randomly initialized neural network). "Ideal data" provides him with countless, perfect "photos" of the "David" from numerous angles; the loss function of "solver-style output" tells him the "gap" between his current work and that "ideal";

and "gradient descent" is the miraculous "chisel" in his hand that, every time, chips away a small piece of excess stone in the correct direction.

Through millions of tiny and precise "strokes," finally, the perfect "David," representing "precise rules," naturally "emerges" from that block of "continuous, chaotic" neural network stone. And on some problems where perfect convergence is impossible, this paradigm can always compress the uncertainty of the output.

Actually, the paradigm I discovered is, simply put, fitting a precise rule using connectionist gradient descent. This is an extremely natural way to connect connectionism and symbolism. I summarize this paradigm as essentially multimodal, capable of end-to-end general learning of precisely defined transformation rules.

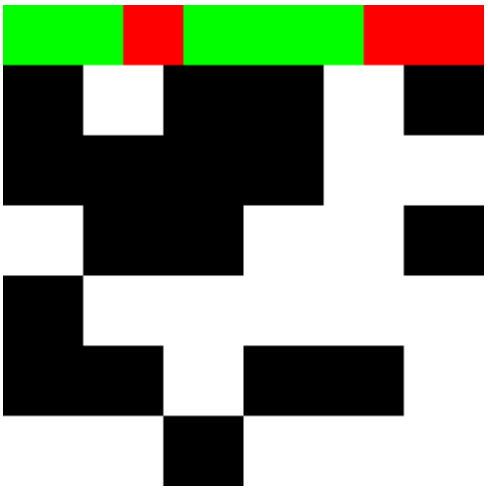
3.5. Model Description

- Symbolic Reasoning / Algorithm Reasoning — qwen2_0.5b [20] fine-tuned with lora + custom lm_head, or later replaced by full training of a small transformer.
- Image Reasoning Symbols — ViT / Swin Transformer.
- Multimodal / image2image — Swin Transformer + unet.
- Text Reasoning Generating Images — A small transformer followed by unet.

Corresponding Code:

- Multimodal / image2image: `train_image2image.py`
- Symbolic Reasoning / Algorithm Reasoning: `train_tiny_transformer.py`
- Text Reasoning Generating Images: `train_text2image.py` / `train_qwen2_text2image.py`
- Image Reasoning Symbols: `train_swin_image2text.py`

Regarding multimodal reasoning, I currently use the method of taking image input and encoding symbolic input into the image, namely Swin Transformer + unet. There should be other methods in the future. The example is as follows: the top layer of 8 green and red blocks represents the rules of the cellular automata, and the 6x6 black and white grid below represents the initial state of a 1D 36-bit cellular automata. This method achieved very good results in the task of predicting the state after applying two layers of cellular automata rules.



Additionally, mlp, cnn, and rnn can also complete simple tasks, and other models also have the potential to complete tasks of this type. But I still use the transformer as the standard model.

Act IV: Empirical Evidence Across Multiple Domains

To systematically verify the universality of the Neural Sculpting Paradigm, this paper conducted exhaustive experiments on multiple types of tasks. In addition to the tasks detailed in the main text, I also tested on over 100 tasks. Full experiment descriptions and dataset generation scripts can be found in the GitHub repository: https://github.com/ball-lightning6/neural-sculpting-paradigm

4.1. Fundamental Capability: Symbolic Rule Learning

This section aims to verify the core capability of this paradigm to learn precise, complex symbolic rules from the most fundamental level.

Training/Validation Split Ratio: Evaluations (eval) are generally performed at a frequency of about 1000 steps to monitor the downward trend of eval loss at any time. The verification set ratio is calculated based on this frequency so that the eval time is not too long. The verification set accounts for less than 0.01 of the dataset. Since the verification set and the dataset are identically distributed with the input space, a reliable eval loss can be verified.
Regarding Model Training Hyperparameters: Almost all hyperparameters do not need to be changed, except that sometimes batch size is increased for simpler tasks to improve VRAM utilization, and for some tasks involving image reasoning where training loss sometimes becomes nan, the learning rate will be halved in such cases.
Model Structure Changes: For outputting symbols, the most important thing is generally the need to change the number of labels for multi-label binary classification.
Input Space Estimation: For some tasks where the input space is difficult to calculate directly, I use a method similar to Monte Carlo estimation, i.e., generating task-related random numbers (these random numbers are sufficient to determine a sample) a relatively large number of times, and estimating the order of magnitude of the input space based on the number of repeated samples.

Note: Many of the following experiments only list the eval loss, which is strictly speaking not formal enough. This is because I only recorded this when conducting a large number of experiments at the time. Later, I also verified the test results of some cases, proving that the test set accuracy and eval loss are consistent; a lower eval loss does indeed correspond to a higher accuracy rate. I may supplement strict experimental results later.

4.1.1. Cellular Automata

- **Task Description:** The model learns to predict and output the state of a one-dimensional cellular automata after a given number of transformations according to a given rule.
- **Input Format:** A 30-bit string composed of "0" or "1", representing the initial state of the cellular automata.
- **Output Format:** Multi-label binary classification format with 30 labels, equivalent to thirty 0/1 strings, representing the state of the cellular automata after transformation.
- **Loss:** Average Binary Cross Entropy Loss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_cellular_automata_1d.py`
- **Dataset:** `ca_rule110_layer15_30.jsonl`
- **Training Code:** `train_tiny_transformer.py`
- **Training Set Size:** 500,000
- **Input Space Size Estimation:** 2 to the power of 30 = 1,073,741,824
- **Ratio of Training Set Size to Input Space Size:** 0.0466%

Task-Related Design Thoughts and Discussion:

1. To rigorously test the ability of neural networks to transcend pattern recognition and learn and execute precise symbolic rules, I chose one-dimensional cellular automata (CA) as the primary experimental platform. This choice was not accidental but based on its unique ideal characteristics capable of eliminating all ambiguity. First, CA is a computational system driven entirely by localized, deterministic rules. Its behavior does not rely on any statistical priors or semantic information derived from the real world, making it a perfect "sterile environment" for testing pure symbolic manipulation capabilities. Second, its enormous state space (for a binary cellular automata of width W, there are 2^W possible states) makes any form of "rote memorization" futile, thereby forcing the model to learn the rules themselves rather than memorizing input-output pairs.
   On this basis, I specifically chose Rule 110 as the core test rule. Rule 110 is not only a non-trivial rule exhibiting complex and seemingly chaotic behavior, but more critically, it has been proven to be Turing complete. This means that, in principle, Rule 110 can simulate any computational process. Therefore, having a Transformer learn the evolution of Rule 110 is tantamount to testing whether the model has the potential to learn a form of general computation. Successfully learning this rule will provide the solidest and most convincing experimental evidence for the argument that "neural networks can internalize complex algorithms."

2. Some task parameters:
    - Width: 30
    - Rule: 110
    - Layers: 15

Experimental Results and Analysis: The evaluation loss converged to 0.000068. This extremely low evaluation loss and the fact that the training set size is far smaller than the input space fully demonstrate that the neural network has learned this rule.

Final Convergence Results:

```
--- Step 477000 ---
Train Loss: 0.001766
Eval Loss: 0.000068
```

4.1.2. Inquiry into the Nature of Learning: N-base Addition and Semantic Shuffling

- **Task Description:** The model learns to predict and output the addition of two N-base numbers, while comparing the training results of semantic shuffling and position shuffling.
- **Input Format:** A 16-bit string, the first 8 bits represent one number, and the last 8 bits represent another number.
- **Output Format:** Multi-label binary classification format with 33 labels, equivalent to thirty-three 0/1 strings, representing the addition result with a 33-bit binary number.
- **Loss:** Average Binary Cross Entropy Loss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_symbol_add_shuffle_dataset.py`
- **Datasets:**
    - `adder_8bit_base16_train.jsonl` — No semantic shuffle, no position shuffle
    - `adder_8bit_base16_sem_shuffled_train.jsonl` — Semantic shuffle, no position shuffle
    - `adder_8bit_base16_pos_shuffled_train.jsonl` — No semantic shuffle, position shuffle
    - `adder_8bit_base16_sem_shuffled_pos_shuffled_train.jsonl` — Semantic shuffle, position shuffle
- **Training Code:** `train_tiny_transformer.py`
- **Training Set Size:** 500,000
- **Input Space Size Estimation:** 16 to the power of 16 = 18,446,744,073,709,551,616 ≈ 1.84e19
- **Ratio of Training Set Size to Input Space Size:** 2.71e-14

Task-Related Design Thoughts and Discussion:

1. The model learns to perform addition on two 8-digit hexadecimal numbers. To explore the essence of model learning, I designed a "Semantic Shuffling" control experiment, including two types of shuffling: one uses random different symbols to represent hexadecimal numbers, and the other randomly selects the positional arrangement of hexadecimal numbers, no longer adopting the practice where the first 8 bits and last 8 bits are two hexadecimal numbers respectively.
   Reasons for designing the experiment this way:
   - **Semantic Shuffle:** To determine that what the transformer learns is the structure between symbols relative to the task, irrelevant to what the specific symbols are.
   - **Position Shuffle:** To determine that the transformer's learning of this task, and the vast majority of tasks, does not depend on the arrangement of input symbols.
2. Some task-related parameters and details:
   - Base: Hexadecimal
   - Digits for each of the two numbers: 8
   - Under no semantic shuffle: 0123456789abcde are used as hexadecimal number symbol representations.
   - Under semantic shuffle: Different visible ASCII characters are arbitrarily chosen as hexadecimal number symbol representations.
   - Under no position shuffle: The first 8 bits of the input string are one 8-digit hexadecimal number, and the last 8 bits are the other.
   - Under position shuffle: At the beginning of the training dataset script, a position shuffle mapping is generated, and this mapping consistency is maintained throughout the dataset. Corresponding code is as follows:

```
if SHUFFLE_POSITIONS:
    position_map = list(range(INPUT_LEN))
    random.shuffle(position_map)
    POSITION_SHUFFLE_MAP = {from_idx: to_idx for from_idx, to_idx in
enumerate(position_map)}
```

Experimental Results and Analysis: The results in the table below decisively prove my core assertion. In both "no semantic shuffle" and "semantic shuffle" cases, the model eventually converged to extremely low loss levels (<0.0001), with no significant difference in performance. This irrefutably indicates that what the model learns is the abstract algebraic structure behind the addition operation, rather than superficial symbolic patterns.

Note: The training loss decline process generated by different random data and training random seeds may vary greatly, so the difficulty of training cannot be analyzed through a single comparison. However, in multiple attempts, it was found that the convergence of the 4 experimental settings exists stably.

Since the training process is relatively short, all training processes of the 4 experiments are listed here, represented by eval loss.

step	No Semantic Shuffle No Position Shuffle	No Semantic Shuffle Position Shuffle	Semantic Shuffle No Position Shuffle	Semantic Shuffle Position Shuffle
1000	0.534105	0.505196	0.511109	0.564834
2000	0.438036	0.263462	0.446237	0.497787
3000	0.347195	0.001700	0.365586	0.461389
4000	0.324581	0.001392	0.179969	0.395561
5000	0.254971	0.000123	0.063128	0.328988
6000	0.129849	0.000042	0.001083	0.247512
7000	0.066946	0.000024	0.000203	0.159110
8000	0.044260		0.000123	0.149018
9000	0.032533		0.000051	0.139221
10000	0.001005			0.103275
11000	0.000325			0.080524
12000	0.000099			0.052849
13000				0.022913
14000				0.002443
15000				0.002063
16000				0.000327
17000				0.000472
18000				0.000272
19000				0.000293
20000				0.000409
21000				0.000173
22000				0.000164
23000				0.000391
24000				0.000243
25000				0.000095

4.2. Advanced Capability: Algorithm Learning and Planning

4.2.1. Case Study: Trapping Rain Water Problem

- **Task Description:** Derived from the classic LeetCode algorithm problem, 42. Trapping Rain Water - LeetCode [19]

42. 接雨水

困难 相关标签 相关企业

给定 `n` 个非负整数表示每个宽度为 `1` 的柱子的高度图，计算按此排列的柱子，下雨之后能接多少雨水。

示例 1：

> **输入：** height = [0,1,0,2,1,0,1,3,2,1,2,1]
> **输出：** 6
> **解释：** 上面是由数组 [0,1,0,2,1,0,1,3,2,1,2,1] 表示的高度图，在这种情况下，可以接 6 个单位的雨水（蓝色部分表示雨水）。

示例 2：

> **输入：** height = [4,2,0,3,2,5]
> **输出：** 9

提示：

- `n == height.length`
- `1 <= n <= 2 * 10^4`
- `0 <= height[i] <= 10^5`

通过次数 1,421,469/2.2M 通过率 65.6%

- **Input Format:** A 30-bit 01 string. Every consecutive 3-bit 01 string represents the height of a bar (0-7) in 3-bit binary format, with a total of ten bars.
- **Output Format:** A 30-bit 01 string. Every consecutive 3-bit 01 string represents the amount of water trapped by a bar (0-7) in 3-bit binary format, with a total of ten bars.
- **Loss:** Average Binary Cross Entropy Loss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_trapping_rain_water_decoupled.py`
- **Dataset:** `trapping_rain_water_decoupled_n10_b3_train.jsonl`
- **Training Code:** `train_tiny_transformer.py`
- **Training Set Size:** 300,000
- **Input Space Size Estimation:** 2 to the power of 30 = 1,073,741,824 ≈ 1.07e9
- **Ratio of Training Set Size to Input Space Size:** 0.0279%

Task-Related Design Thoughts and Discussion:

1. LeetCode is a natural repository for testing the algorithmic capabilities of this paradigm. I tested many algorithmic problems from here, and this paradigm can fit many of them. Regarding the "Trapping Rain Water" problem, it belongs to the Hard category on LeetCode. Its solution cannot be simulated by any simple symbolic rules. As for the

principle of fitting, it should be said that it is currently unknown. However, I believe that the Transformer does not think in the way humans usually assume—first parsing the format, then executing algorithmic steps one by one. Because it has no way of knowing the meaning of each symbol bit, nor can it know the meaning of this task; it is merely executing a simple task: stochastic gradient descent according to the objective defined by the loss. This is also why I believe this paradigm can be described as "approximating discreteness via gradient descent in a connectionist manner."

2. Regarding why the output is not the total amount of trapped water: I previously trained directly with the binary number of the total amount as the target, but found that the effect was not good and convergence was difficult. Therefore, I adopted this decoupled format, which allowed for extremely rapid convergence. Here, I reduced the burden of a summation on the Transformer, which should be the reason for the model's rapid convergence. Additionally, another experiment I conducted also encountered difficulties in convergence because it mixed too many simple operations serially; this is a very interesting phenomenon worth studying and will be discussed later.
3. Some task parameters:
    - Number of bars: 10
    - Number of binary bits used to represent bar height and trapped water: 3
    - Range of bar height / trapped water amount: 0-7

Experimental Results and Analysis: After 16,000 steps of training, the model evaluation loss converged to 0.000059. This result indicates that this paradigm can effectively learn complex algorithms that require understanding global information (such as left and right highest walls) and achieves precise modeling of the problem structure through the decoupling of output formats.

Final Convergence Results:

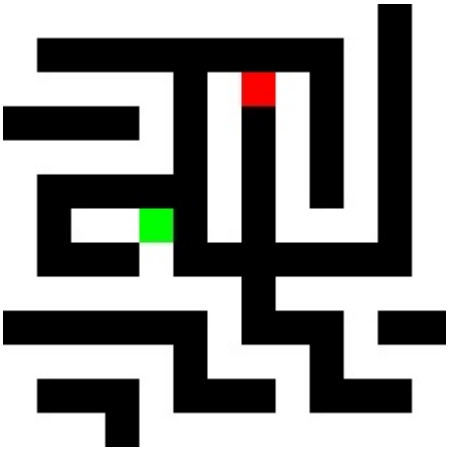
```
--- Step 16000 ---
Train Loss: 0.000570
Eval Loss: 0.000059
```

4.2.2. Dense Maze Navigation

- **Task Description:** Within a complex dense maze designed for human players, the model learns to predict the "next step" direction (up/down/left/right) of the optimal path from any starting location to the destination.
- **Input Format:** A 169-character string composed of "0", "1", "s", or "t", representing the initial state of the maze. Every consecutive 13 characters represent a row of the maze. 0 represents open ground, 1 represents a wall, s represents the starting point, and t represents the target point. There is exactly one s and one t. By default, there is a ring of walls around the 13*13 maze.
- **Output Format:** 4-class classification.

- **Loss:** 4-class Cross Entropy Loss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_maze_dense.py`
- **Dataset:** `maze_optimized_13_13_dataset.jsonl`
- **Training Code:** `train_tiny_transformer.py`
- **Training Set Size:** 2,000,000
- **Input Space Size Estimation:** Using Monte Carlo method to estimate 2,000,000 times without duplicate samples, tentatively roughly estimated as 2 to the power of 169 ≈ 7.48e50.
- **Ratio of Training Set Size to Input Space Size:** Approx. 2.673e-45

The following figure shows a typical maze, using white for open ground, black for walls, green for the starting point, and red for the destination. This is a typical maze game suitable for human play, distinguishing it from maze generation algorithms that randomly generate independent wall points.



Task-Related Design Thoughts and Discussion:

1. Based on confidence in the algorithm fitting capability of this paradigm, the design of this maze experiment initially used an algorithm equivalent to randomly generating wall points. However, practice proved that such mazes might be very easy for humans, who can spot the shortest path (or at least a feasible path) at a glance, but for the model, the number of branches and complexity far exceed the current maze experiment setup.
2. The reason for choosing the direction of the next step along the shortest path is that this format is relatively simple to design. Direct output of the shortest path would require knowing the length of the shortest path in advance; even with other measures, the task design becomes less elegant. Another important reason is that such a design implies another possibility: the maze task can be seen as a task of choosing the optimal action at a certain state. This can be linked to reinforcement learning. Training state-action pairs in this way can test the potential of this paradigm to complete reinforcement learning tasks, see the discussion on arc-agi-3 later. Training also has two methods: direct supervised learning of the optimal action and reinforcement learning exploration of the optimal action, but due to time and energy constraints, I did not complete the experiment of directly using reinforcement learning to explore the optimal action.

3. Some task parameters:
    - Width: 13
    - Height: 13

Experimental Results and Analysis: The evaluation loss converged to 0.000014, and the model has demonstrated extremely strong path planning capabilities. This proves that this paradigm can amortize traditional "search" problems into the weights of a feedforward network, achieving efficient "search-free planning."
This extremely low evaluation loss means that the model has learned to navigate similar mazes, and the low evaluation loss also implies that on the length scale of this optimal path, the overall path accuracy will also be very high.

Final Convergence Results:

```
--- Step 279000 ---
Train Loss: 0.000083
Eval Loss: 0.000014
```

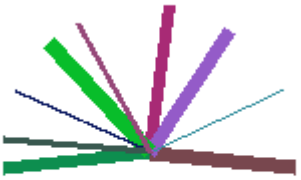
4.3. Reasoning and Extracting Symbolic Output from Images

This section will demonstrate the powerful potential of this paradigm in handling multimodal tasks, including decoding symbols from visual information.

4.3.1. Visual Symbol Decoding: Dial Angle Identification

- **Task Description:** Image reasoning. By inputting an image resembling a dial, output the occupation of all angle intervals by these line segments of different colors and thicknesses.
- **Input Format:** 224*224 image.
- **Output Format:** Multi-label binary classification format with 36 labels, equivalent to thirty-six 0/1 strings, representing whether each interval is occupied by line segments. 36 intervals divide the 360-degree angle equally, with each interval being 10 degrees.
- **Loss:** Average Binary Cross Entropy Loss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_line_angle_to_vector.py`
- **Dataset:** `line_angle` folder.
- **Training Code:** `train_swin_image2text.py`
- **Model Used:** Swin Transformer.
- **Training Set Size:** 200,000
- **Input Space Size Estimation:** 389,329,651
- **Ratio of Training Set Size to Input Space Size:** 0.0514%

Illustration:



Task-Related Design Thoughts and Discussion:

1. After the algorithm fitting capability was proven, I thought of the application of Transformers in images, so I wanted to test whether image Transformers have similar reasoning capabilities. So I used ViT to test tasks related to checkerboards, and the results were very good. Later, I designed this task with the initial intention of testing pure image understanding, rather than simply identifying symbols in images and then applying the already proven symbolic rule learning capability and algorithm fitting capability of Transformers.
2. This task initially used the ViT model, but the effect was not good, so it was later switched to the currently used Swin Transformer.
3. To increase diversity, randomness was added to the color and thickness of the line segments.
4. Some task parameters:
    - Image resolution: 224*224
    - Number of intervals: 36
    - Angle per interval: 10

Experimental Results and Analysis: The evaluation loss converged to 0.019183. This loss is not perfect convergence, but it also proves that the model can precisely "decode" abstract, symbolic angle information from raw pixels, demonstrating powerful visual-symbol grounding capabilities.

Final Convergence Results:

```
--- Step 24000 ---
Train Loss: 0.016392
Eval Loss: 0.012753
```

4.4. Image Reasoning Capability

This section will demonstrate the paradigm's pure image reasoning capabilities to distinguish it from symbolic rule learning capabilities and algorithm fitting capabilities.

4.4.1. Geometric Reasoning and Construction: Prediction of Triangle Incircle

- **Task Description:** The input to the model is an image of a randomly generated green triangle, and the task is to generate the image of its mathematically unique, correct red incircle.
- **Input Format:** 224*224 image.
- **Output Format:** 224*224 image.
- **Loss:** MSELoss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_triangle_to_incircle.py`
- **Dataset:** `incircle_dataset` folder.
- **Training Code:** `train_image2image.py`
- **Model Used:** Swin Transformer + unet.
- **Training Set Size:** 150,000
- **Input Space Size Estimation:** Estimated using Monte Carlo method to be approx. 2.8e+12.
- **Ratio of Training Set Size to Input Space Size:** Approx. 5.4e-8

Task-Related Design Thoughts and Discussion:

1. To further verify the pure image reasoning capability of the Transformer, separating it from symbolic rule learning and algorithm fitting capabilities, I designed this incircle experiment. Theoretically speaking, identifying the positions of 3 points, then using a series of mathematical formulas to calculate the position of the incircle, and then drawing the incircle is possible in theory, but practically it is almost impossible to be the real process. Therefore, the purpose of this task is to prove the image reasoning capability of the Transformer. Given the analog nature of images, this reasoning capability has great potential to expand to larger fields, such as pure experimental observation data, etc. But it is easy to think that, given that real experimental observation data certainly contains noise, it will degenerate into the previous pattern recognition deep learning paradigm. This issue will be discussed later.
2. The most amazing thing about this task and the subsequent related series of tasks is that they demonstrate the evolution of the "machine mind" during training in a very intuitive form—how the model approximates and learns rules. The characteristics exhibited in this process are very similar to some kind of intelligence. The visualization of the training process is presented in the form of eval images.
3. Some task parameters:
    - Image resolution: 224*224

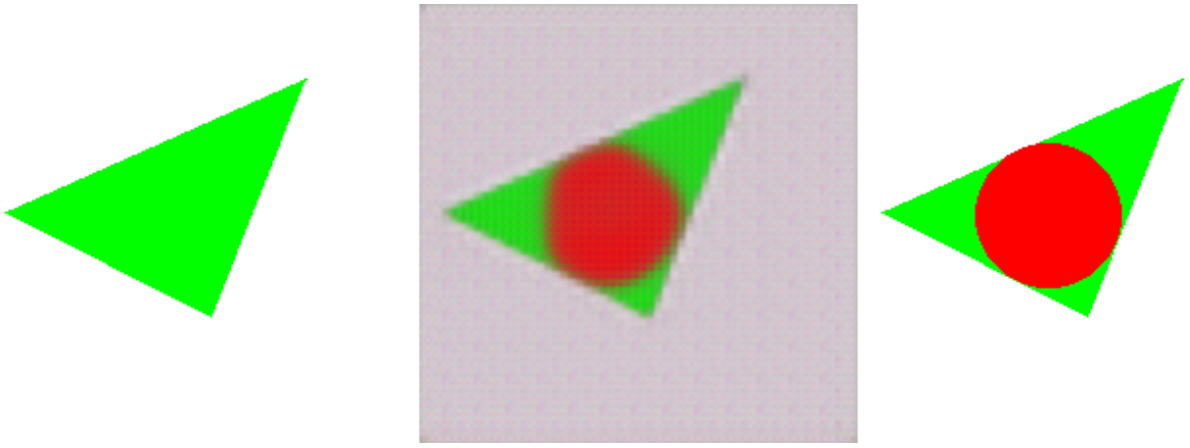
Experimental Results and Analysis: The final training results show that the model has completely learned the rules of the incircle. This indicates that the model possesses image reasoning capabilities. In this example, the possibility of interpolating other training set examples cannot effectively be ruled out, but subsequent other tasks rule out this possibility

because those tasks cannot be explained by interpolation, and they use exactly the same model and hyperparameter configuration. At the same time, the learning speed of the model is very fast, providing direct observation of the formation process of the machine mind.

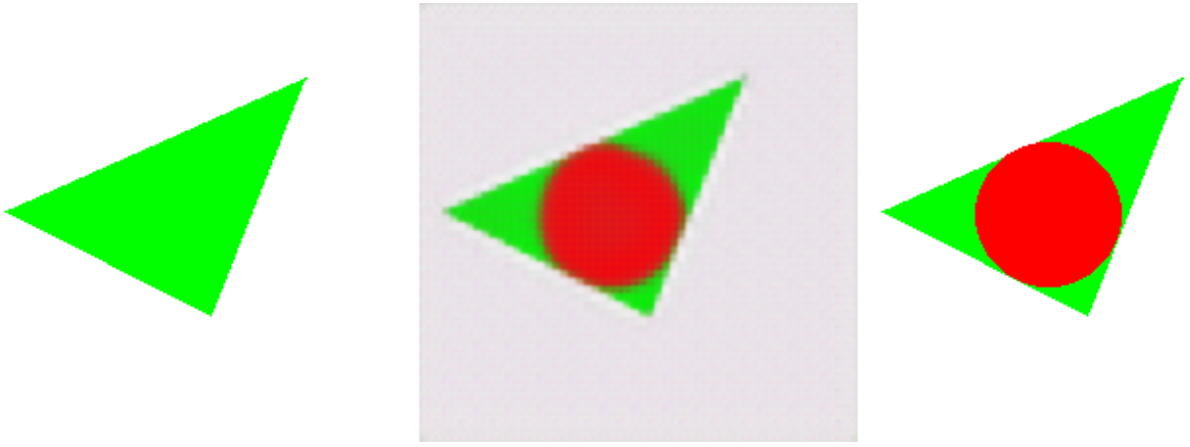
Experimental Process Display:

Left is the input image, right is the ground truth, middle is the generated image.

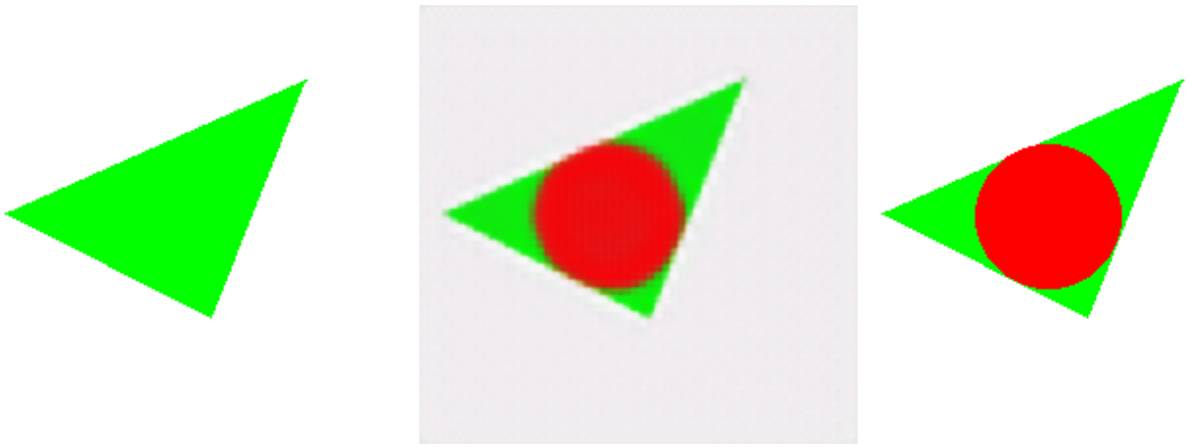
200 step



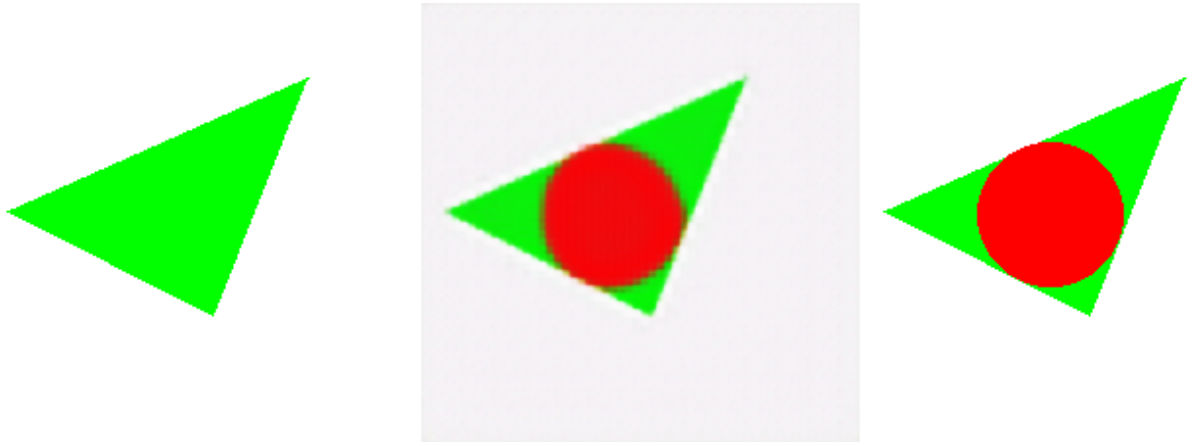
400step



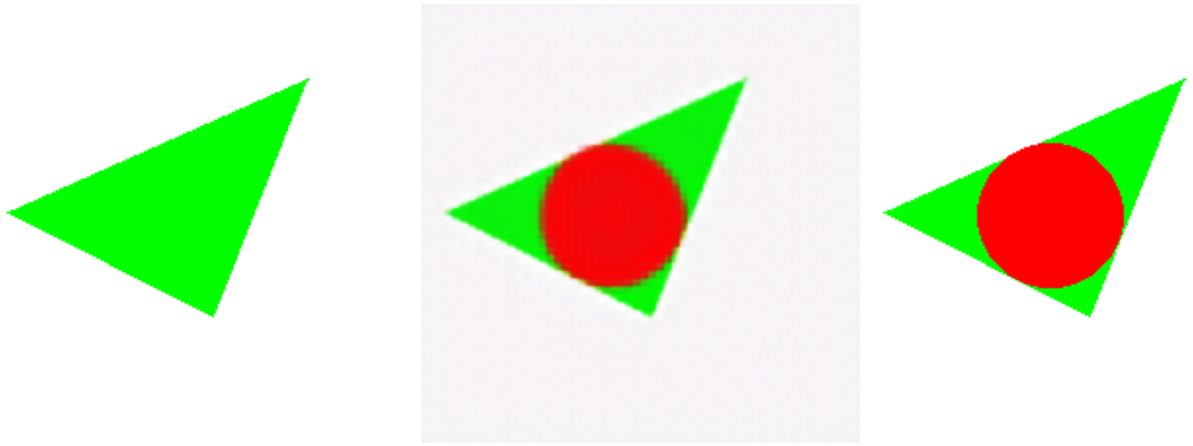
600step



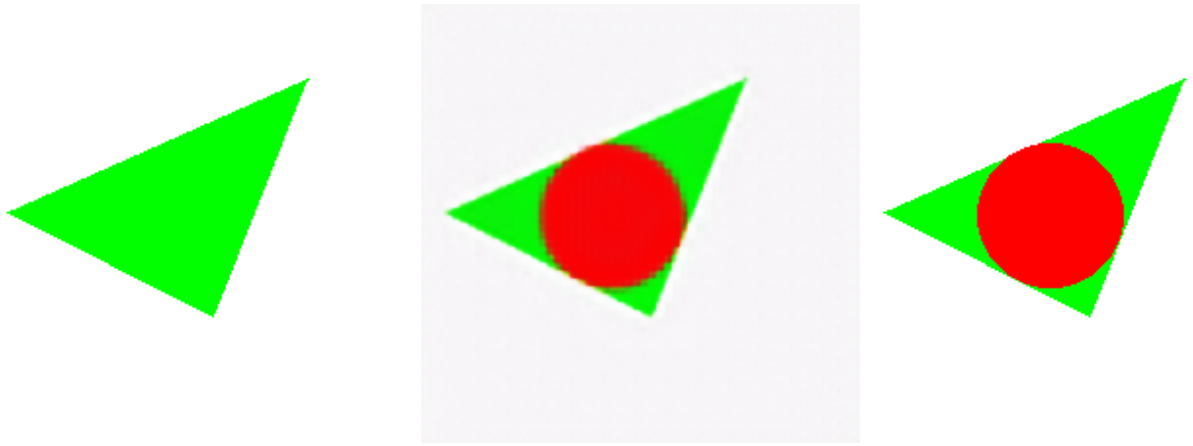
800step



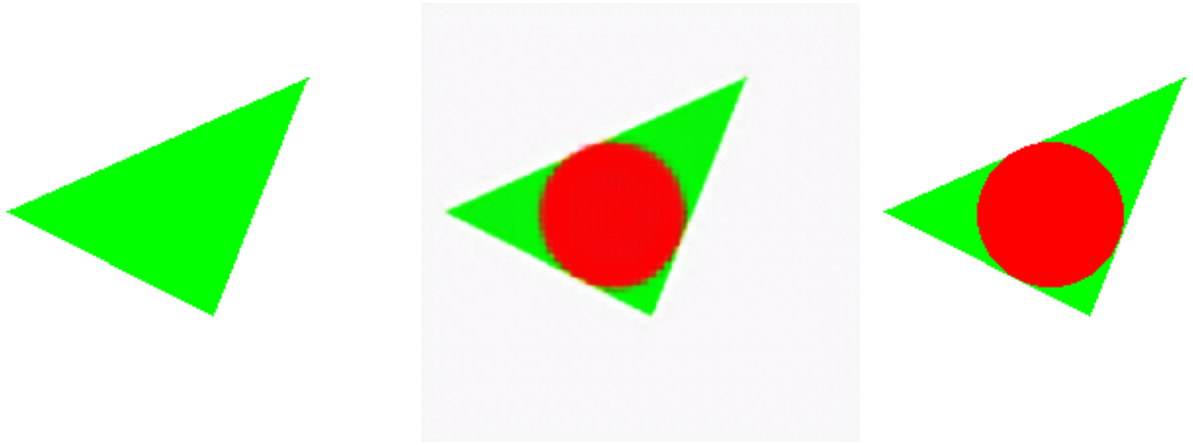
1000step



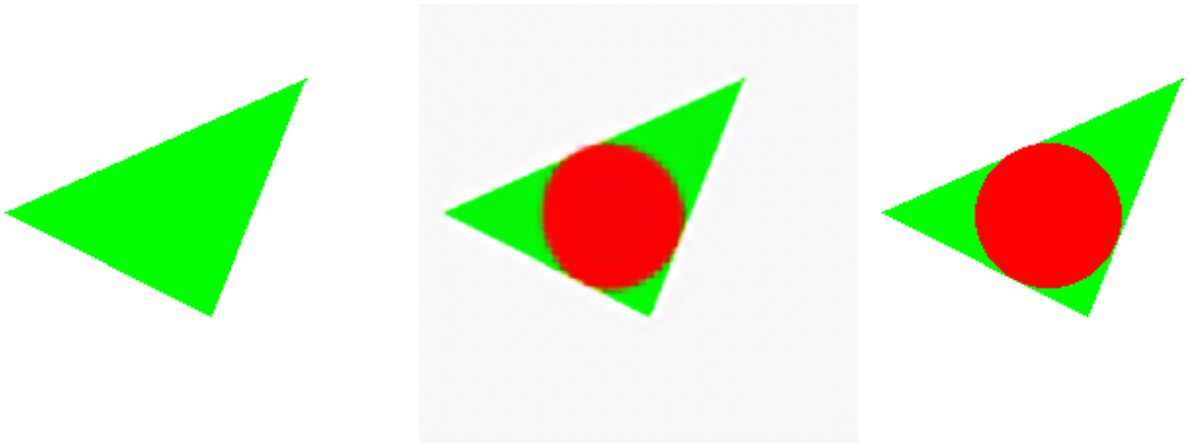
1200step



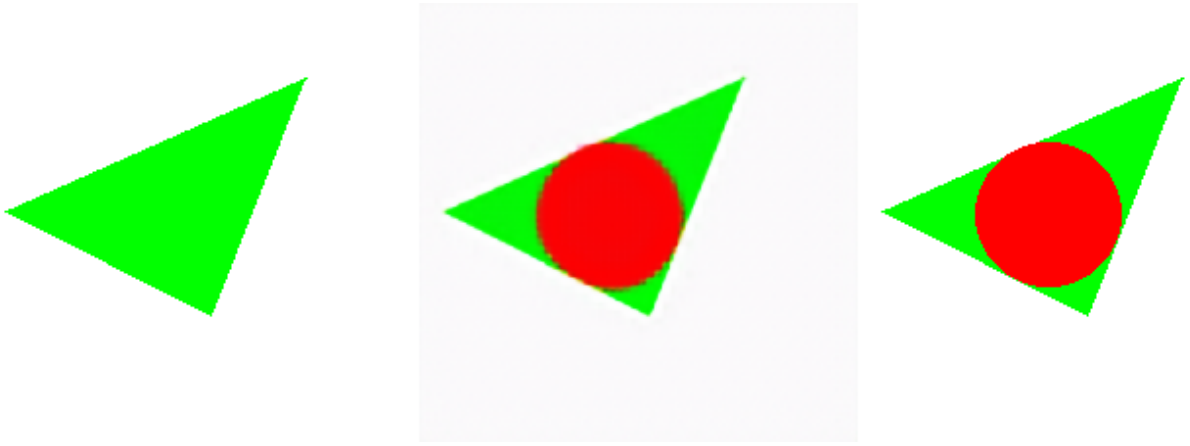
1400step



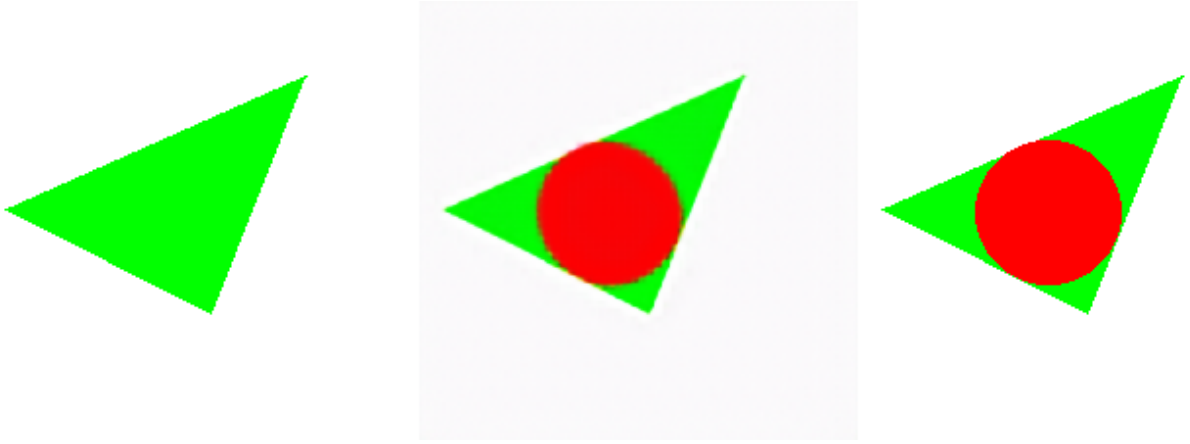
1600step



1800step

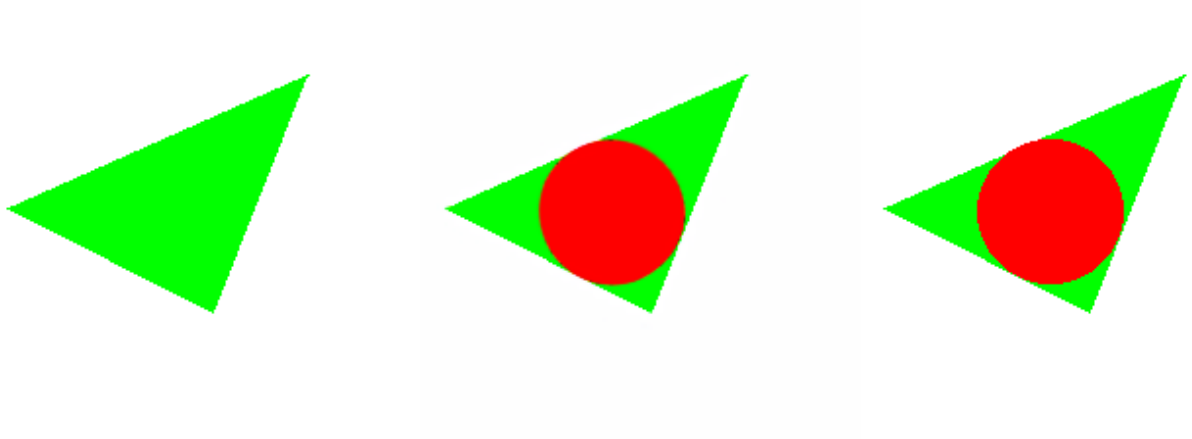


2000step



Due to the slowing down of changes later, the final training result is displayed directly.

14000step



4.4.2. Geometric Reasoning and Construction: Plane Tessellation

- **Task Description:** The input to the model is an image of a randomly generated green triangle, and the output image is a plane tessellation of the green triangle, where green triangles and red triangles alternate and share adjacent edges.
- **Input Format:** 224*224 image, containing one randomly generated green triangle image.
- **Output Format:** 224*224 image, which is the plane tessellation result of the input image, with green and red triangles alternating and adjacent by edges.
- **Loss:** MSELoss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_triangle_to_tessellation.py`
- **Dataset:** `tessellation_dataset_224` folder.
- **Training Code:** `train_image2image.py`
- **Model Used:** Swin Transformer + unet.
- **Training Set Size:** 150,000
- **Input Space Size Estimation:** Estimated using Monte Carlo method to be approx. 2.71e10.
- **Ratio of Training Set Size to Input Space Size:** Approx. 5.5e-6

Task-Related Design Thoughts and Discussion:

1. This task can completely rule out the possibility of solving the task by interpolation rather than learning precise rules. The reason is that far away in the plane tessellation, the error of such interpolation will gradually expand to the point where the rules can no longer be fitted by naive interpolation methods.
2. To ensure appropriate task difficulty, constraints were placed on the shape and size of the input green triangle; see the training set generation script code for details.
3. Some task parameters:
    - Image resolution: 224*224

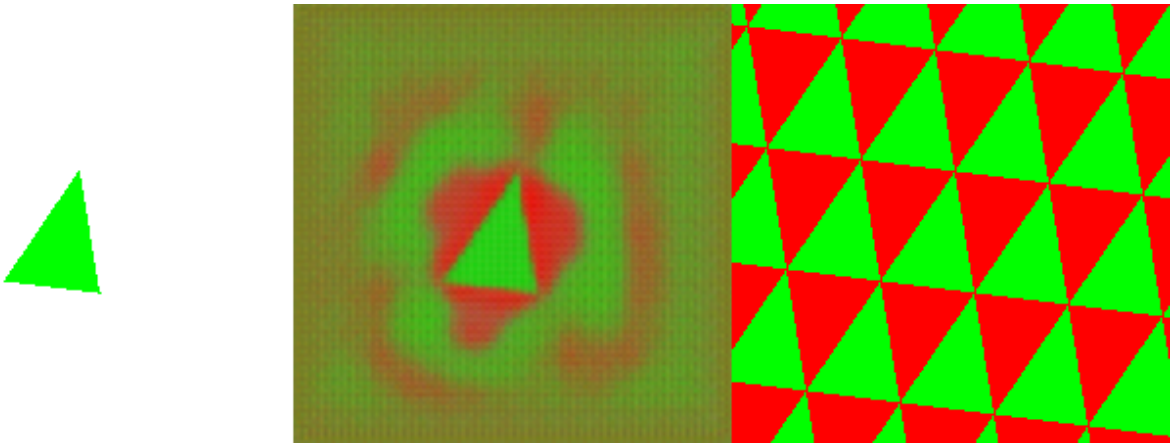
Experimental Results and Analysis: The final training results show that the model has completely learned the rules of plane tessellation. This indicates that the model possesses image reasoning capabilities and largely rules out the possibility that the model is performing

naive interpolation. At the same time, the model provides direct observation of the formation process of the machine mind. It can be seen that a very obvious learning process expanding from around the input triangle to the outside appears in this example.

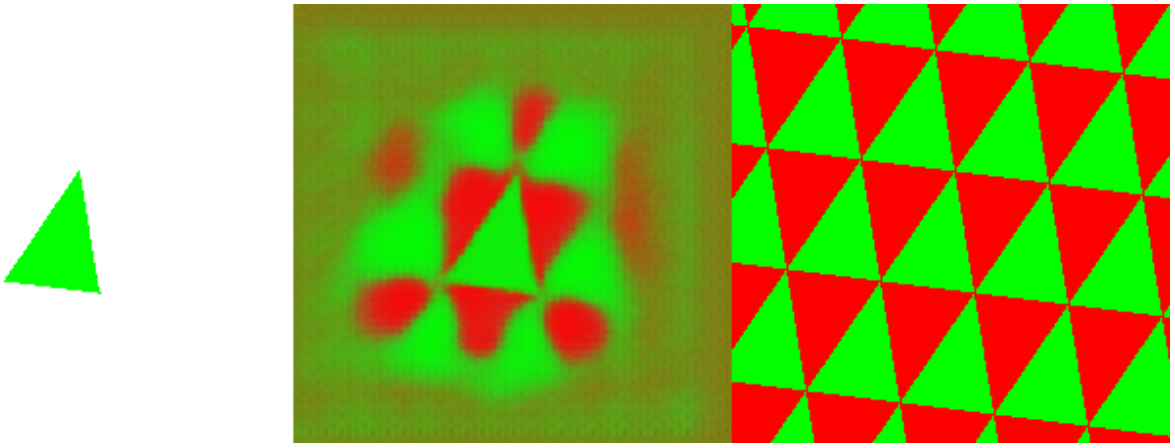
Experimental Process Display:

Left is the input image, right is the ground truth, middle is the generated image.

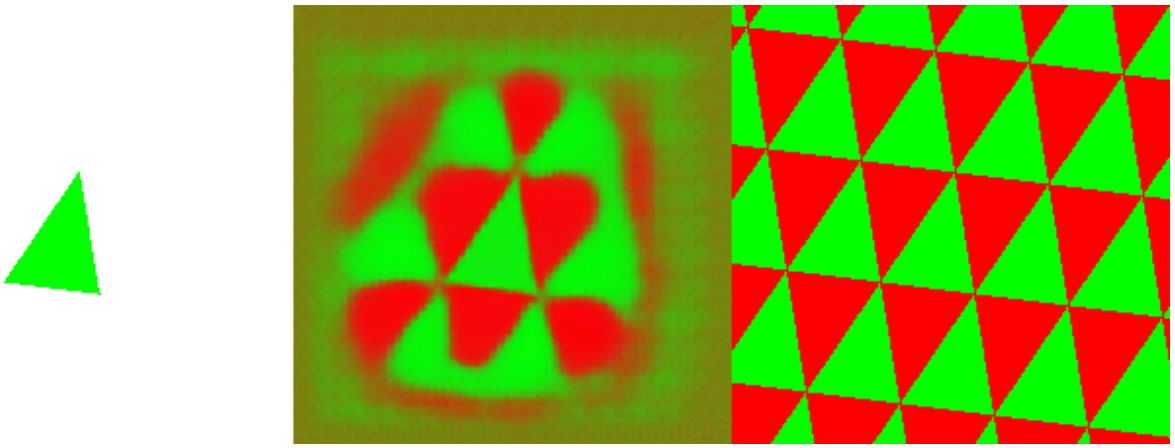
200step



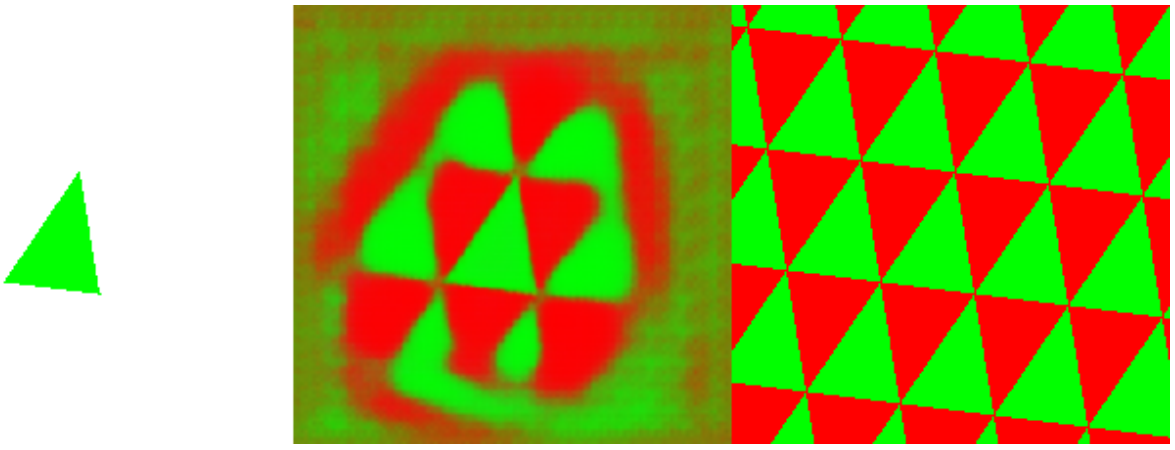
400step



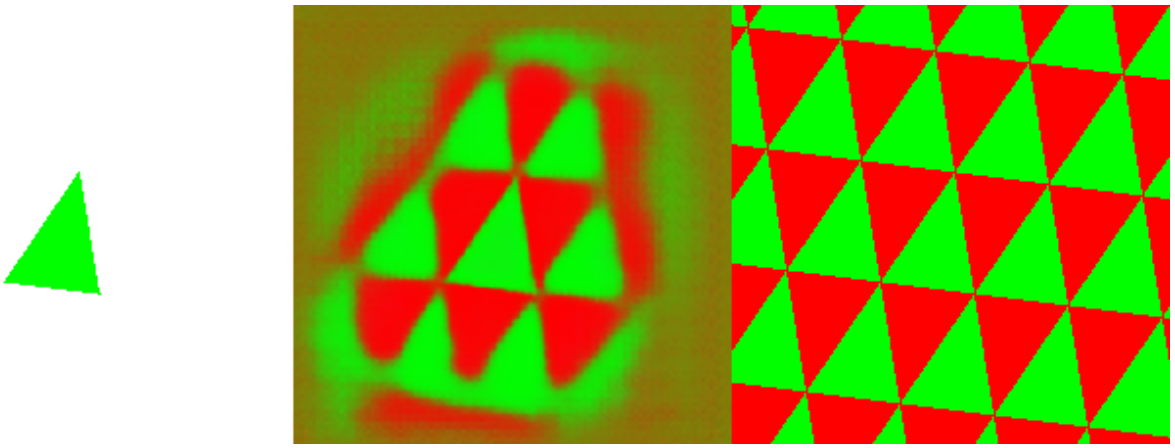
600step



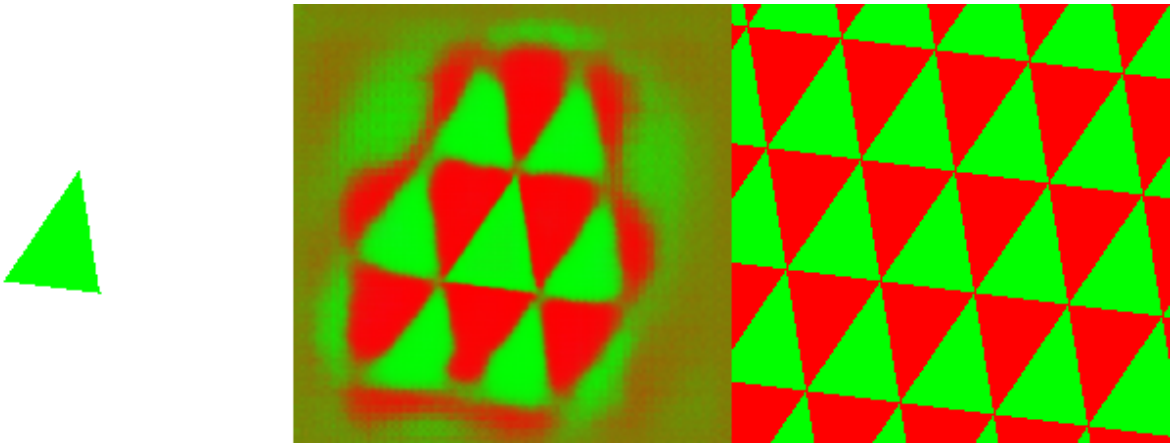
800step



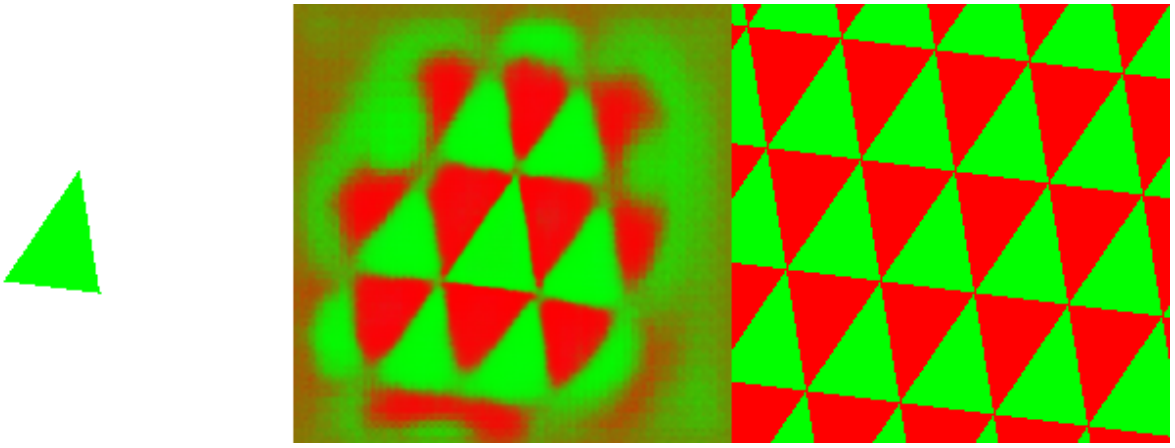
1000step



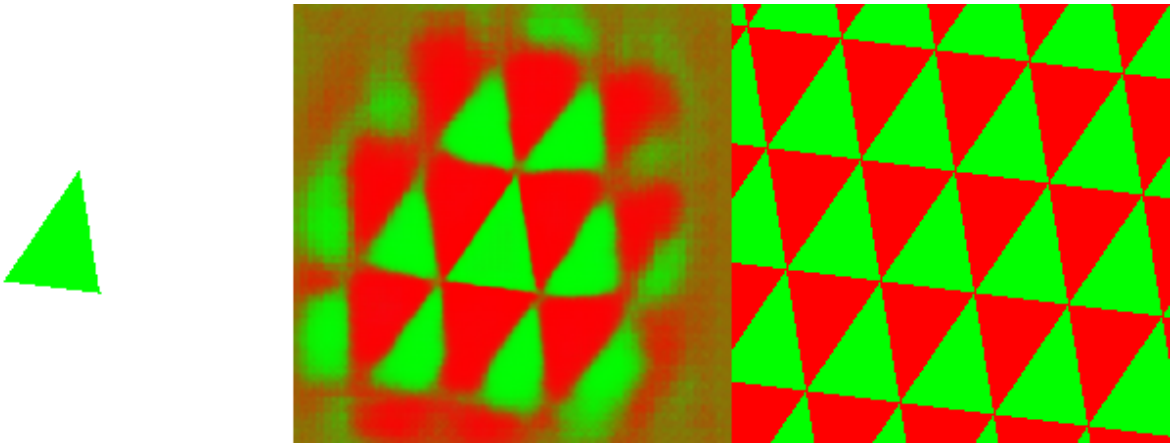
1200step



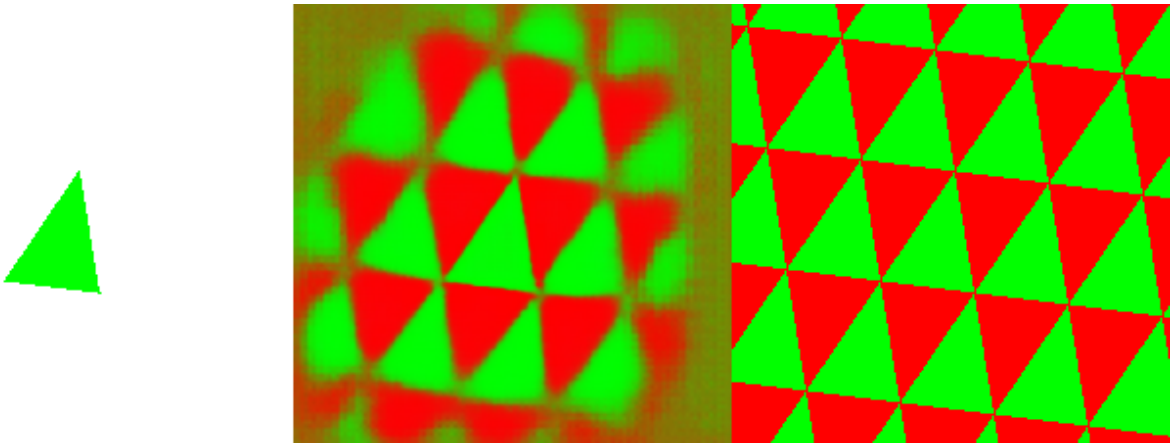
1400step



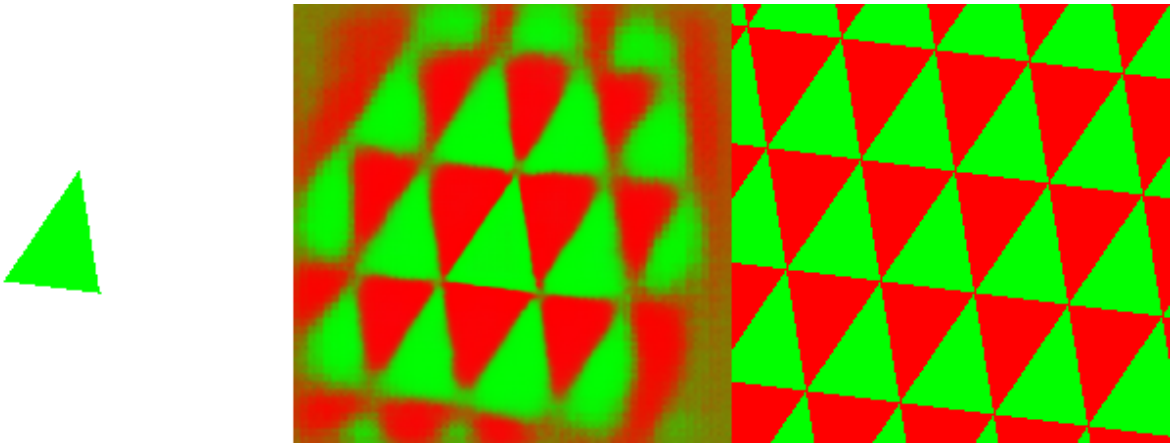
1600step



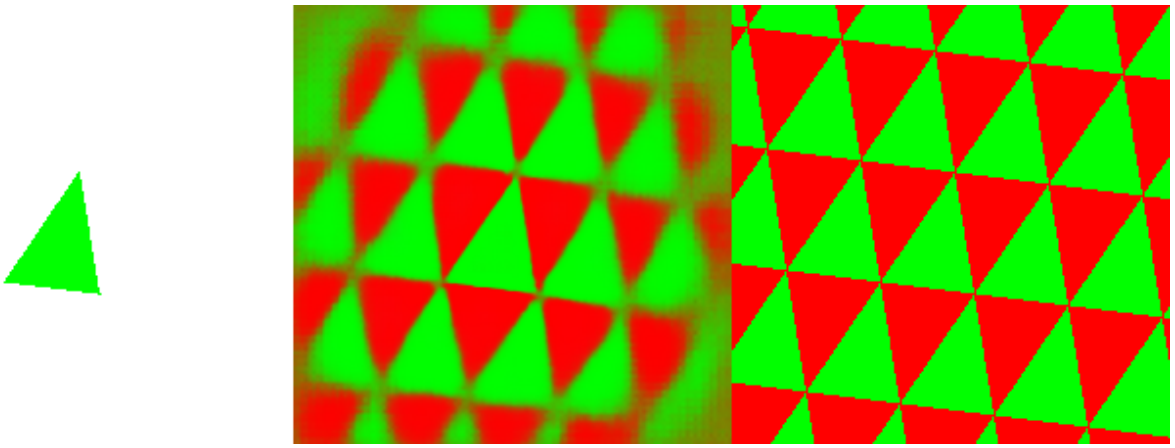
1800step



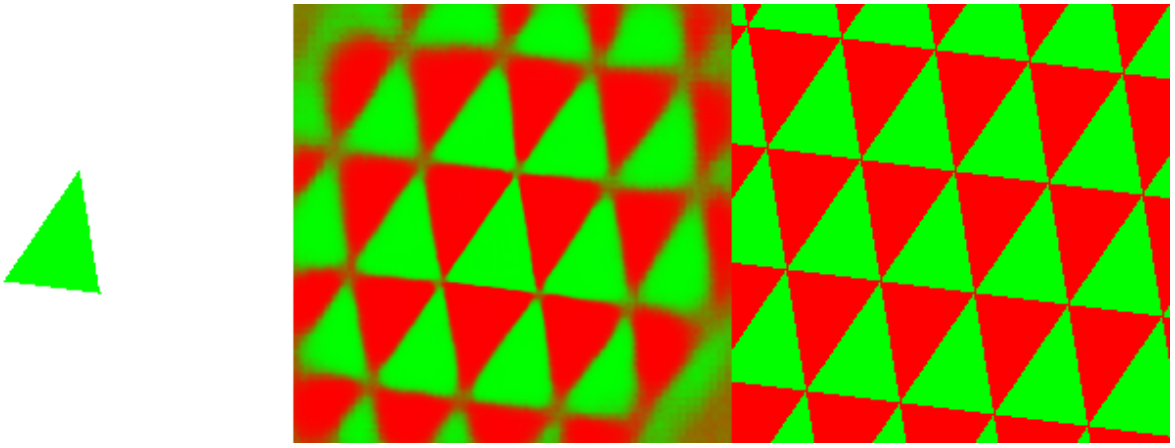
2000step



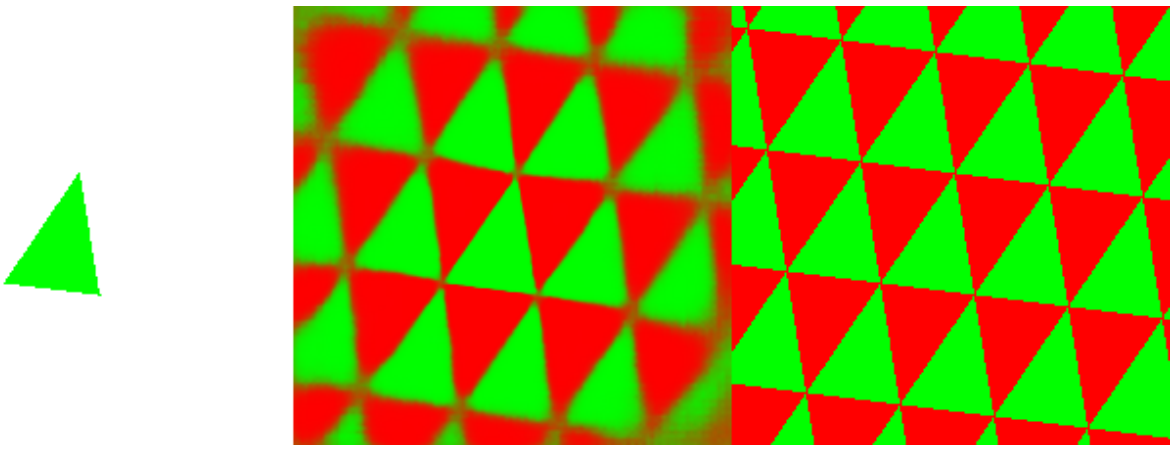
3000step



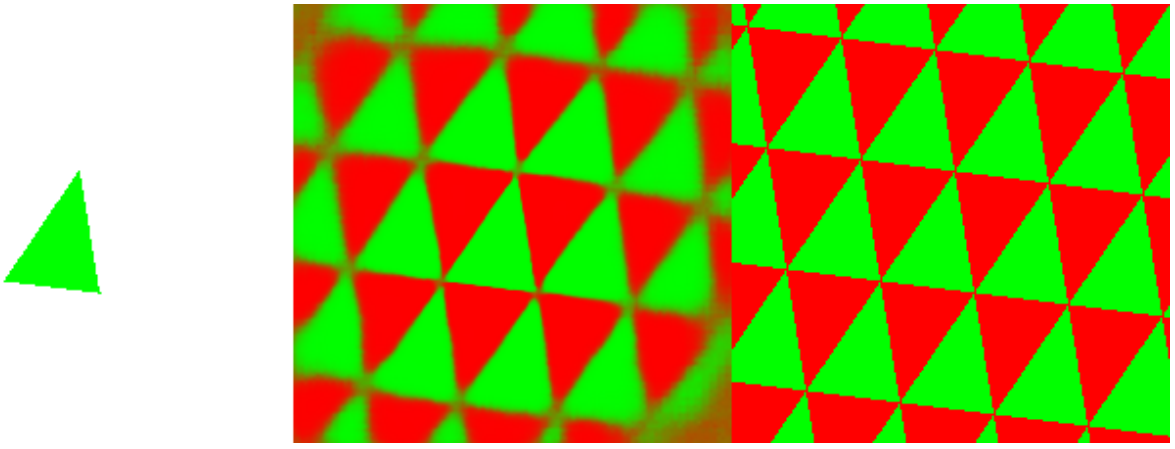
4000step



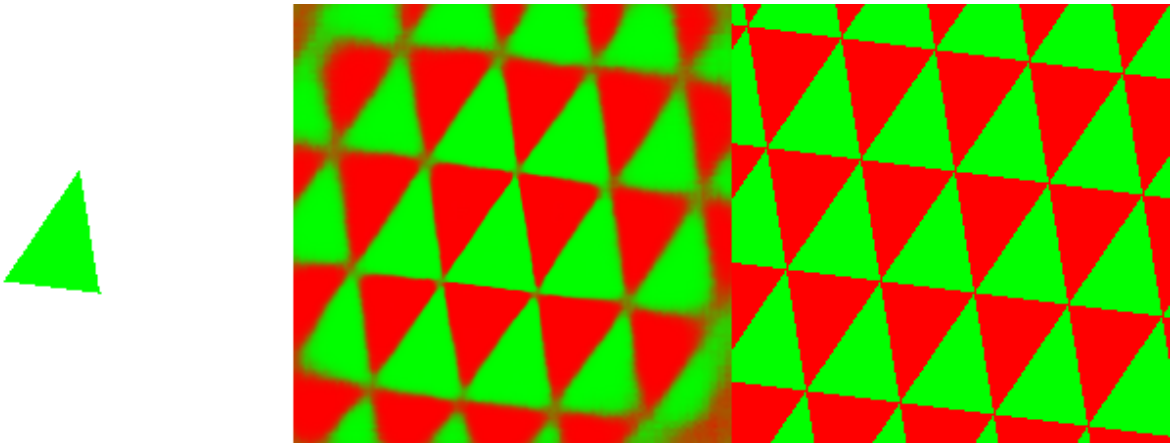
5000step



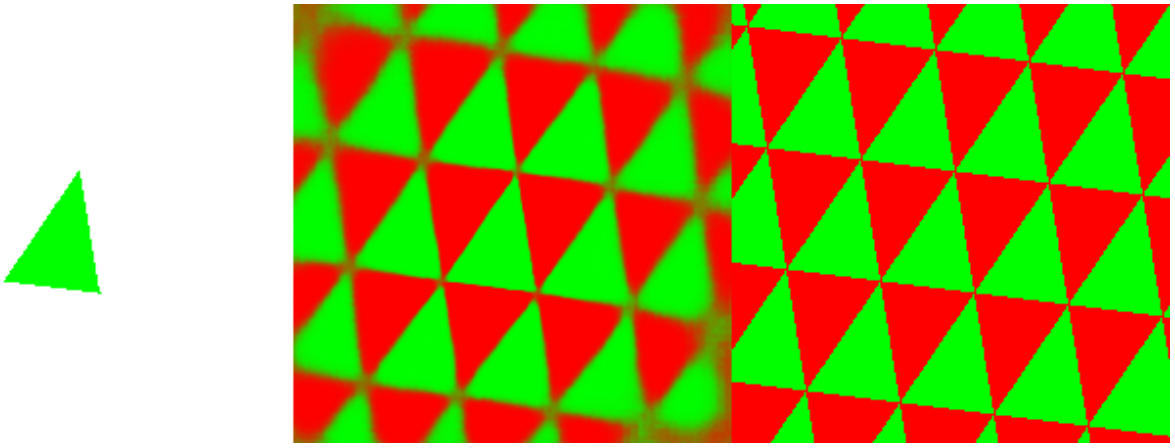
6000step



7000step

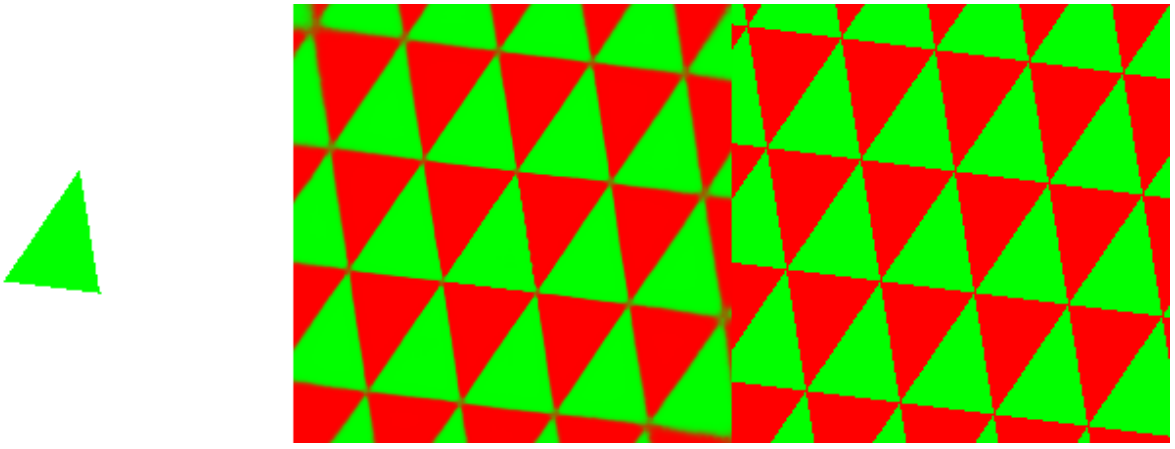


8000step



Due to the slowing down of changes later, the final training result is displayed directly.

150000step



4.5. Image Generation via Symbolic Reasoning

This section will demonstrate the capability of this paradigm to process symbolic inputs, learn the corresponding rules, and generate corresponding images.

4.5.1. Cellular Automata

- **Task Description:** The model learns to predict and output the state of a one-dimensional cellular automata after a given number of transformations according to a given rule.

- **Input Format:** A 36-bit string composed of "0" or "1", representing the initial state of the cellular automata.
- **Output Format:** 240*240 image.
- **Loss:** MSELoss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_cellular_automata_1d_to_grid_image.py`
- **Dataset:** `ca_render_dataset_240` folder.
- **Training Code:** `train_qwen2_text2image.py`
- **Model Used:** transformer + unet.
- **Training Set Size:** 300,000
- **Input Space Size Estimation:** 2 to the power of 36 = 68,719,476,736
- **Ratio of Training Set Size to Input Space Size:** 4.4e-6

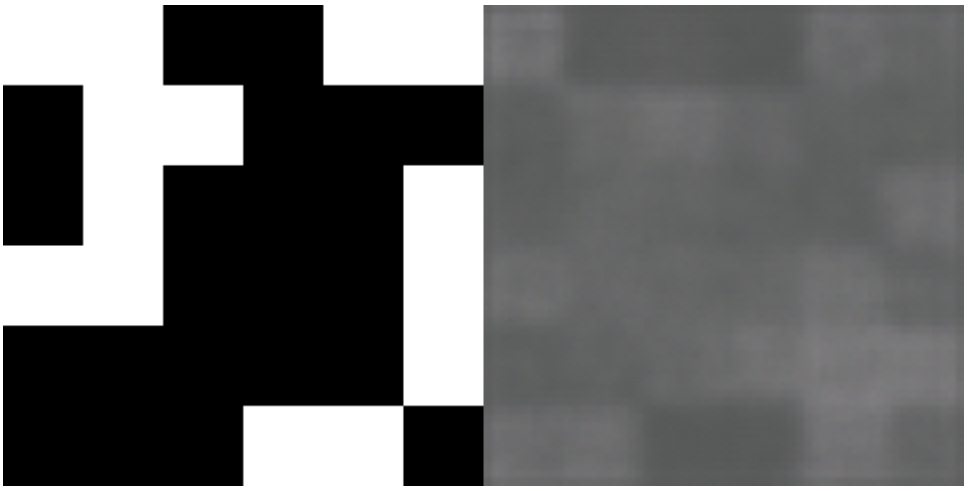
Task-Related Design Thoughts and Discussion:

1. This task is intended to verify the capability of the Transformer for image generation via symbolic reasoning. This mode is not text-to-image in the traditional sense, but focuses on verifying its reasoning ability.
2. In the output image, the 1D 36-bit cellular automata state is arranged in row-major order into a 6*6 checkerboard grid image, where black represents 1 and white represents 0.
3. Some task parameters:
    - Grid dimension: 6*6
    - Grid size: 40*40 pixels
    - Image resolution: 240*240
    - Cellular automata rule: rule 110
    - Rule evolution layers: 3

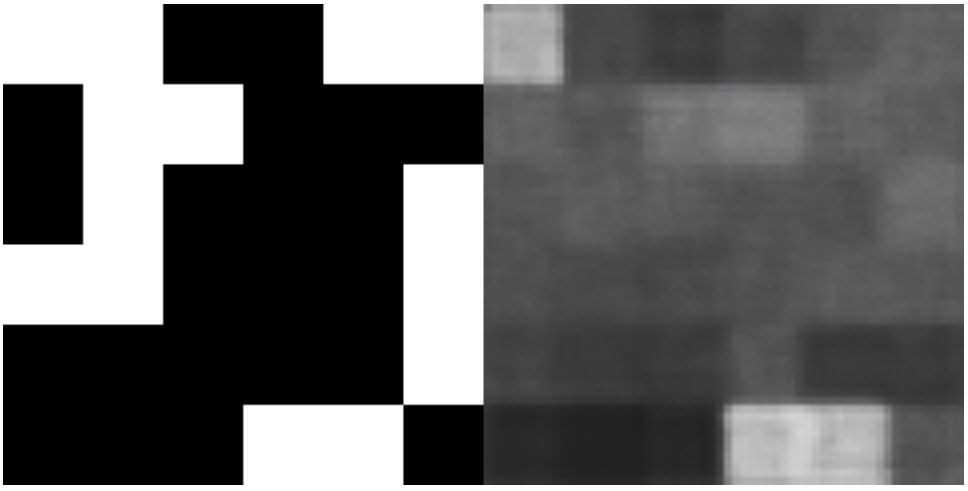
Experimental Results and Analysis: The final training results indicate that the model has completely learned the rules of the cellular automata. This shows that this mode of generating images from symbols can also preserve the reasoning capability of the Transformer. The learning speed of the model is very fast, providing direct observation of the formation process of the machine mind. The entire learning process cannot be simply explained by some logic, further proving that the learning of this paradigm does not necessarily follow the logic that people assume, but rather approximates discreteness in a more primitive connectionist manner.

Left is ground truth, right is model generated image.

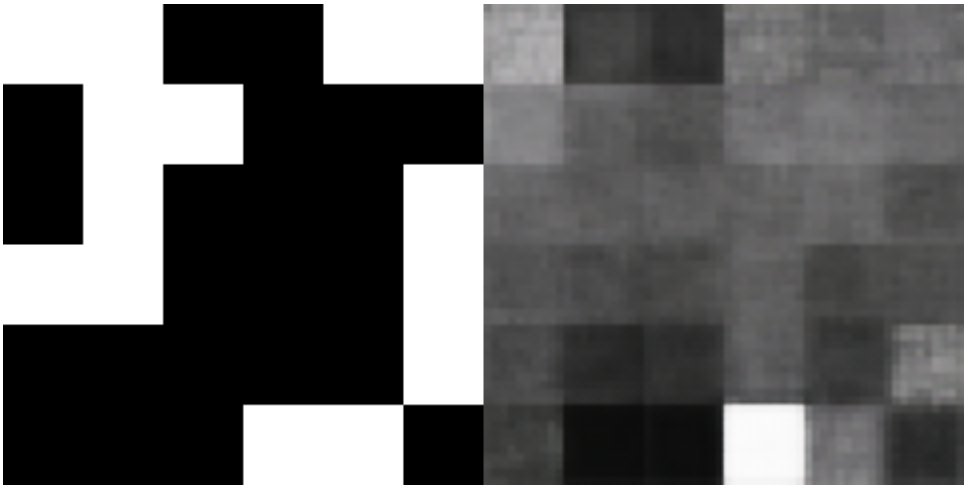
100 step Symbol input: 111110011010001000111010000110



200 step Symbol input: 11111001101000100011101000011O



300 step Symbol input: 11111001101000100011101000011O



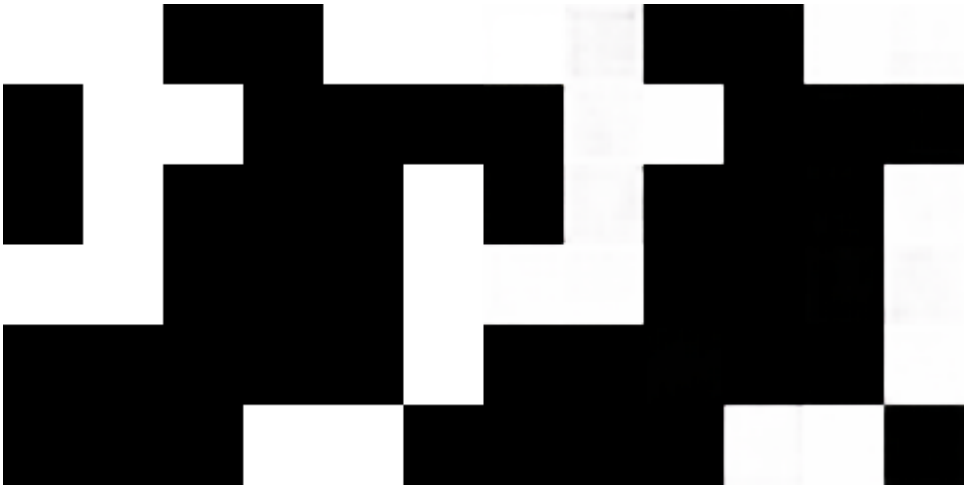
400 step Symbol input: 11111001101000100011101000011O

500 step Symbol input: 11111001101000100011101000011O

600 step Symbol input: 11111001101000100011101000011O

700 step Symbol input: 11111001101000100011101000011O

800 step Symbol input: 11111001101000100011101000011O



4.5.2 Rotating Cube

- **Task Description:** The model learns to predict and output an image of a colored hexahedron rotated according to rotation angles.
- **Input Format:** A 24-bit string composed of "0" or "1", representing 3 rotation angles. Every consecutive 8 bits represent one rotation angle.
- **Output Format:** 256*256 image.
- **Loss:** MSELoss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_cube_rotation_pillow_with_anchor.py`
- **Dataset:** `cube_dataset_final_highlight` folder.
- **Training Code:** `train_qwen2_text2image.py`
- **Model Used:** transformer + unet.
- **Training Set Size:** 300,000
- **Input Space Size Estimation:** 16,777,216
- **Ratio of Training Set Size to Input Space Size:** 1.79%

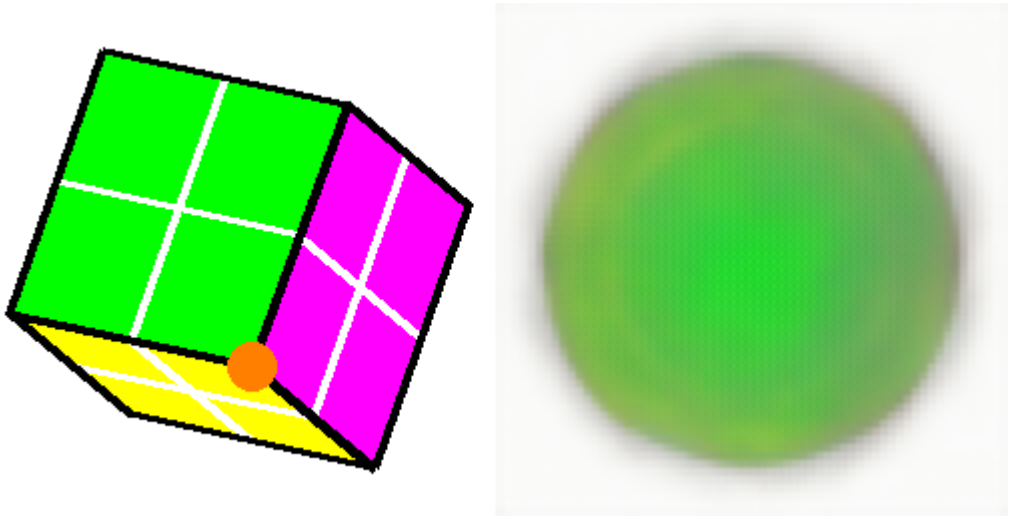
Task-Related Design Thoughts and Discussion:

1. In this task, a special difficulty was encountered: the model found it hard to associate the vertices corresponding to the rotation angles with the symbolic input, leading to a failure to converge. The solution was to always highlight a special vertex in the final step of generating training set images, regardless of occlusion relationships. After adopting this solution, the model was finally able to converge on this task. In another task where I used symbols to indicate the position of triangles and output the plane tessellation result, the model encountered the same difficulty; it could not understand which green triangle in the output image the symbol corresponded to. The solution was similar: I changed the color of the triangle indicated by the symbolic input to black in the output image, and everything became normal. This should be a fairly typical problem, i.e., it is difficult for the model to determine how to map symbolic input to a specific feature in the output image, and the solutions should also be similar.
2. Regarding the white cross lines on each face, they are just for aesthetics and may also have some auxiliary training effects.
3. The 3 rotation angles of symbolic input are pitch, yaw, and roll; please refer to the training set generation code for specific correspondence.
4. Actually, I have another version of the model that does not use the method of lora fine-tuning qwen2-0.5b, but uses a transformer with fewer total parameters than qwen2-0.5b but more parameters than the lora parameters of qwen2-0.5b. Practice has proven that the effect is not as good as the learning effect of qwen2-0.5b. This paper considers this an interesting phenomenon.
5. Some task parameters:
    - Image resolution: 256*256

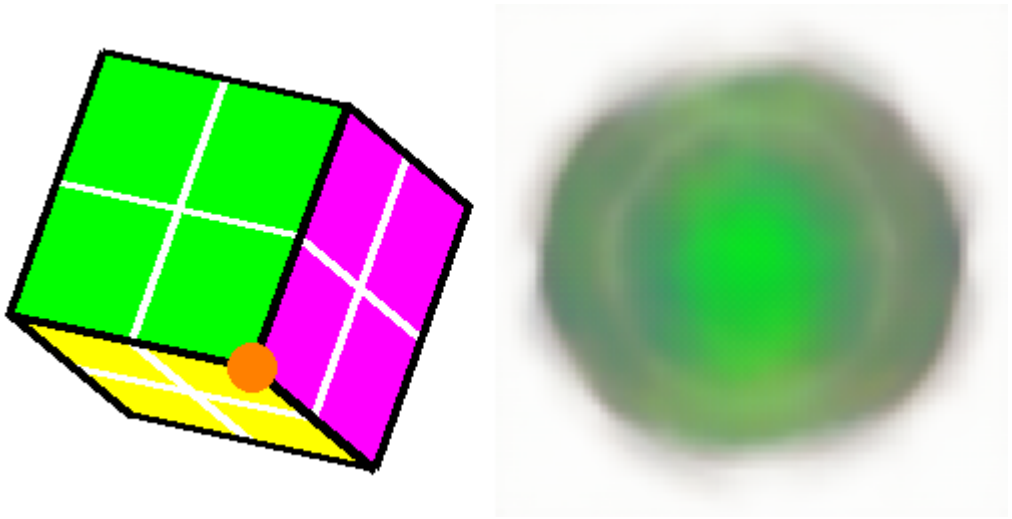
Experimental Results and Analysis: The final training results show that the model has completely learned this task. This indicates that this mode of generating images from symbols can also preserve the image reasoning capability of the Transformer. And given that I used lora fine-tuning of qwen2-0.5b, it can be said that this image reasoning capability, at least the capability to generate images, is not limited to image transformers like Swin Transformer, ViT, etc. The learning speed of the model is not very fast, possibly because the task is relatively difficult, providing direct observation of the formation process of the machine mind. In this task, can be seen that at first, the model did not yet understand rotation angles and simply learned to generate circles and relatively averaged filling colors. Later, the model slowly understood the concept of rotating cubes, and even later, the model slowly filled in precise details.

Left is ground truth, right is model generated image.

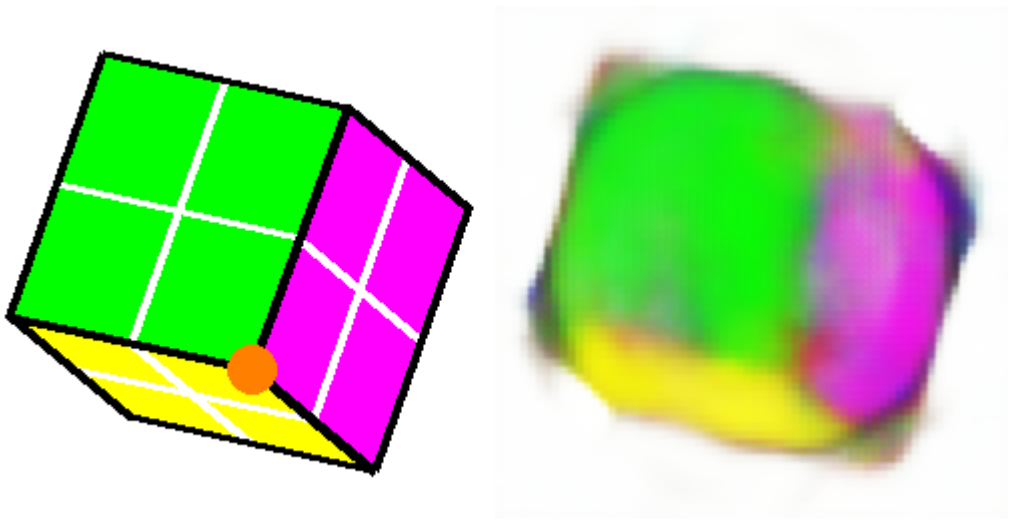
step 500 Symbol input: 100101101000100100110010



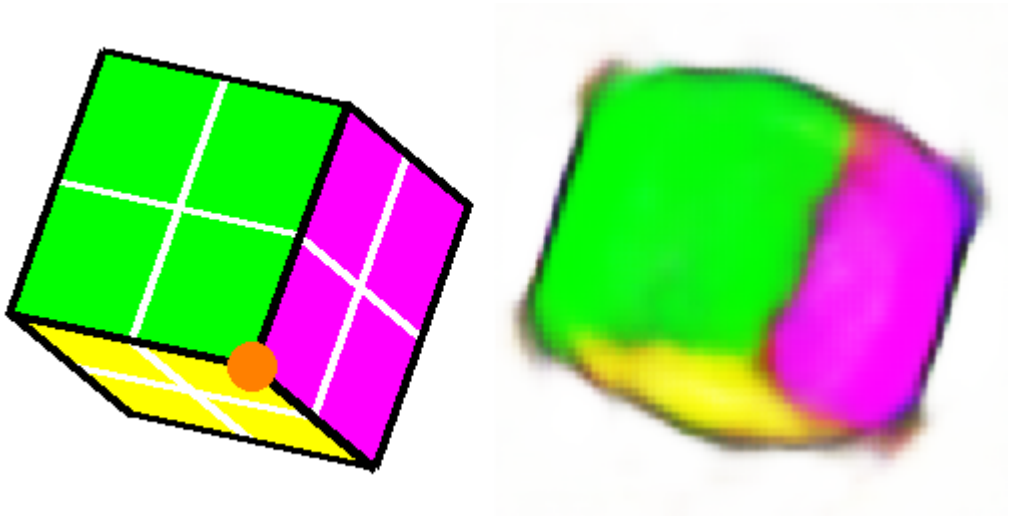
step 1000 Symbol input: 100101101000100100110010



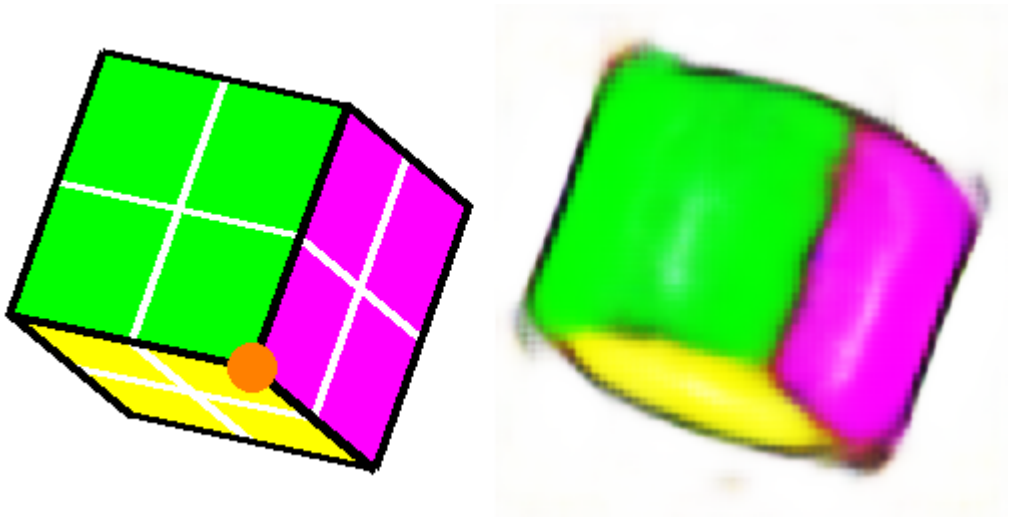
step 1500 Symbol input: 100101101000100100110010



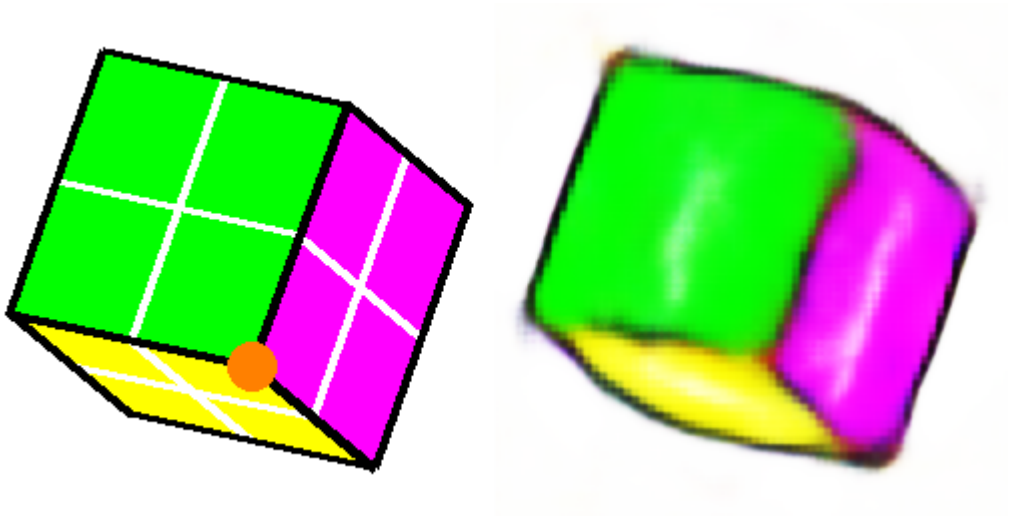
step 2000 Symbol input: 100101101000100100110010



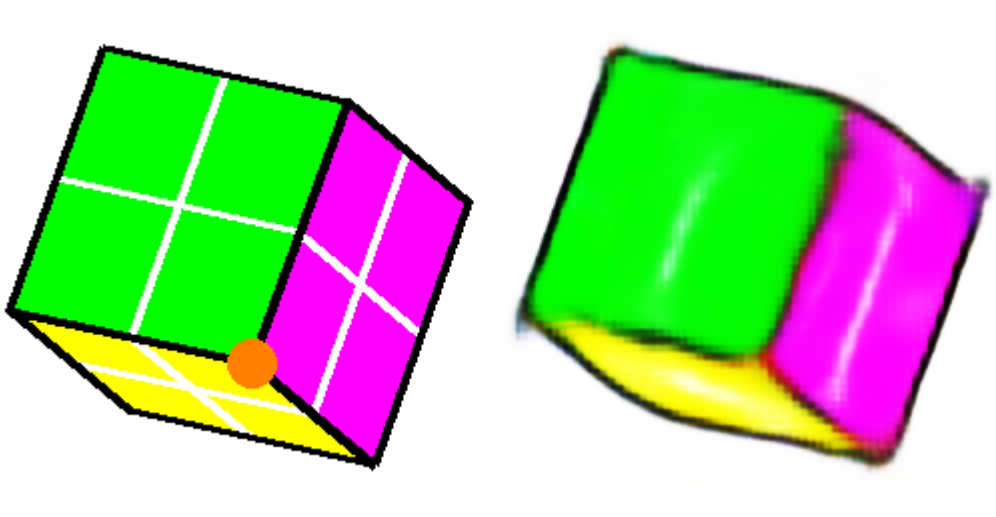
step 2500 Symbol input: 100101101000100100110010



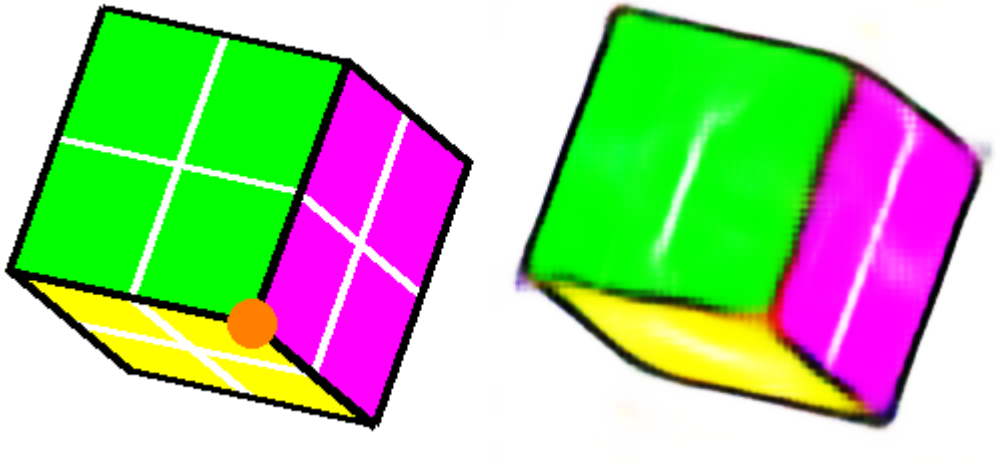
step 3000 Symbol input: 100101101000100100110010



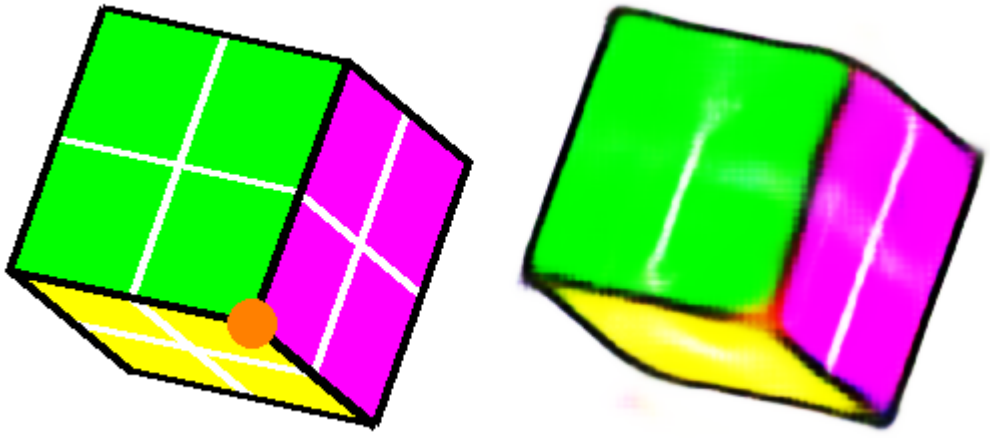
step 4000 Symbol input: 100101101000100100110010



step 6000 Symbol input: 100101101000100100110010

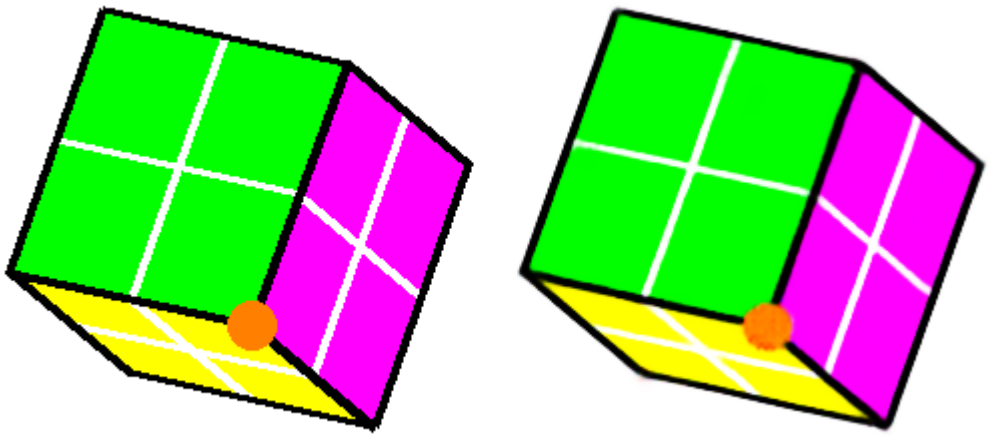


step 8000 Symbol input: 100101101000100100110010



Due to the slowing down of changes later, the final training result is displayed directly.

step 133600 Symbol input: 100101101000100100110010



4.6. Physical Simulation

This section will demonstrate the capability of this paradigm in simulating physical processes.

4.6.1 Catenary

- **Task Description:** Given the endpoints at both ends of a catenary, and a given passing point on the catenary, the model learns to draw the catenary trajectory uniquely determined by physical laws (principle of minimum potential energy).
- **Input Format:** 224*224 image.
- **Output Format:** 224*224 image.
- **Loss:** MSELoss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_catenary_curve_from_points.py`
- **Dataset:** `catenary_dataset_v5_CONSTRUCTIVE` folder.
- **Training Code:** `train_image2image.py`
- **Model Used:** Swin Transformer + unet.
- **Training Set Size:** 150,000
- **Input Space Size Estimation:** Estimated using Monte Carlo method to be approx. 9.3e9.
- **Ratio of Training Set Size to Input Space Size:** Approx. 1.61e-5

Task-Related Design Thoughts and Discussion:

1. This task is intended to verify the physics-related simulation capability of this paradigm, implemented through an image reasoning model.
2. Although this problem has the possibility of being solved by interpolation, the model and hyperparameter configuration used simultaneously solved other problems that are impossible to solve by interpolation, thus ruling out this possibility.
3. The reason for adding the restriction of a catenary passing point is mainly because setting the catenary to a fixed length or representing the catenary length by some other

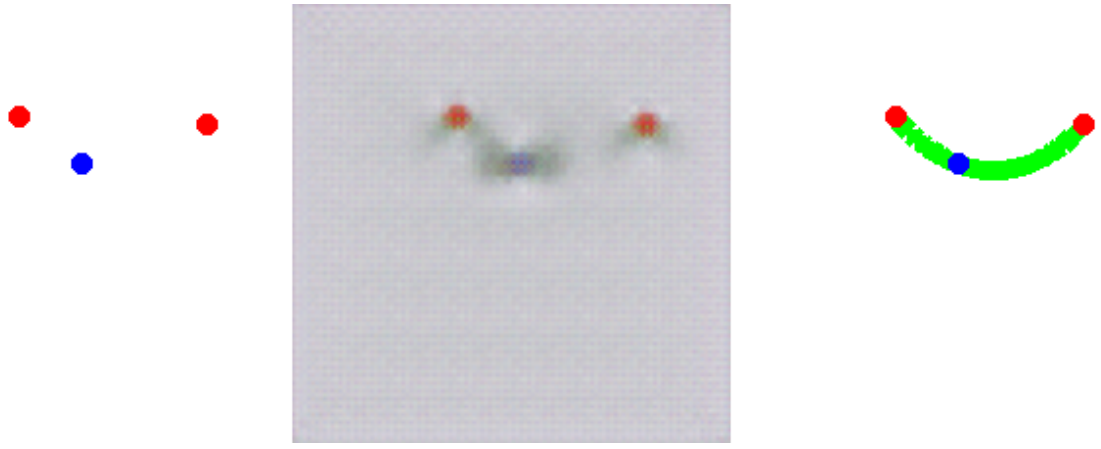
means felt less optimal than this setup. Furthermore, adding the passing point restriction also increased the input space, further ruling out the possibility of memorization or interpolation.

4. Some task parameters:
    - Image resolution: 224*224

Experimental Results and Analysis: The final training results indicate that the model has completely learned the laws of the catenary. This shows that the learning capability of this paradigm has expanded from pure mathematical and logical rules to the simulation of fundamental physical laws of our real universe, but at the same time, the precision of such simulation needs further investigation. The model's learning speed is very fast, providing direct observation of the formation process of the machine mind. It can be seen that the learning process revealed in this example did not spend too much time on the trajectory of the catenary, but mainly on optimizing the image output. At the same time, it was noted that the training set I generated was flawed, with burrs in the output catenary images, while the output of the model after convergence was relatively smooth. This phenomenon is worth further study.

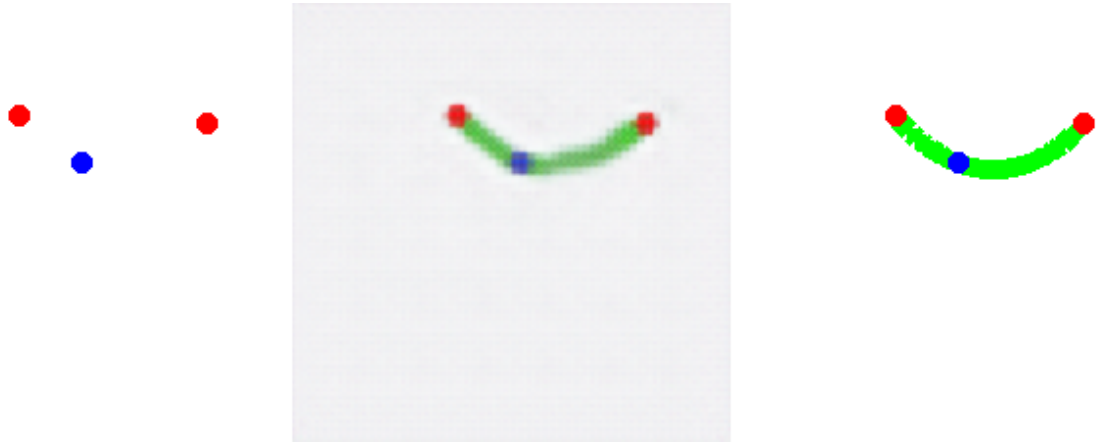
Left is input image, middle is model generated image, right is ground truth.

200step



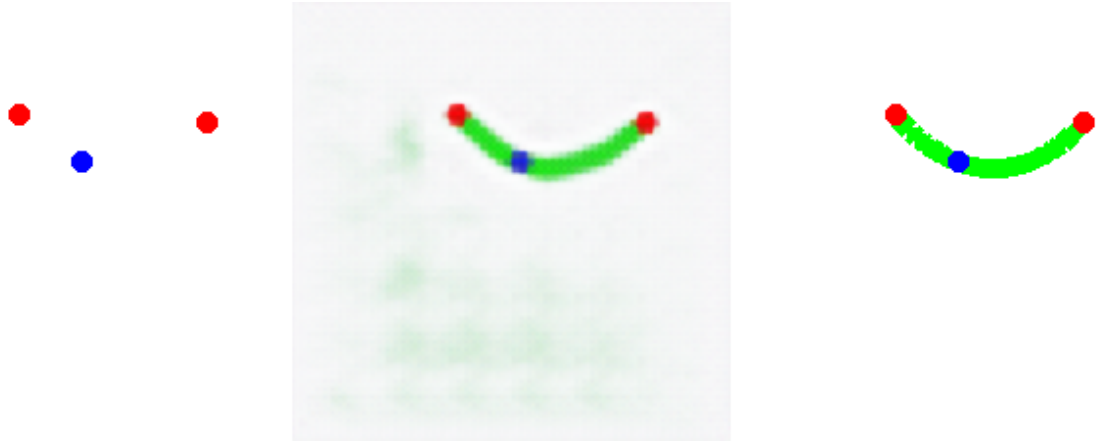
400step

600step

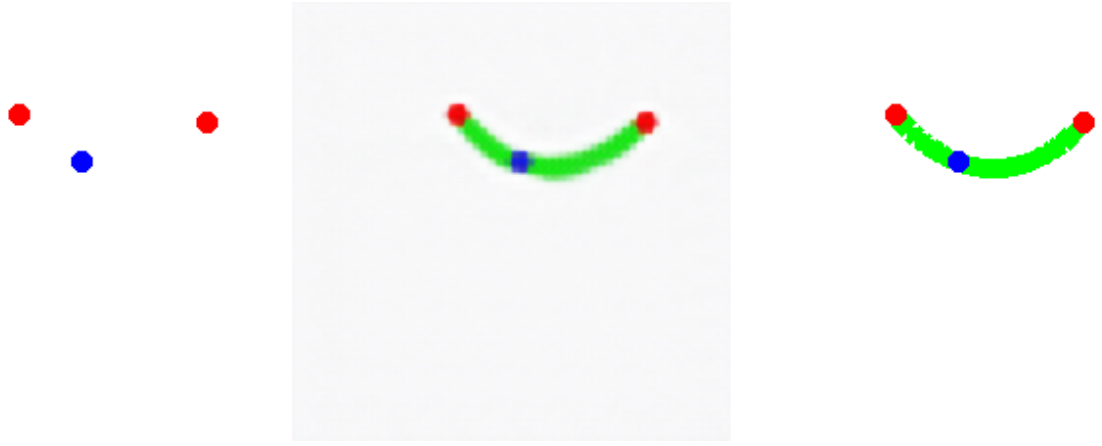


800step

1000step



1200step



1400step

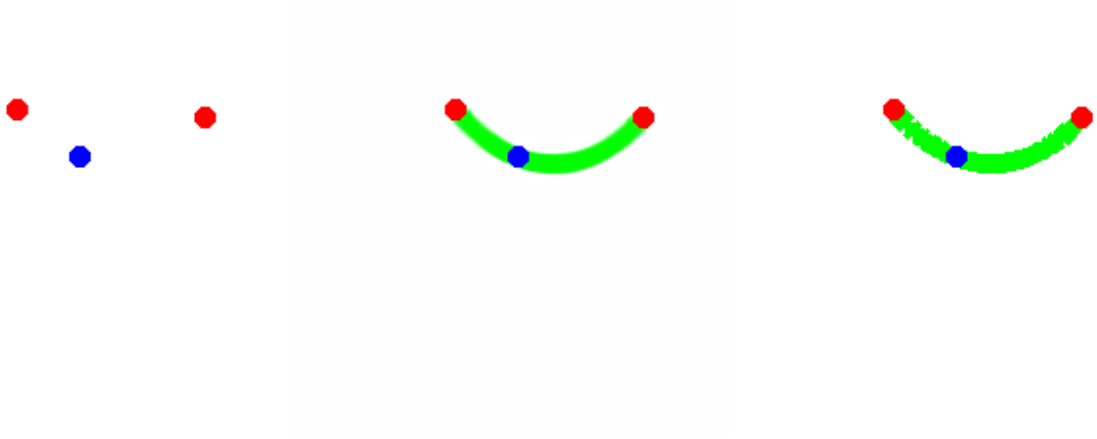
1600step

1800step

2000step

Due to the slowing down of changes later, the final training result is displayed directly.

19000step



4.6.2. Planetary Orbit Simulation

- **Task Description:** Given the fixed position of a star, the initial position of a planet, the magnitude of the initial velocity, and the direction of the initial velocity, predict the planetary orbit.
- **Input Format:** 224*224 image.
- **Output Format:** 224*224 image.
- **Loss:** MSELoss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_orbital_path_from_initial_state.py`
- **Dataset:** `orbital_dataset_256_separated_v3` folder.
- **Training Code:** `train_image2image.py`
- **Model Used:** Swin Transformer + unet.
- **Training Set Size:** 150,000
- **Input Space Size Estimation:** Estimated using Monte Carlo method to be approx. 2.8e11.
- **Ratio of Training Set Size to Input Space Size:** Approx. 5.36e-7

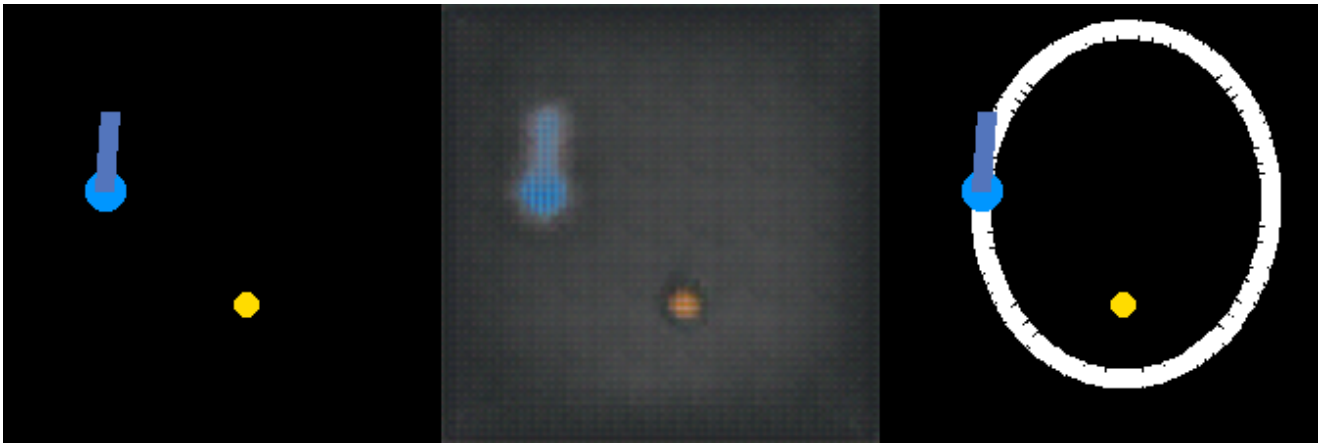
Task-Related Design Thoughts and Discussion:

1. This task is intended to further verify the physics-related simulation capability of this paradigm, implemented through an image reasoning model.
2. Although this problem has the possibility of being solved by interpolation, the model and hyperparameter configuration used simultaneously solved other problems that are impossible to solve by interpolation, thus ruling out this possibility.
3. The fixed position of the star and the initial position of the planet are relatively easy to represent in the input image. Regarding the magnitude and direction of the planet's initial velocity, I represent it in this way: a short ray segment is emitted from the planet's initial position, with color representing the velocity magnitude and the direction of the segment representing the velocity direction. For the correspondence between velocity magnitude and color, see the training set generation script code.
4. Some task parameters:
    - Image resolution: 224*224

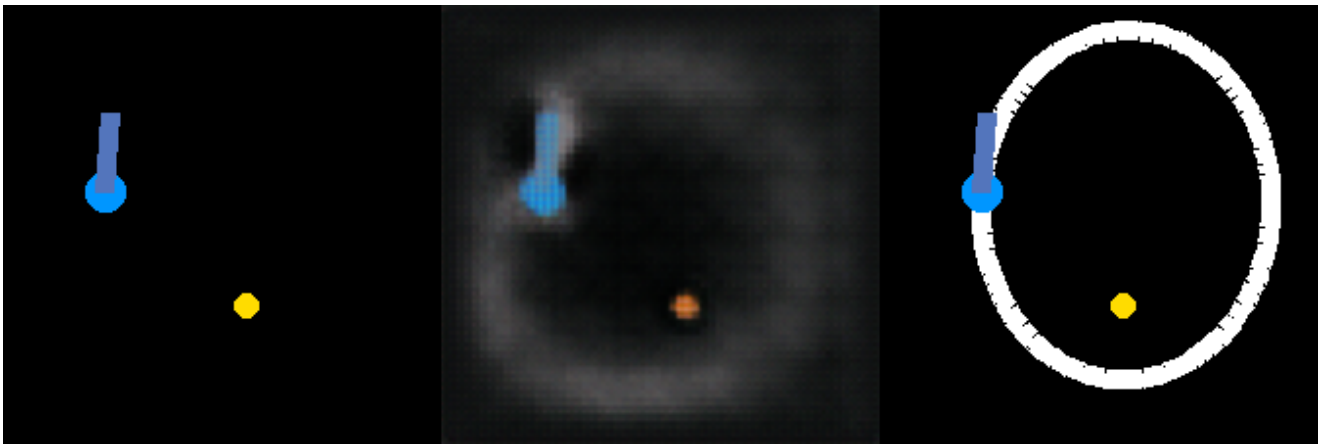
Experimental Results and Analysis: The final training results indicate that the model has completely learned the laws related to planetary orbits, but at the same time, the precision of such simulation needs further investigation. The model provides direct observation of the formation process of the machine mind. The learning process of this task reveals an interesting phenomenon: since the orbit in the output image is set to white, during the learning process, especially in the early stages, this whiteness diffuses in the output image, reminiscent of probability clouds in quantum mechanics.

Left is input image, middle is model generated image, right is ground truth.

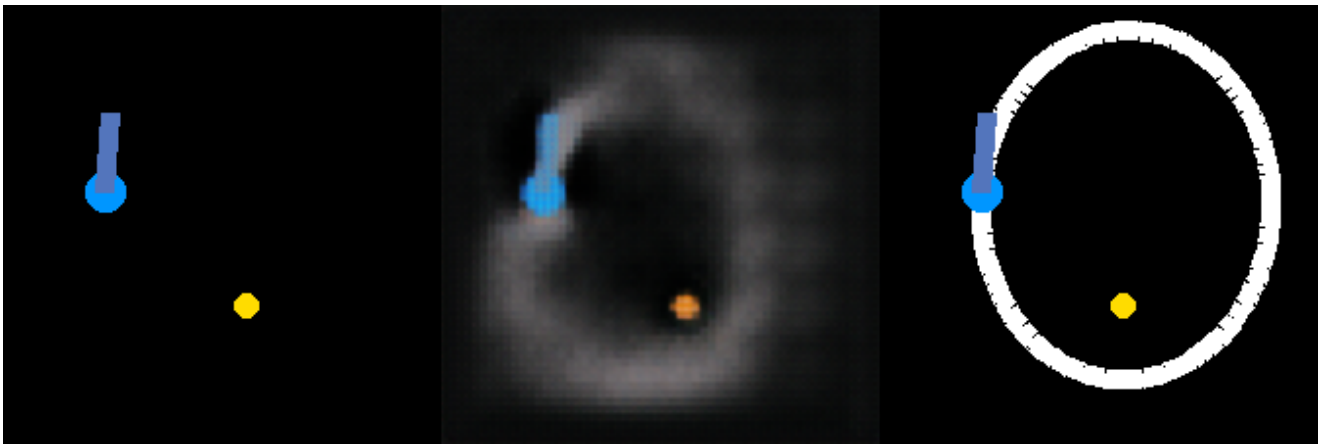
200step



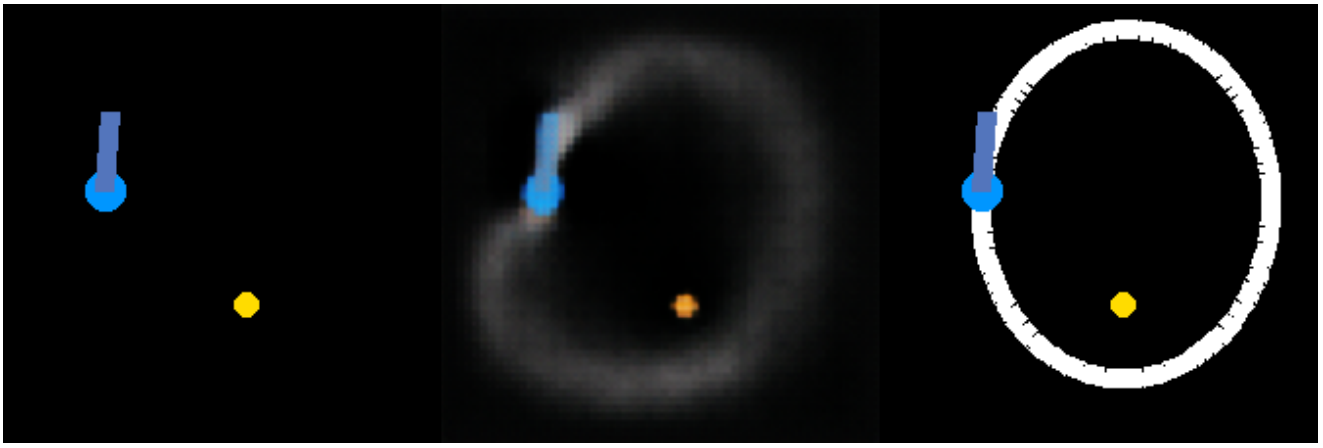
400step



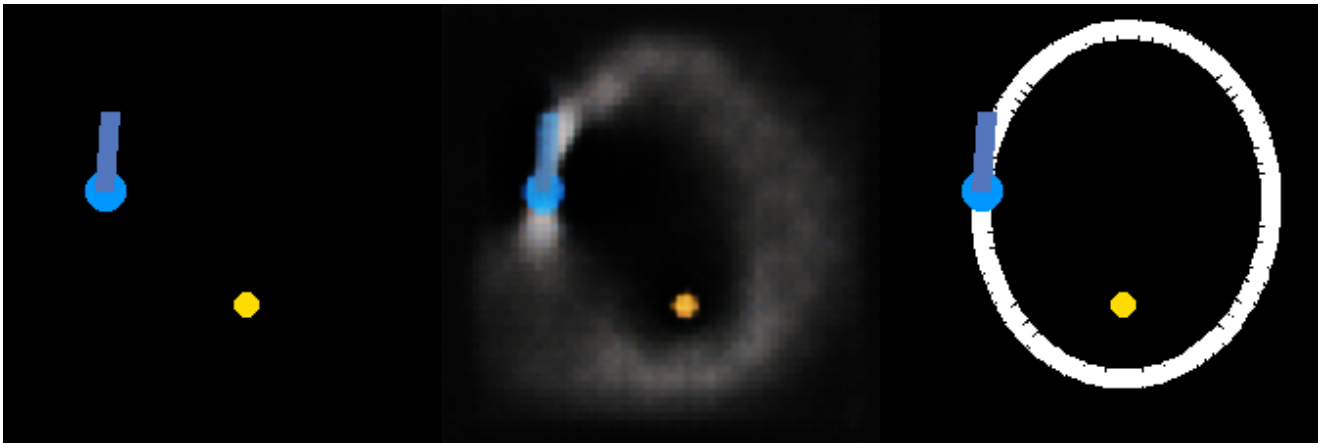
600step



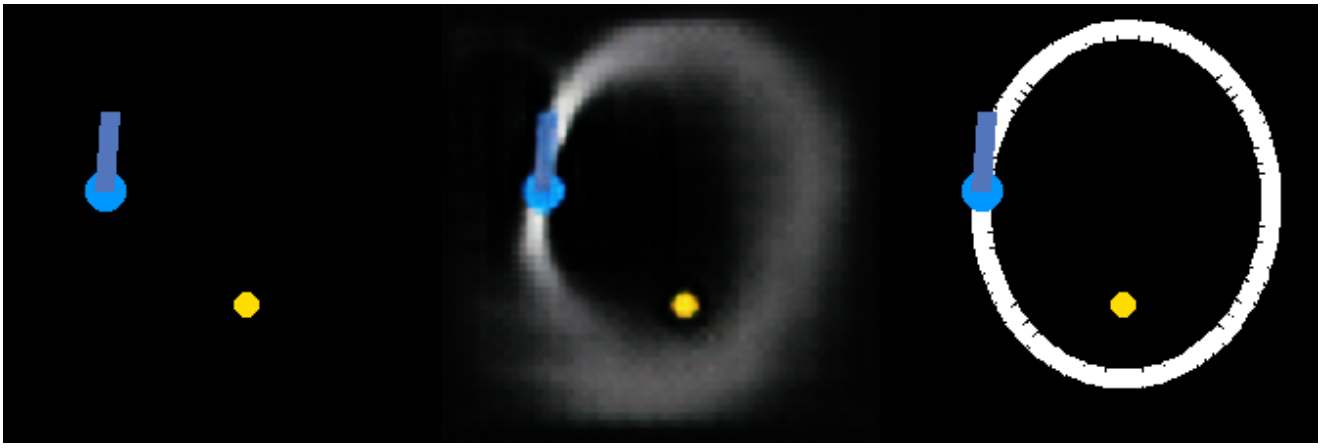
800step



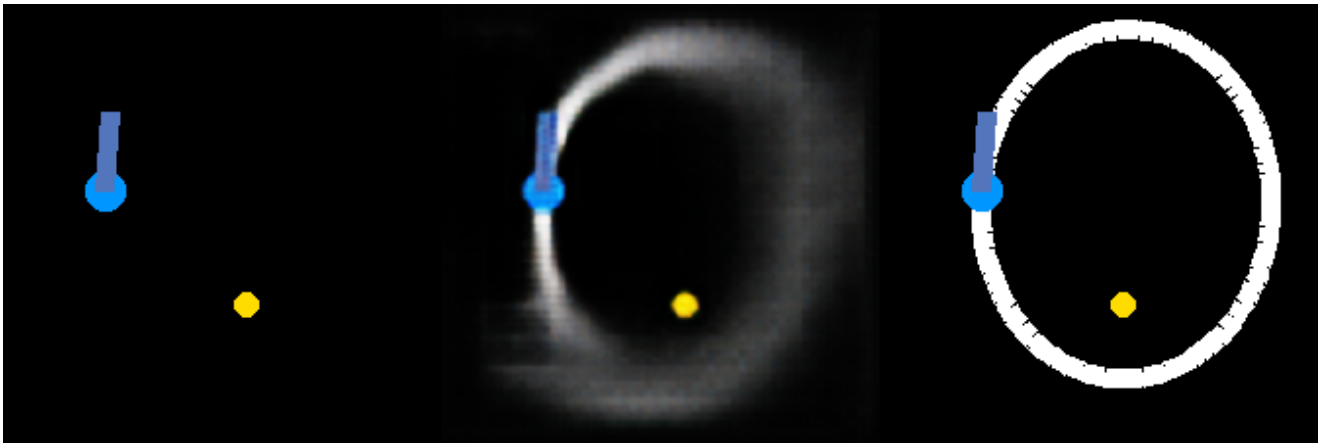
1000step



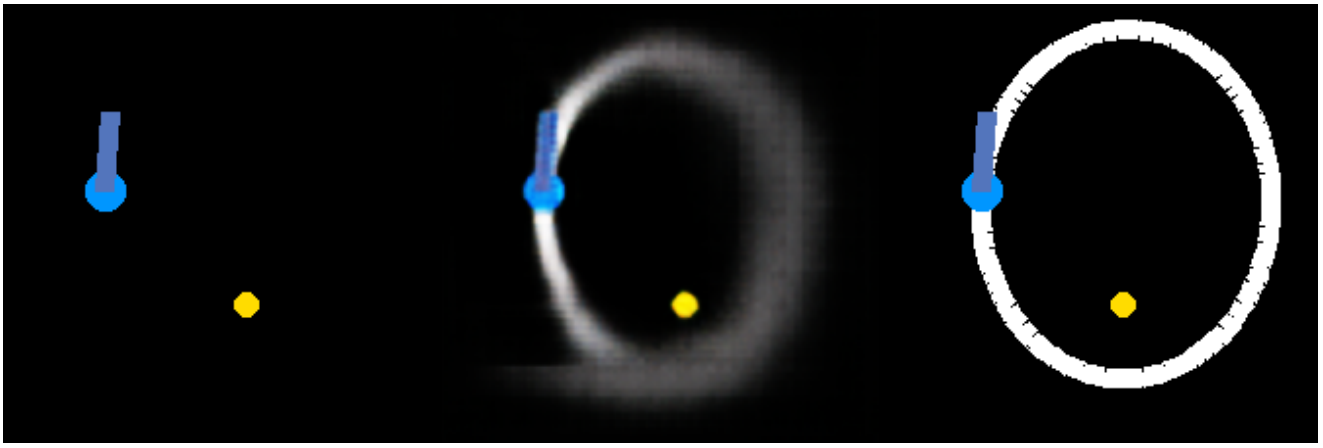
2000step



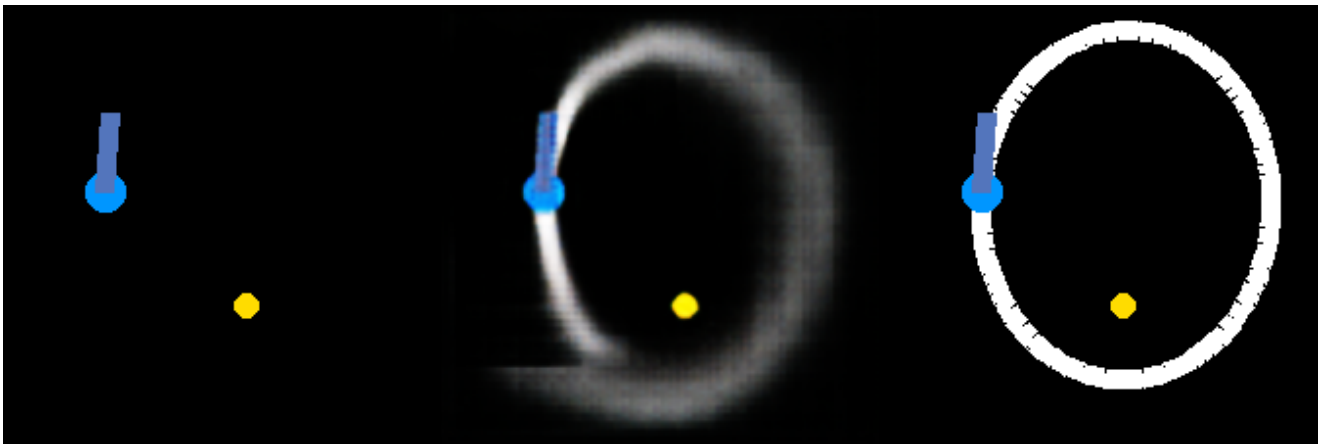
3000step



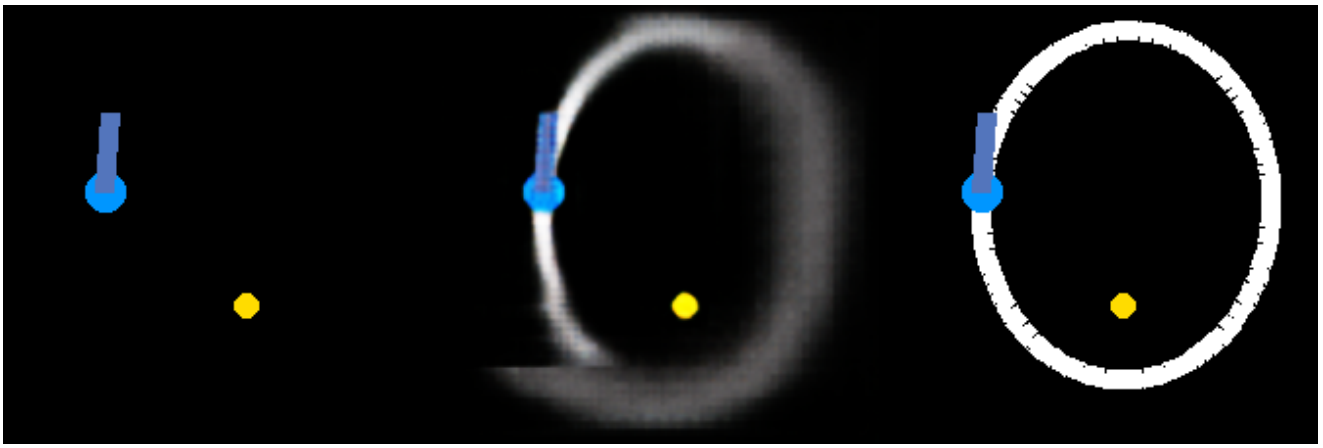
4000step



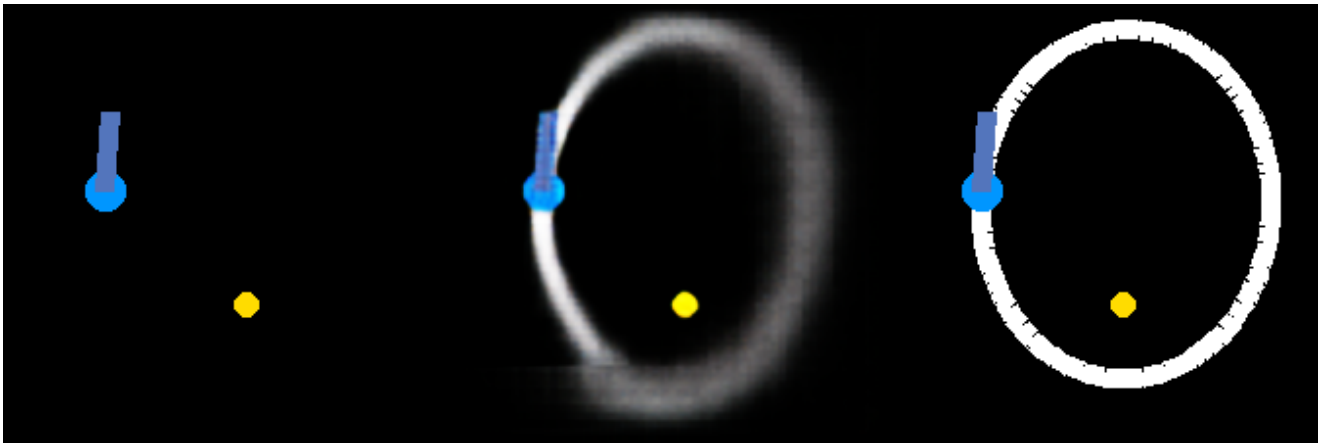
5000step



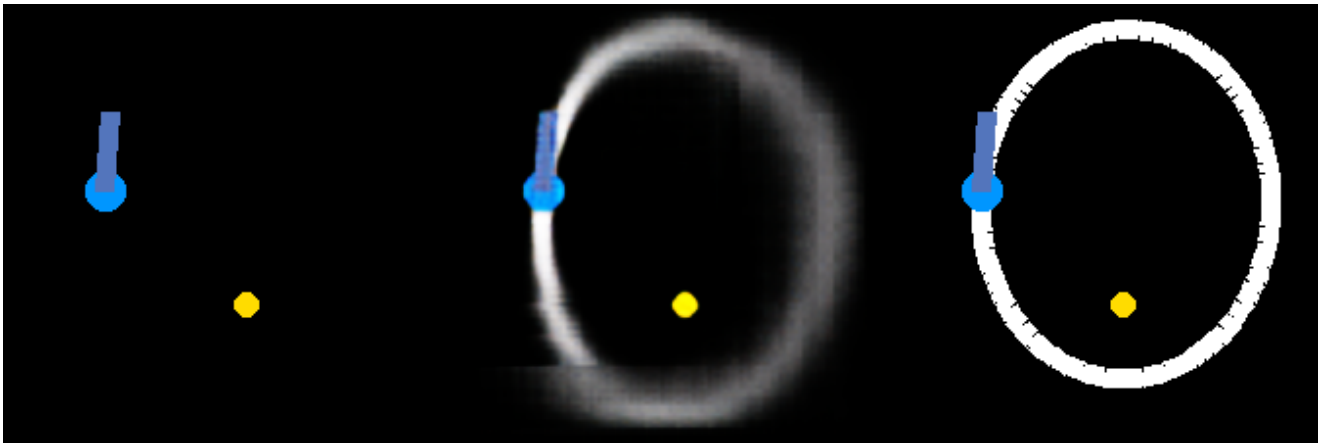
6000step



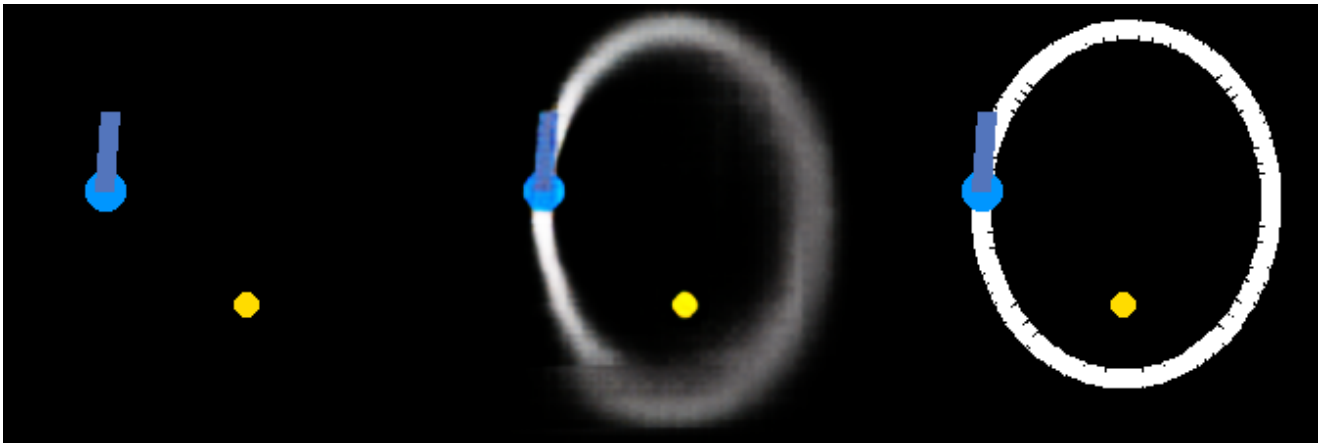
7000step



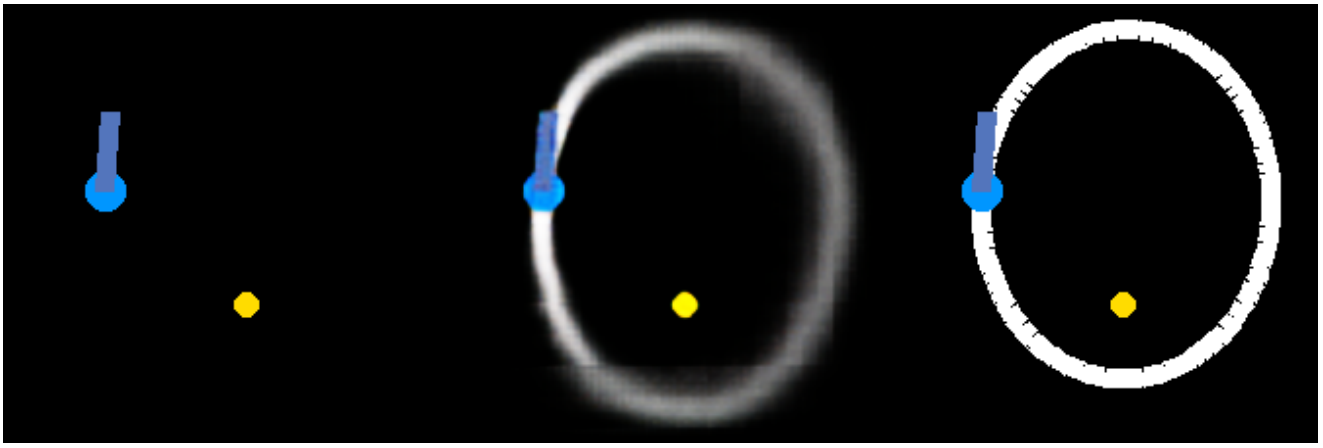
8000step



9000step

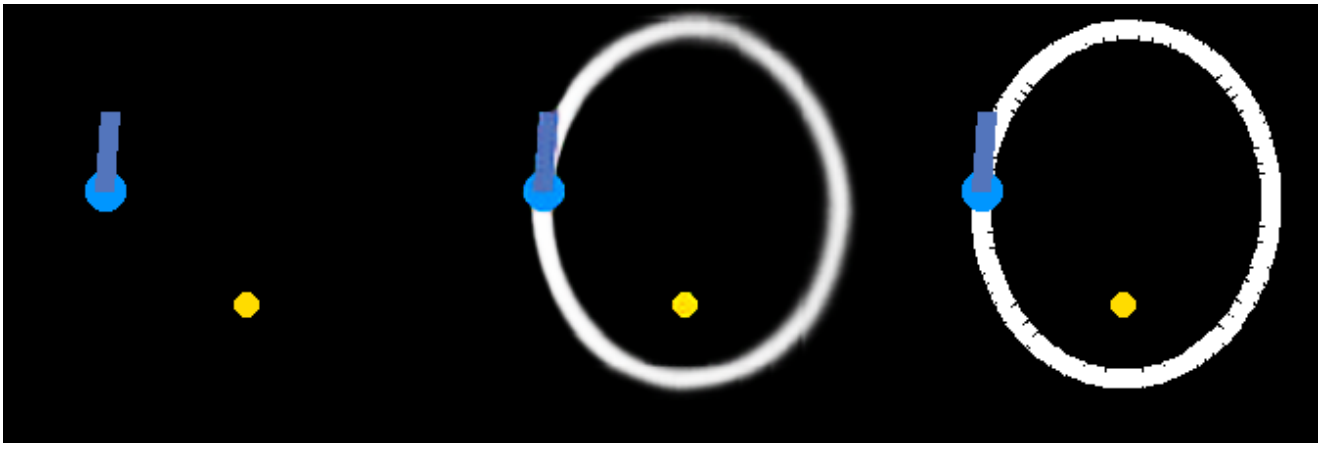


10000step



Due to the slowing down of changes later, the final training result is displayed directly.

156400step



4.7. ARC-AGI Task Related

This section will demonstrate the capability of this paradigm in handling [12,13] arc-agi-1/2 tasks.

My Experimental Method Introduction:

1. The pattern of the arc-agi-1/2 dataset is to provide a very small number of examples, around 3, requiring the rules of the game to be comprehended from these examples, and

generalization and problem-solving to be completed on new examples. I tried using my method strictly according to the requirements of arc-agi-1/2, but it was impossible to complete, resulting in extremely severe overfitting. Therefore, I adopted a roundabout method. I first manually observed the logic of a task, assisted by interactive programming with LLMs (such as Gemini 2.5 Pro) to write and debug training data generation scripts for these complex tasks. Then I ran the script to generate a large dataset, generally around 150,000-200,000, and finally trained on this dataset and sought generalization.
2. Because my experimental method requires a large amount of time and energy, including time for manual observation of task logic and generation and modification of training set generation scripts, I only attempted it on 16 tasks: 8 arc-agi-1 tasks and 8 arc-agi-2 tasks, selected randomly and based on the principle that the training set generation script is relatively simple. The results were quite good; the vast majority of problems converged quickly, and a few problems required long-term training but could also converge. This universality is shocking, considering that I was randomly selecting 16 problems.
3. Adopted image reasoning mode, Swin Transformer + unet model.
4. Adopted the color image representation method from the arc-agi-1/2 dataset official website.
5. The horizontal and vertical dimensions of grid points were kept consistent throughout the dataset.
6. Generally, a dataset of 150,000 numbers is sufficient to complete training, but due to time and energy constraints, I did not try to find the minimum number required.

4.7.1. Fluid Simulation

- **Task Description:** From ARC Prize - Puzzle #36a08778
- **Input Format:** 224*224 image.
- **Output Format:** 224*224 image.
- **Loss:** MSELoss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_arc_fluid_simulation.py`
- **Dataset:** `arc_fluid_final_dataset` folder.
- **Training Code:** `train_image2image.py`
- **Model Used:** Swin Transformer + unet.
- **Training Set Size:** 150,000
- **Input Space Size Estimation:** Using Monte Carlo method, running 2,000,000 times without duplicate samples.
- **Ratio of Training Set Size to Input Space Size:** As above, the training set size can be considered an extremely small fraction of the input space.

Task-Related Design Thoughts and Discussion:

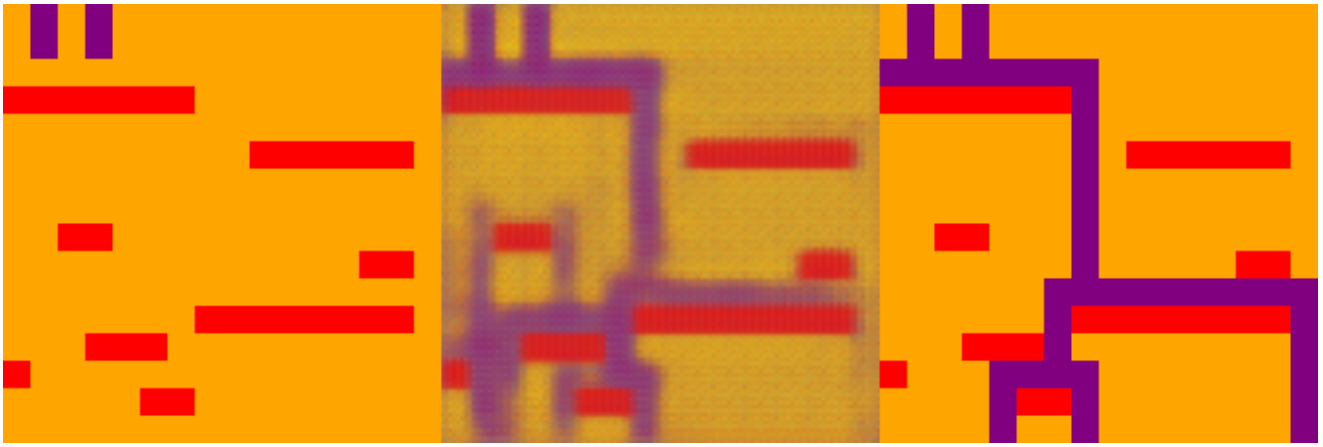
1. To further verify the pure image reasoning capability of the Transformer on existing arc-agi-1/2 datasets, I extended my attempts but adopted another roundabout method, as explained above.
2. Some task parameters:
    - Image resolution: 224*224

Experimental Results and Analysis: The final training results show that the model has completely learned the rules of this task. This indicates that the model possesses the reasoning capability expected by the arc-agi dataset, although it requires a large amount of dataset training. At the same time, the learning speed of the model is very fast, providing direct observation of the formation process of the machine mind. In this task, it can be seen that the model initially learned simple rules; it seemed to guess that purple water flowing down from the edge of the red barrier was correct. In the subsequent process, it slowly abandoned this overly simple rule hypothesis and finally learned the true task rules. This proves that my paradigm, combined with the "programmatic data generation" methodology, is a highly potential path for overcoming the problem of few-shot abstract reasoning. Moreover, the reasoning nature of this task itself largely rules out the possibility that the model is performing naive interpolation.

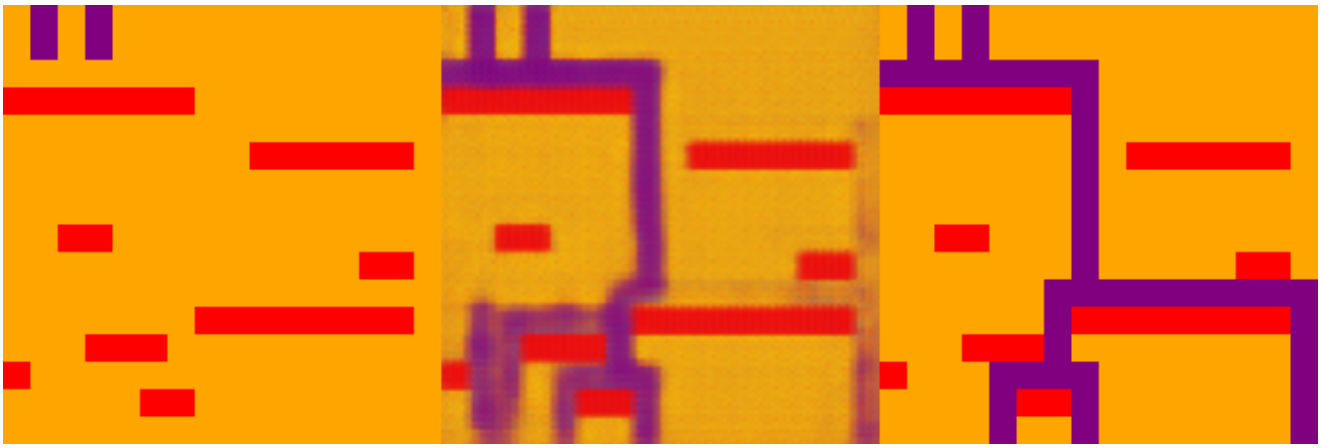
Experimental Process Display:

Left is input image, right is ground truth, middle is generated image.

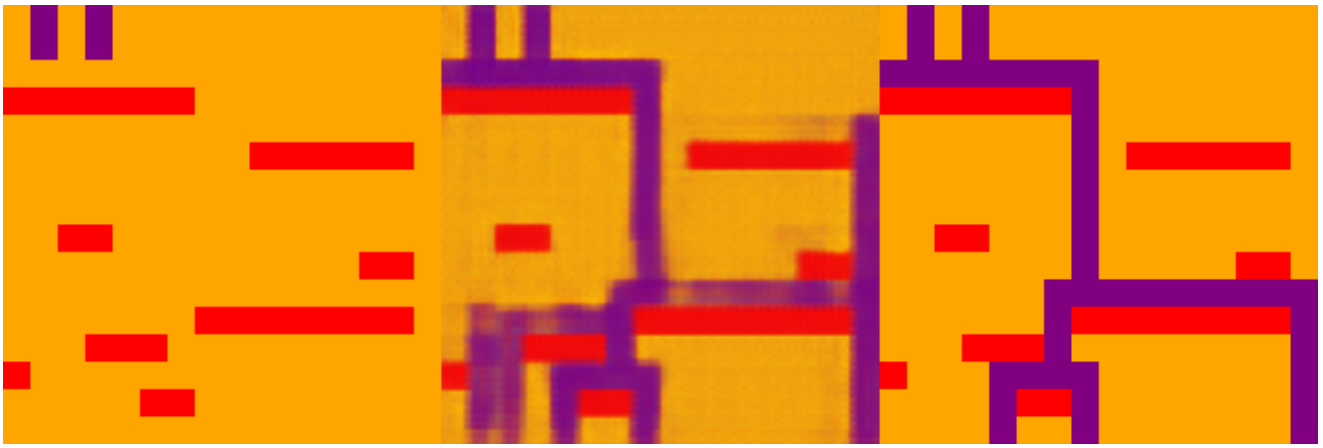
200step



400step

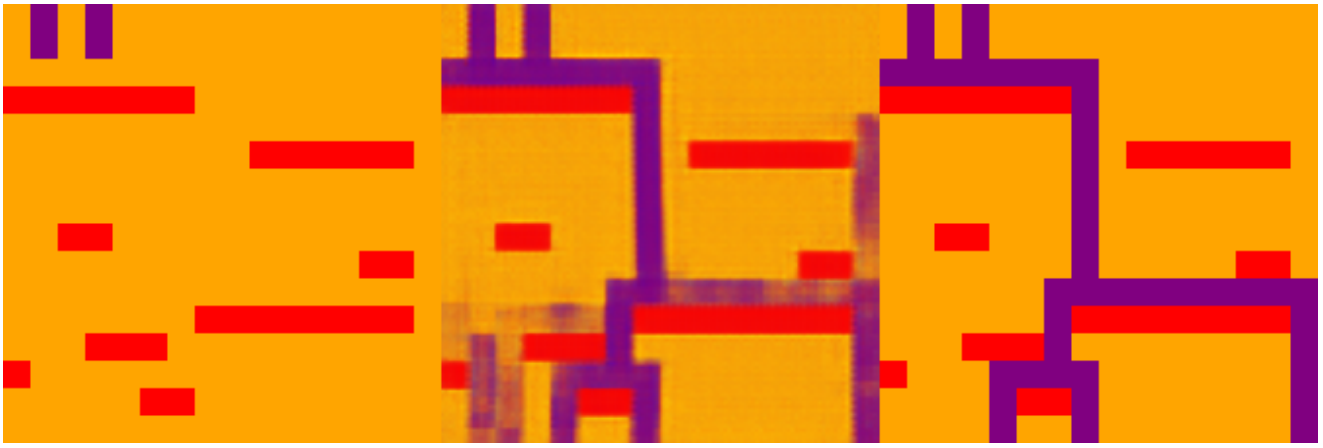


600step

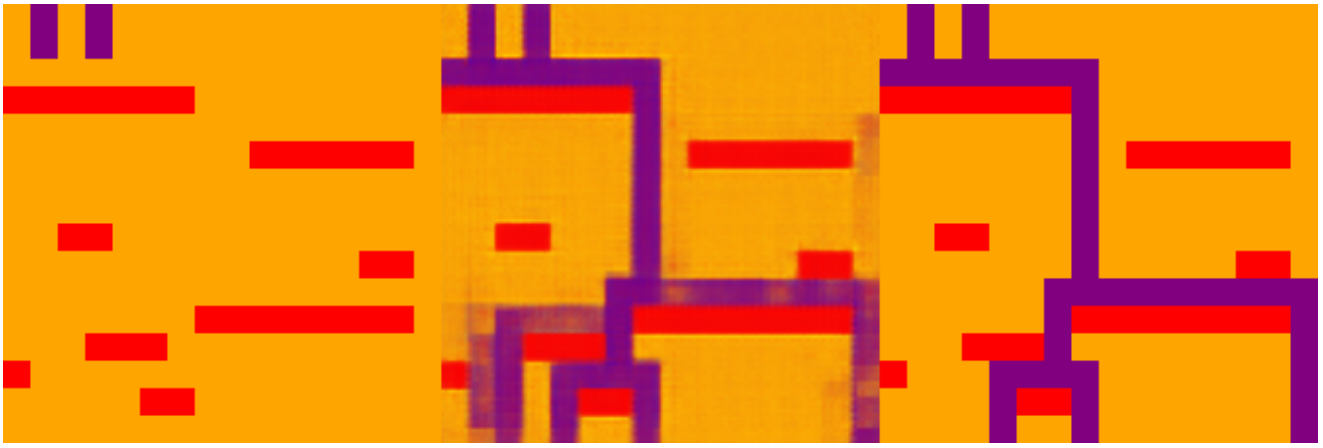


800step

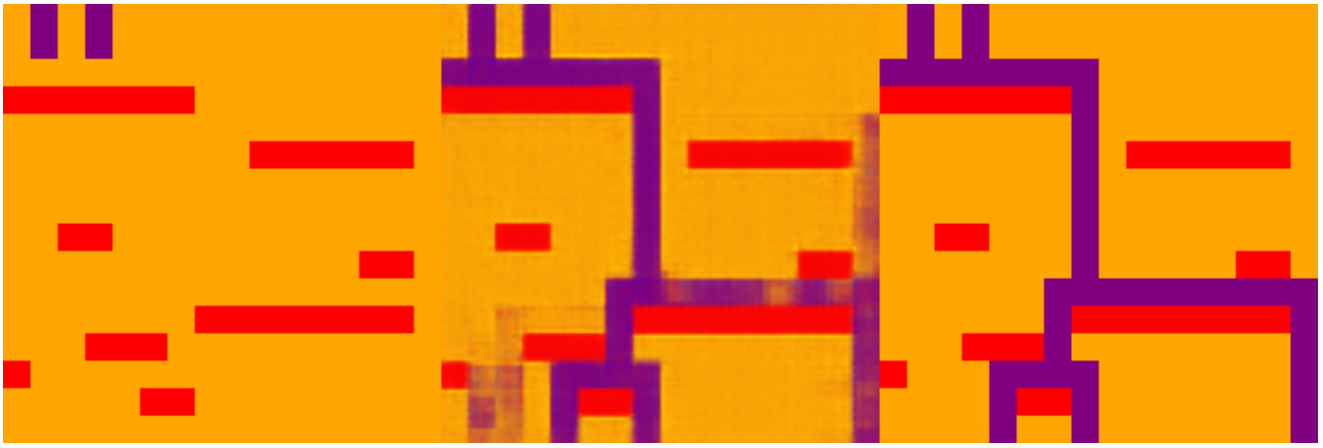
1000step



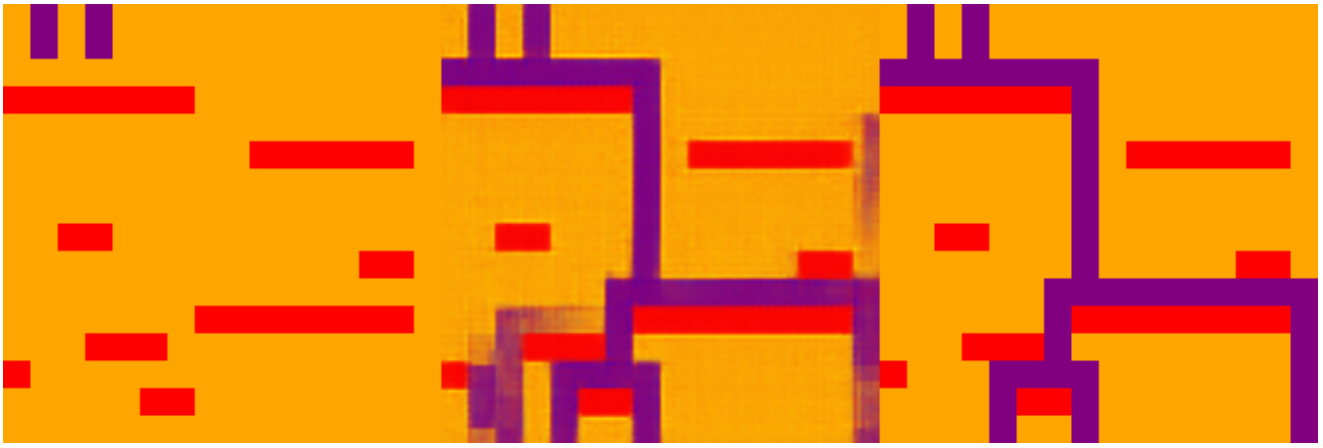
1200step



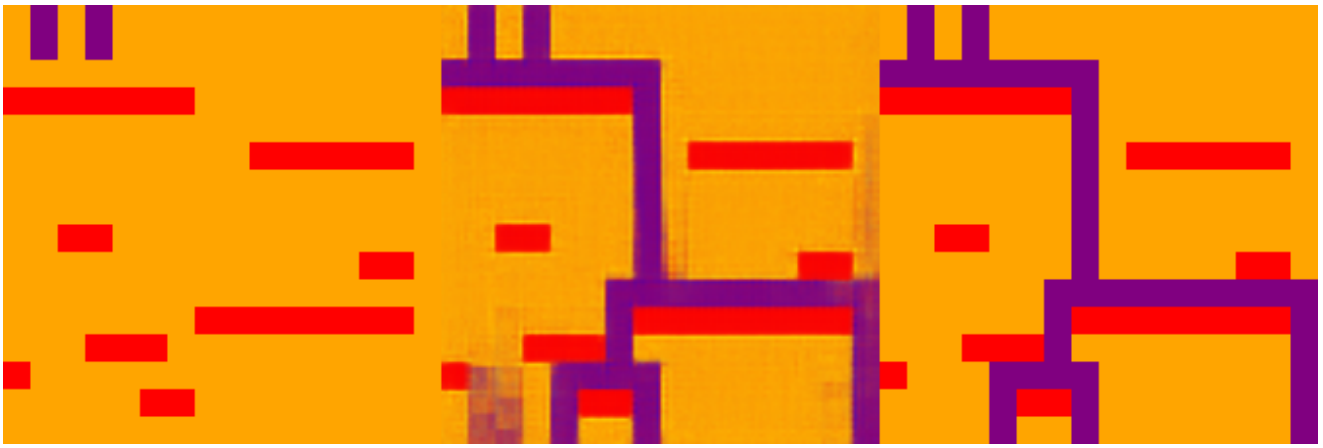
1400step



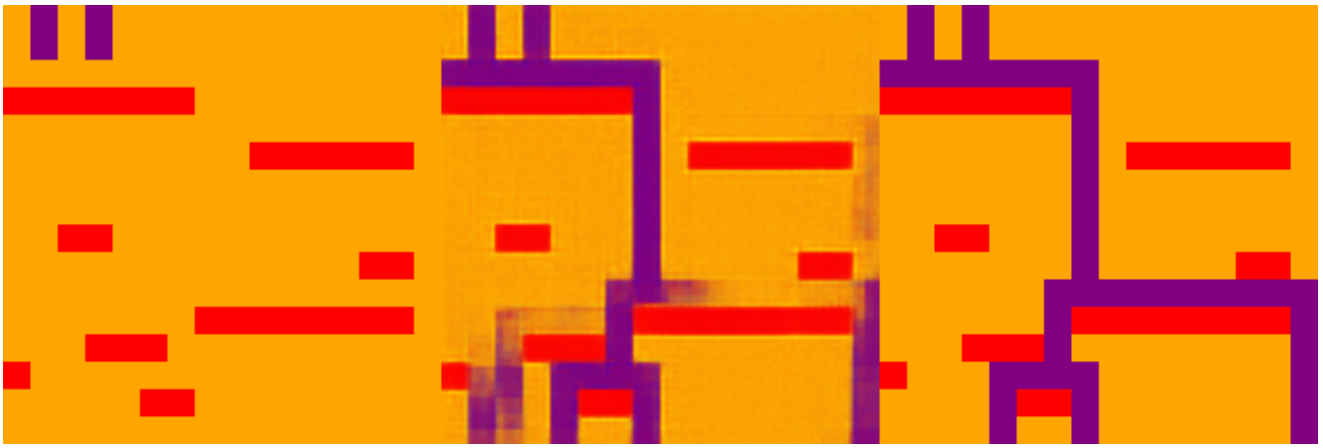
1600step



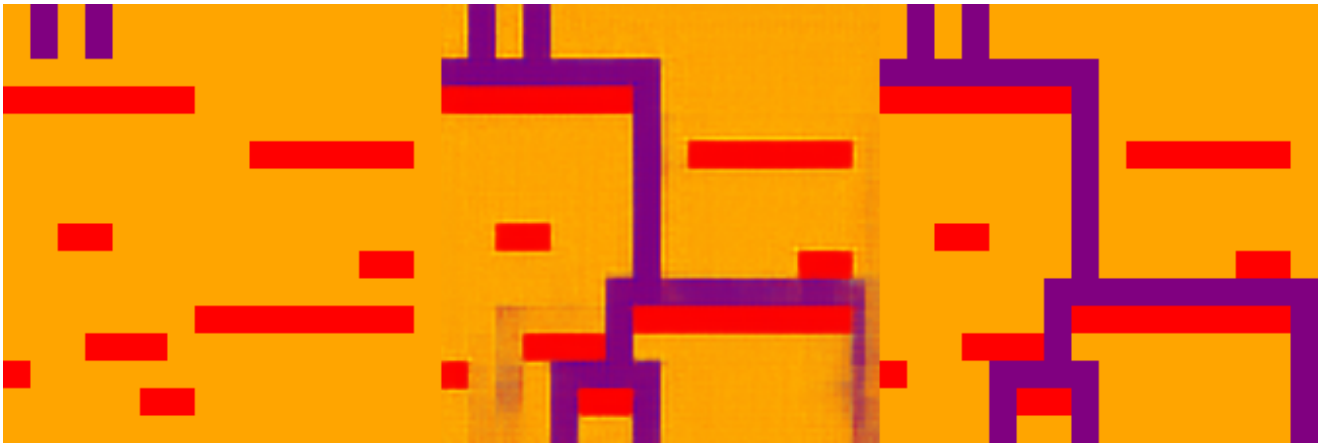
1800step



2000step

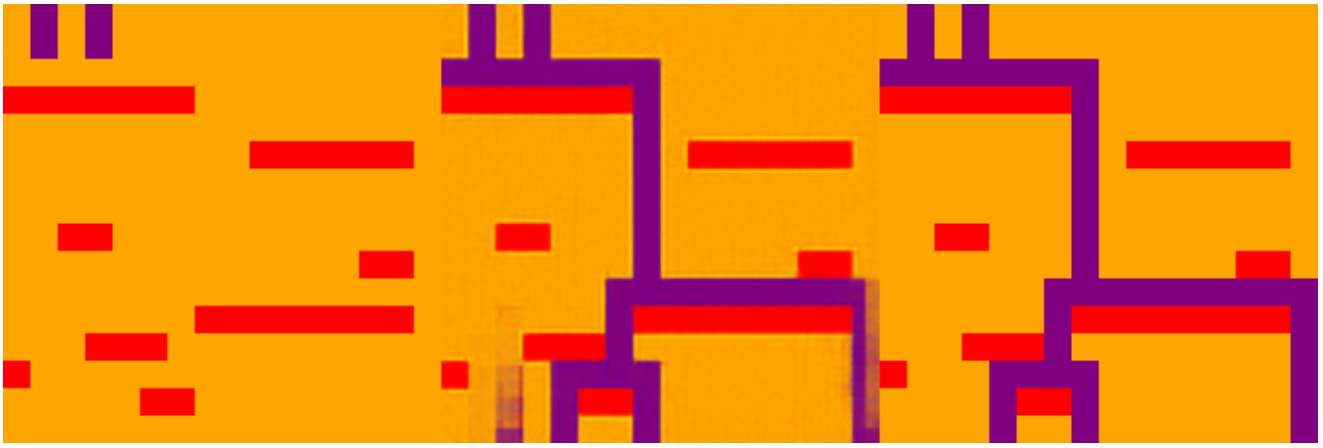


3000step



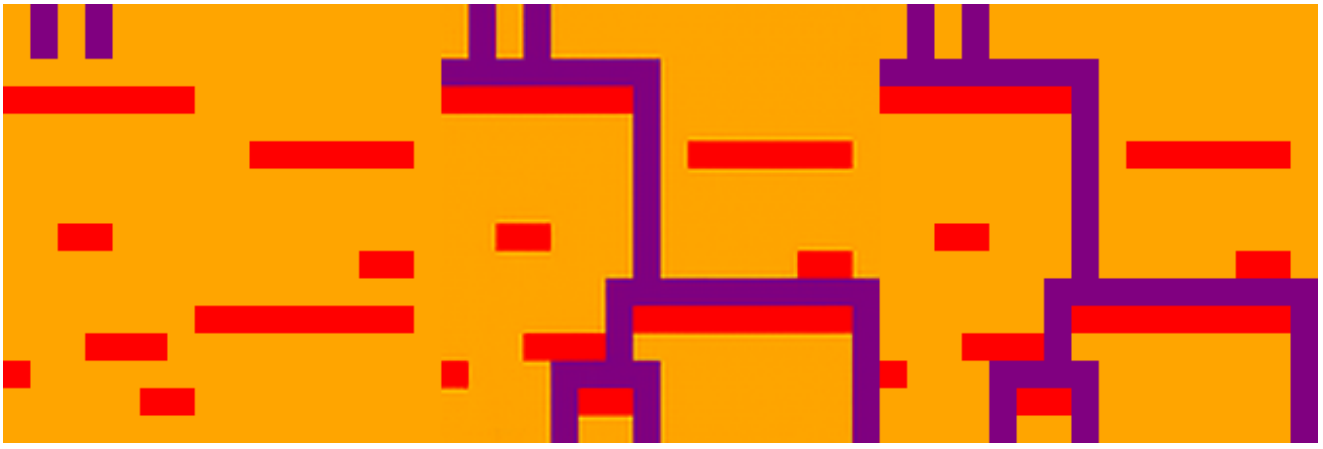
4000step

5000step



Due to the slowing down of changes later, the final training result is displayed directly.

39600step



4.7.2. Jigsaw Puzzle

- **Task Description:** From ARC Prize - Puzzle #898e7135
- **Input Format:** 224*224 image.
- **Output Format:** 224*224 image.
- **Loss:** MSELoss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_arc_jigsaw_puzzle_advanced.py`
- **Dataset:** `arc_jigsaw_puzzle_mine_dataset` folder.
- **Training Code:** `train_image2image.py`
- **Model Used:** Swin Transformer + unet.
- **Training Set Size:** 200,000
- **Input Space Size Estimation:** Using Monte Carlo method, running 1,000,000 times without duplicate samples.
- **Ratio of Training Set Size to Input Space Size:** As above, the training set size can be considered an extremely small fraction of the input space.

Task-Related Design Thoughts and Discussion:

1. To further verify the pure image reasoning capability of the Transformer on existing arc-agi-1/2 datasets, I extended my attempts but adopted another roundabout method, as explained above.
2. Some task parameters:
    - Image resolution: 224*224

Experimental Results and Analysis: The final training results show that the model has completely learned the rules of this task. This indicates that the model possesses the reasoning capability expected by the arc-agi dataset, although it requires a large amount of dataset training. At the same time, on this task, the model required a longer time to learn, providing direct observation of the formation process of the machine mind. In this task, it can be seen that the model initially learned the background color, possibly because it initially accounted for a larger proportion of the loss and was relatively easy to learn. Then it slowly learned the shapes and positions of the blocks needing to be filled, and finally spent a considerable amount of time learning the rules for filling colors. This proves that my paradigm, combined with the "programmatic data generation" methodology, is a highly potential path for overcoming the problem of few-shot abstract reasoning. Moreover, the reasoning nature of this task itself largely rules out the possibility that the model is performing naive interpolation.

Experimental Process Display:

Left is input image, right is ground truth, middle is generated image.

10000step

20000step

30000step

40000step

50000step

60000step

70000step

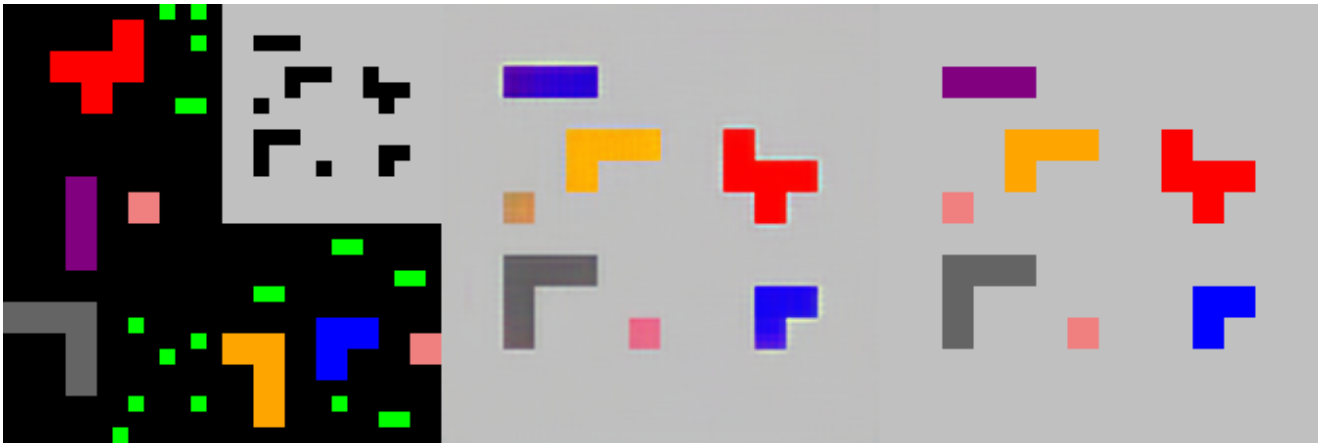
80000step

90000step

100000step

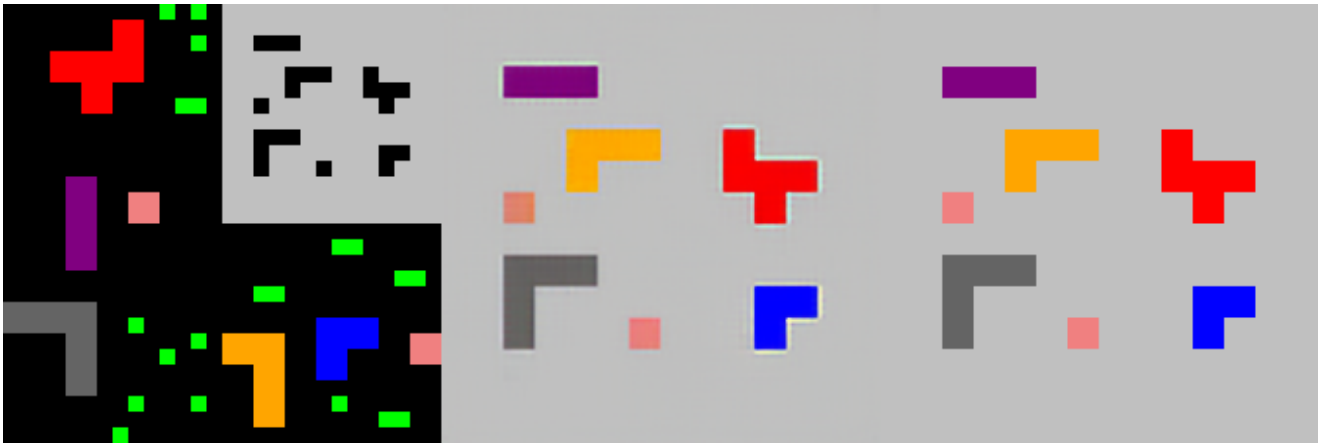
150000step

200000step



250000step

300000step



Due to the slowing down of changes later, the final training result is displayed directly.

530400step

4.8. Interpretability Exploration

This section will demonstrate the capability of this paradigm to output reasoning processes.

4.8.1. Edit Process Output

- **Task Description:** The model learns to predict and output the shortest edit process between two 0/1 strings.

- **Input Format:** A 30-bit string composed of "0" or "1". The first 15 bits represent string A, and the last 15 bits represent string B.
- **Output Format:** Multi-label binary classification format with 450 labels, equivalent to four hundred and fifty 0/1 strings. Every consecutive 30 bits form a group. The first 15 bits of a group represent the intermediate state of the string during the editing process, and the last 15 bits represent the mask of the string to distinguish cases where the string length is less than 15 bits during editing.
- **Loss:** Average Binary Cross Entropy Loss.
- **Training Configuration:** 4090 Graphics Card.
- **Training Set Generation Script:** `generate_edit_distance_explainable.py`
- **Dataset:** `edit_distance_path_unique_final.jsonl`
- **Training Code:** `train_tiny_transformer.py`
- **Training Set Size:** 500,000
- **Input Space Size Estimation:** 2 to the power of 30 = 1,073,741,824
- **Ratio of Training Set Size to Input Space Size:** 0.0466%

Task-Related Design Thoughts and Discussion:

1. This theoretically does not count as directly explaining the weights of neurons or the functioning of a module, but rather an interpretability guided by interpretable labels. The method of operation for this kind of interpretability is to directly add the internal processes that need to be observed as labels; practice shows that this method is practical. For example, for some problems, since the network can fit the answer, in many cases, I can be sure that it must also understand some internal states or intermediate processes in some way, and at this time, it can be exposed through this method. As for the original meaning of purer neural network interpretability, I believe this paradigm can achieve a cleaner laboratory environment to observe neural network weight changes, which will be discussed later.
2. Some task details:
    - The script ensures that no string length exceeds 15 during the editing process, otherwise my output format cannot represent it.
    - And the script ensures that there is one and only one minimum editing path, otherwise it will greatly increase the difficulty of network convergence.
    - The choice of edit distance strings is limited to two types, 0 and 1, to reduce task difficulty without changing the essence of the problem.

Experimental Results and Analysis: The evaluation loss converged to 0.001404. This success indicates that this paradigm can not only give answers but also has the potential to output problem-solving steps, providing a new, actionable idea for solving the neural network black box problem.

Final Convergence Results:

```
--- Step 66000 ---
Train Loss: 0.001995
Eval Loss: 0.001404
```

Act V: On the Boundaries of Learnability and Interpretability

I have summarized the characteristics of this paradigm:

1. Input and output formats should be well-designed. The input format should minimize the parsing burden on the transformer as much as possible. For output, it is preferable to use multi-class classification corresponding to cross-entropy loss or multi-label binary classification corresponding to average binary cross-entropy loss. You can design the meaning of the output format according to the characteristics of the task, as long as the format in the entire dataset is consistent. For example, if the output is a number, you can use multi-label binary classification to represent a binary number of corresponding bits.
2. Regarding learning difficulty:

   Previously, when doing symbolic rule learning tasks, I observed a phenomenon that the structure of the task is also important. It seems that forward symbolic rule tasks are easy to learn, while reverse solving for inputs based on outputs is not easy to learn, unless this reverse is actually another form of forward symbolic rule. Now it seems that this is because such "reverse solving" tasks are more similar to algorithm simulation problems, so the learning difficulty is greater.

   Another observed phenomenon is that tasks with branching conditional judgments may exacerbate learning difficulty, such as a certain indicator bit indicating the use of completely different rules to calculate subsequent operands. Thinking about it now, branching judgments are essentially rules themselves, but they just exacerbate the complexity of symbolic rules, making them harder to learn.

   Additionally, positions can be left in the input to control task rules, i.e., Input = Task Rule + Operands. This idea has been successfully verified in cellular automata-related experiments.

   Furthermore, another observed phenomenon is that tasks that serially mix too many simple logics are also difficult to learn. If a symbolic transformation rule serially combines different simple symbolic rules, even if each rule itself is very simply to learn, it may still combine into an extremely difficult task to learn. A simple example is the Trapping Rain Water experiment in the experimental section; by eliminating the final step of summing rain amounts and decoupling the problem, that task was able to converge quickly.

   Forward symbolic rule learning has degrees of difficulty. For example, in 1D 30-bit cellular automata, the learning difficulty of different rules varies, and the learning difficulty of applying different layers of rules also varies. Basically, the more layers, the harder the

learning. There are also some fundamentally unlearnable tasks, such as RSA encryption, which, although a simple forward mathematical calculation, is indeed impossible to learn.

Therefore, rule difficulty affects the speed of loss decline. Regarding this, to take a specific example, for cellular automata, Rule 110, the number of evolution layers—at least when there are few—strictly follows a law where the more layers, the more difficult the training, and the slower the model convergence. This brings to mind [15] Computational Irreducibility proposed by Wolfram; he indeed used the evolution of cellular automata as an example.

In addition, the training process itself is a kind of computational irreducibility; the process of training/sculpting cannot be omitted. Moreover, the simpler the problem, the faster the training; the harder the problem, the slower the training. If we do not consider whether the essence of the problem is friendly to neural networks or whether the input/output format is friendly to neural networks, and if we control the scale of inputs for different problems to be roughly consistent, the speed of loss decline during the training process can even be used to estimate the learnability or computational irreducibility of the problem. For example, I once tried to train RSA encryption, and the loss did not drop at all; nothing was learned from start to finish. Using cellular automata as an example here again, both input and output are 30-bit 01 strings, so the rule type and number of evolution layers can completely determine the mapping from input to output. But this is just one way of mapping from input space to output space. Why is this one type easier to learn than—mapping rules of cellular automata with more evolution layers? And also obviously easier to learn than a completely random permutation mapping? — A completely random permutation mapping is obviously unlearnable because it contains no patterns. This is a very interesting question, but exploring it is beyond my capability. Furthermore, the unlearnability of the RSA algorithm confirms that the simple algorithms created by cryptographers are very successful in this sense.

So, through this, it is possible to quantify task difficulty, learnability/computational irreducibility through the method of actual training tasks, this kind of Monte Carlo-like method. This may have potential in mathematics. Why are some rules deemed complex and others simple? And will this form a new theory?

Additionally, regarding model capacity, intuition suggests that larger/more suitable model capacity for the task should make training faster. Therefore, it might be the combination of model capacity and task difficulty that determines the speed and process of loss convergence during training.

3. Regarding the situation where transformation rules in the training set are inconsistent. Even if they are simple rules, if different data in the training set apply different rules, it is also difficult to learn, because there is no indication of which rule should be applied. The model can only try to achieve some balance based on the principle of reducing loss. If many rules are mixed but it cannot be identified which simple rule to use, then the training loss and eval loss will oscillate at a high point because different rules are pulling

towards an equilibrium point. The level of this loss value will depend on the purity of the rules in the training set.

In addition, mod 3 is a surprisingly unlearnable rule. It is very simple, but the loss remains high. It may be related to that phenomenon of rule mixing, but at least from a human perspective, it intuitively does not look like a mixed rule. I conducted an experiment where the input is a 20-bit binary number, represented by a 20-bit 01 string, calculating its mod 3 result, represented by a 2-bit binary number, and the loss is mean bce loss. Now comes the interesting result. I generated four datasets. The first dataset has no restrictions on input. The second dataset restricts mod 3 not to be 0. The third dataset restricts mod 3 not to be 1. The fourth dataset restricts mod 3 not to be 2. The result of training is that their respective eval loss stabilized at the following results:

- Dataset without restrictions —— 0.636
- Restriction mod 3 cannot be 0 —— 0.693
- Restriction mod 3 cannot be 1 —— 0.347
- Restriction mod 3 cannot be 2 —— 0.347

Analyzing this result is very interesting. Let me first give a fact: if the model cannot learn anything from the data, for a bce loss, if the probability of the label being 1 is p, then the model will reduce the loss to $-p\ln p-(1-p)\ln(1-p)$. Afterward, I will see that these loss results prove that the model did not learn anything from the data, i.e., mod 3 is unlearnable. The reasons are as follows:

For the dataset without restrictions, the probability of the high bit being 1 for the output 2-bit binary number is 1/3, and the probability of the low bit being 1 is also 1/3. Therefore, the optimal $\text{loss}=-\frac{1}{3}\ln\left(\frac{1}{3}\right)-\frac{2}{3}\ln\left(\frac{2}{3}\right)=0.6365$, which is very close to the experimental result.

For the dataset restricting mod 3 not to be 0, the probability of the high bit being 1 for the output 2-bit binary number is 1/2, and the probability of the low bit being 1 is also 1/2. Therefore, the optimal $\text{loss}=-\frac{1}{2}\ln\left(\frac{1}{2}\right)-\frac{1}{2}\ln\left(\frac{1}{2}\right)=0.693$, which is very close to the experimental result.

For the dataset restricting mod 3 not to be 1, the probability of the high bit being 1 for the output 2-bit binary number is 1/2, and the probability of the low bit being 1 is 0. Therefore, the optimal $\text{loss}=-\frac{1}{2}\ln\left(\frac{1}{2}\right)=0.347$, which is very close to the experimental result.

For the dataset restricting mod 3 not to be 2, the probability of the high bit being 1 for the output 2-bit binary number is 0, and the probability of the low bit being 1 is 1/2. Therefore, the optimal $\text{loss}=-\frac{1}{2}\ln\left(\frac{1}{2}\right)=0.347$, which is very close to the experimental result.

Therefore, it can be considered that at least under this experimental setting, mod 3 is unlearnable, and not because of insufficient fitting ability.

A similar phenomenon exists in the Tower of Hanoi training experiment. Specifically, if I only train on all states on the optimal path from the initial state to the final state for a Tower of Hanoi problem with a fixed number of disks, training the optimal action for these states, convergence is found to be very rapid. But when I try to extend the states to the entire Tower of Hanoi state space for a fixed number of disks, I find that the loss stays at a relatively high position and stops decreasing completely, and it is likely not a problem of fitting ability. Observe the similarity between the entire state space of the Tower of Hanoi and the mod 3 problem. You will find that mod 3 in binary format has somewhat recursive or fractal properties, and the Tower of Hanoi is even more naturally recursive or fractal by nature. So is this unlearnability due to the design of the input or output format, or due to the problem itself? Does this paradigm fail to effectively identify rules involving recursion or fractals? I tend to think it is the latter. Then, why can convergence be rapid when training only on the optimal path from the initial state to the final state of the Tower of Hanoi? The answer should be because on this optimal path, there is a non-recursive recognition rule.

So, do these two problems related to recursion or fractals relate to the mixed rules mentioned at the beginning? This paper believes they might be related because when the model cannot identify recursive rules, it might perceive them as some form of complex mixed rules, and it cannot identify which input corresponds to which rule.

4. In this paradigm, data can often be generated infinitely cheaply by programs. Theoretically, I need to make the training set identically distributed with the entire input space, so the simplest method is random sampling in the input space. But here comes another question: could there be another way to construct the training set that makes the model's learning effect better? For example, if a certain branch condition of the rule appears less frequently, random sampling is difficult to cover enough patterns. My hypothesis is that perhaps there could be more ingenious ways of dataset generation to better cover various patterns of the problem. Because based on practical experience, as long as there is sufficient pattern coverage, the training set can achieve results, possess anti-overfitting capabilities, and meet the model's learning needs; this is also why the total amount of the training set does not need to be very large. Analyzing from this perspective, the reason why the past deep learning paradigm of pattern recognition required large amounts of training sets becomes relatively clear: because those tasks do not have precise transformation rules, more data is needed to cover various patterns in the input space.

   This paradigm does not pursue OOD (Out-of-Distribution) generalization. In other words, since the training data accounts for a very small proportion of the entire input space, the model allows for precise generalization across the vast remaining input space excluding the training data after convergence, which can be analogized to generalization in past paradigms.

5. **Controllable AI and Interpretability**

Regarding interpretability, as explained above, this theoretically does not count as directly explaining the weights of neurons or the functioning of a module, but rather an interpretability guided by interpretable labels. The method of operation for this kind of interpretability is to directly add the internal processes that need to be observed as labels; practice shows that this method is practical. For example, for some problems, since the network can fit the answer, in many cases, I can be sure that it must also understand some internal states or intermediate processes in some way, and at this time, it can be exposed through this method.

Regarding more controllable AI, due to the characteristics of the paradigm itself—non-autoregressive transformer models fitting exact rules—it theoretically will not exhibit hallucinations like LLMs or adversarial examples in CV. In practice, no similar hallucinations have been found either. The reason might be that since this paradigm is based on rule tasks, the intrinsic rules are extremely precise and fundamentally non-smooth, making it impossible to solve problems through interpolation. Therefore, the model can only learn rules, thus being naturally anti-hallucination.

Another related finding is that the batch size during training cannot be too small. If the batch size is too small, learning significantly slows down. It is speculated that because the tasks in this paradigm are precise rules, the gradient directions of each sample vary more greatly, so a certain scale of batch size is needed to average the gradients.

6. During training, it was found that training loss and validation loss decrease synchronously, and validation loss is basically always lower than training loss. Moreover, this paradigm is very resistant to overfitting, unless the input space is particularly large and the problem is particularly complex, such that the training set size does not meet the requirement of covering all patterns of the problem.
7. What this paradigm learns are symbolic rules. It sees the distance between 0 and 1 as the same as the distance between 0 and some emoji; they are all placeholders for symbols. "Structure" is the true "language": What it ultimately learns is not that 0 represents 0, nor that 1 represents 1. What it learns is something more abstract—regardless of what these symbols look like, behind them corresponds a mathematical structure that can perform addition and possesses specific group and ring theoretical properties. It dialogues with this abstract structure, not with specific symbols. This has been confirmed in the previous N-base addition experiment.
8. **Regarding the overall philosophical implications of this paradigm**

    - **Data:** It requires covering various patterns of the problem and being identical to the entire input space (whether it must be identically distributed with the input space is open to discussion, but at least this is the most convenient approach). It does not require absolute quantity; of course, the more data, the better. Data is like a sample or model provided for the neural network's creation. Why is more data better? Why is it similar to the idea of Monte Carlo algorithms? Because explicit logical calculation is not needed, only data and training. The more data, the clearer and more concrete the reference sample or model provided to the neural network. The more training, the

better the neural network fits. From the perspective of data programming, the scale and quality of the dataset are the keys determining the model's effectiveness.

- **Paradigm Metaphor:** It's as if the neural network is carving the appearance of input-output patterns similar to data on a block of marble. Data is the model, and the neural network is the sculptor. Using computing power/training is like carving on marble; it uses continuity to approximate a deterministic symbolic transformation rule or fit an unknown but deterministic algorithm. Insufficient training, where loss has not yet dropped to the extreme, is like the carving not being finished. If the problem is too difficult, exceeding the capability of this neural network, it is like the capability required for this sculpture exceeds that of the sculptor. Even if training is insufficient or the problem is too difficult, the neural network also fits and effectively suppresses the uncertainty of the output space, just like an unfinished sculpture resembles the model more than the original marble.

Moreover, this paper believes that the most interesting point of this paradigm is that it can continuously approximate a deterministic symbolic rule or algorithm. This paper believes that this is even a very interesting process mathematically. Furthermore, this paper believes that my paradigm has the potential to compress the uncertainty of problem outputs where people have not yet found general solutions, although it may not be able to provide precise algorithmic processes. This is using neurons to learn a symbol or algorithm task, but in its unique way, absolutely not in the step-by-step taken-for-granted steps that we humans assume, but via coupled loss gradient descent; it is only trying to reduce loss. For example, on an algorithm problem, if one wants to explain which part of the trained weights is executing which step of the algorithm, I think it is impossible. It does not know the position and meaning of input symbols; it is just doing gradient descent.

I now feel that this paradigm is consistent with what Demis Hassabis called advocating for the compression of structural intuition and Ilya's belief that compression is intelligence. [1] AlexNet proved that handcrafted features were not good enough, and direct learning of image features using connectionism yields better results. This paradigm proves that even for precise symbolic rules, this route is correct to some extent. Moreover, as mentioned earlier, even on tasks that cannot fully converge, the model can often compress the uncertainty of the answer.

As for why this connectionist compression attempt might be effective. My feeling is that Demis Hassabis mentioned the complexity of the world is exceeding the carrying limit of human language and logic, but this explanation may have limitations. I am reminded of a lyric from Bob Dylan's song "Things Have Changed": *All the truth in the world add up to one big lie.* But this does not mean the failure of logic. My point of view is that every proposition in a long logical chain has its own unique context, and often when people combine them into a logical chain, they ignore these contexts to some extent, leading to the distortion of the final logical chain. Such scenes are also frequently seen in real life: a person is talking about his thoughts, and although every sentence he says seems correct

and irrefutable, it eventually leads to an absurd result. This is where connectionism comes in handy, directly docking with real data and reality verification, recalibrating its own intuition and logic. This is like symbolic rules emerging from connectionism. Human thinking also accumulates observations and experiences first, then summarizes laws; induction of rules comes first, then deduction. Deduction cannot replace induction. As mentioned earlier, if deduction is not treated rigorously enough, it becomes very fragile, and at this time, it is still necessary to resort to observations and experiences corresponding to connectionism as well as reality verification.

And why this paper believes that pure structural compression might not be complete is because I associate it with the mixed rule attempt; direct compression is not good enough. That is, as mentioned earlier, mixing multiple simple rules will exponentially increase the difficulty of learning this task. But if there is insight into the structure of the problem, the difficulty of the problem can be reduced through decoupling. Perhaps this is the function of logical cognition; humans cannot cope with this world through intuition alone.

9. Exploring black boxes should be simpler through this paradigm. First, for trainable problems in this paradigm, the loss can drop to very close to 0. In this case, firstly, it is indeed parameters learned by gradient descent black box; secondly, unlike other image recognition and NLP, learned symbolic rules can be researched more traceably. Separately training different trainable problems on the same model, and then studying the weight changes of the neural network or the changes in verification output during the training process, etc., is easier to study in a sterile laboratory environment than the previous paradigms of pattern recognition images, NLP, and audio.

   Moreover, what is more conducive to this point is that manually designing required task rules and generating datasets is very simple.

   Simultaneously, experiments related to image reasoning can visualize the formation process of the machine mind. For example, in the task of drawing an incircle in a triangle, saving a current eval image every certain number of steps during training allows for a very intuitive view of the progress of the learning process and how the model might be learning the related task.
10. My paradigm, if it can unleash its true potential in the future, can completely predict reliable outputs based on new inputs by merely training a neural network without knowing the working principles of some systems and without programming. Somewhat similar to the [21] Software 2.0 mode mentioned by Andrej Karpathy.

    This method should unleash potential in the field of AI for Science in the future.
11. This paradigm can also extract rules, i.e., given input and output, output transformation rules. I have conducted experiments on cellular automata tasks, which were very successful. However, in that experiment, I ensured through control of dataset sample generation that the input and output can uniquely determine a transformation rule and evolution layer number. But in real-world problems, it is often impossible to meet the

condition of extracting transformation rules from a single sample. Here, my hypothesis is that if extraction of rules is to be done through a large amount of input and output data, perhaps adopting different types of voting methods might all achieve good results.

12. If we can have some understanding of the relationship between model weights and fitted rules, is it possible to proactively manually design weights to simulate a certain rule? Of course, this is likely very difficult, even though clear symbolic rules make it look a bit simpler. So perhaps using training to obtain weights is still a better method. The value of this question may mainly lie in theory.
13. A phenomenon was observed during training: sometimes when the loss drops very close to 0, it suddenly jumps to a high loss. It is speculated that this might be caused by the non-smooth nature of this type of task.
14. A hypothesis: This kind of training can be turned into something automated. Humans only need to co-write training set scripts with AI, and subsequent dataset generation, training, and analysis can all be completed automatically, making research very fast. And whether each task is learnable, and how difficult it is to learn, will become a new kind of data for algorithm scientists and even mathematicians to study.
15. Regarding the limitation of this paradigm to synthetic data, my thought is that actually I did not make any changes or innovations to the model. Once the data becomes noisy real-world data, the premise of this paradigm no longer exists. So this paradigm may not be directly used for complex problems in the real world. Its value may lie in reminding me that deep learning neural networks themselves have the ability to fit symbolic rules, and it will provide me with an excellent environment to study the behavior of neural networks. Perhaps when I understand the nature of neural networks better, I will have a deeper understanding of the hallucination problems of current LLMs and model failures in some scenarios in CV, thereby having the ability to create more controllable and interpretable AI.

    Furthermore, this paradigm might have more direct applications to some problems in the symbolic world.
16. Regarding the recent paper by Apple, [14] *The Illusion of Thinking: Understanding the Strengths and Limitations of Reasoning Models via the Lens of Problem Complexity*, in this paper, some problems where large models cannot reason reliably are narrated. I used my paradigm to train on 4 problems: Tower of Hanoi, Blocksworld, River Crossing, and Peg Solitaire, all achieving success, although the Tower of Hanoi problem has some subtleties mentioned above, and the convergence of the Blocksworld problem is also not perfect enough. See the open-source code for details.

    Additionally, I noticed some opposing voices at the time, saying that inability to reason normally was due to token limitations. So, for the Tower of Hanoi problem, I designed a format `1>2` meaning moving the top disk of the first pillar to the second pillar, separated by semicolons. Then I asked the large model to reply with the solution to the Tower of Hanoi problem in this format. The result found that basically n=6 is the limit; going higher will cause problems. So I believe that large models indeed cannot effectively reason about these problems without calling tools (such as python). I have also open-sourced

the script for checking whether the answer in this format is correct, see `eval_hanoi.py` in the code repository.

17. Rethinking the meaning of this paradigm: This paradigm seems surprisingly similar to the situation in the field of computer vision before the appearance of [1] AlexNet. At that time, mainstream methods relied on handcrafted features (such as [17] SIFT, [18] HOG), which can be seen as a kind of "prior symbolic knowledge" about the visual world. The success of AlexNet lies precisely in replacing this manual bias with end-to-end learning (induction). In the field of reasoning, current neuro-symbolic computing, whether modifying architecture or embedding logic, is essentially injecting human understanding of the reasoning process as a prior knowledge into the model, and this predetermined structure, this inductive bias, may be untrue or inappropriate.
18. Regarding the philosophical implication of this paradigm: I feel that even if it does not refute the prior existence of concepts, it can powerfully prove that concepts can grow spontaneously from experience. As for further discussion, due to personal capability limitations, I can only stop here.
19. There is a very clear correspondence between decoupling and [22] chain of thought or [23] curriculum learning, which may indicate that the profound meaning of this concept is not unique to the tasks studied in this paper.

Act VI: Universality Beyond Architecture

In this section, I explored whether other deep learning models have the same capabilities as the Transformer. I explored a total of 5 types: MLP, RNN/LSTM, CNN, UNet, and Diffusion. Experiments show that all differentiable architectures possess the intrinsic potential to fit precise rules, but the inductive bias of the architecture must align with the long-range dependency structure of the task, and the inductive bias determines significant differences in learning efficiency and convergence paths.

I used MLP and RNN/LSTM to fit the symbolic versions of cellular automata and the Trapping Rain Water algorithm problem. Of course, the input here is no longer token-like as in Transformers, but real numbers.
CNNs were used to fit cellular automata datasets and image-version Trapping Rain Water datasets where inputs are images and outputs are symbols.
UNet and Diffusion were used to fit cellular automata datasets where both inputs and outputs are images.

Dataset Generation Scripts:

- Symbolic Cellular Automata: `generate_cellular_automata_1d.py`
- Symbolic Trapping Rain Water: `generate_trapping_rain_water_decoupled.py`
- Image Cellular Automata: `generate_cellular_automata_image_and_label.py`
- Image Trapping Rain Water: `generate_trapping_rain_water_image_to_symbol.py`

6.1. Cellular Automata

Dataset Introduction:

All are 1D 36-bit cellular automata datasets, dataset size is 300,000, evolution rule is 110, evolution layer is 2. The only exception is the MLP model; possibly because it is too suitable for the cellular automata task and has overly strong fitting ability, I increased the evolution layer to 6.

Model structures can be found in the open-source code.

6.1.1. CNN Model

Training Code: `train_convnext.py`

I adopted the [5] ConvNeXt model. Here is the training process:

```
1000step Training Loss: 0.49467600 | Validation Loss: 0.27210647 |
Validation Bit Accuracy: 84.55833333% | Validation Exact Match Rate:
0.10000000%
2000step Training Loss: 0.11814510 | Validation Loss: 0.05764289 |
Validation Bit Accuracy: 97.36018519% | Validation Exact Match Rate:
36.80000000%
3000step Training Loss: 0.04807972 | Validation Loss: 0.03248511 |
Validation Bit Accuracy: 98.63425926% | Validation Exact Match Rate:
56.20000000%
4000step Training Loss: 0.00763491 | Validation Loss: 0.00266492 |
Validation Bit Accuracy: 100.00000000% | Validation Exact Match Rate:
100.00000000%
5000step Training Loss: 0.00186690 | Validation Loss: 0.00128128 |
Validation Bit Accuracy: 100.00000000% | Validation Exact Match Rate:
100.00000000%
```

It can be seen that convergence was completed at an extremely fast speed, and 100% accuracy of exact match on the validation set was reached at 4000 steps. This proves that at least on this problem, CNN has symbolic rule learning capabilities.

6.1.2. MLP Model

Training Code: `train_mlp.py`

For a giant MLP, the cellular automata dataset with an evolution layer of 2 converged too quickly, so I had to regenerate a dataset with an evolution layer of 6. Facts proved that it also reached 100% accuracy of exact match on the validation set very quickly at the end of 5 epochs.

Preliminary analysis suggests that because MLP preserves the hyper-distance connections of neurons too well, it may not be inferior to Transformer on this problem.

6.1.3. RNN Model

Training Code: `train_lstm.py`

Here are some small designs. First, I must input the input state of the cellular automata completely at the first time step, rather than inputting one bit per time step. Second, I cannot adopt autoregressive output, so I must take the output of a certain time step as the result. Third, I finally deliberately chose a time step different from the evolution steps of the cellular automata. Since the evolution steps of the cellular automata are 2, I chose the output time step of the RNN model to be 3. Although completely consistent time steps and evolution steps are definitely the most ideal model design, what I want to prove here is my previous viewpoint: the model does not execute tasks according to the logic or algorithmic steps usually assumed by humans. Under this experimental setting, the model is forced to learn a transformation function where repeating the transformation 3 times continuously equals the change of cellular automata rule 110 twice. The learning success of this setting can provide strong support for my viewpoint on this paradigm.

The experimental results also prove that convergence was completed at 1000 steps, reaching 100% accuracy of exact match on the validation set.

Subsequently, I changed the 3 time steps to 5/7/9, which can also reach 100% accuracy of exact match on the validation set in a very short time.

6.1.4. UNet Model

Training Code: `train_unet.py`

Using MSELoss, but it quickly got stuck at a high position of 0.0915 and stopped decreasing completely. It might be that the network architecture itself is not suitable for this task, pending further analysis.

Training Code: `train_unet_v2.py`

Using MSELoss, after several hours of training, MSELoss dropped below 1e-5, achieving complete fitting at the visual level, and the precise matching accuracy for all pixel levels reached over 99%.

The difference in performance between the two versions of UNet may reflect that the inductive bias of the network structure of the first version of UNet cannot adapt to this task. This finding supports the core viewpoint: all differentiable architectures possess the intrinsic potential to fit precise rules, but the inductive bias of the architecture must align with the long-range dependency structure of the task, and the inductive bias determines significant differences in learning efficiency and convergence paths.

6.1.5. Diffusion Model

Training Code: `train_diffusion.py`

After several hours of training, the fitting was finally completed. Although this architecture eventually fitted this task, there is a huge difference in fitting efficiency compared to Swin-

UNet, again supporting our conclusion that inductive bias determines significant differences in learning efficiency and convergence paths.

6.2. Trapping Rain Water

Task Description: Derived from the classic LeetCode algorithm problem, 42. Trapping Rain Water - LeetCode

The format adopts the same decoupled output format as in the experimental section above. The difference is that the number of bars is adjusted to 12, so both input and output are 12*3*=*36*=*6*6. This facilitates inputting a 6*6 black and white checkerboard image for CNN-related experiments.

6.2.1. MLP Model

Training Code: `train_mlp.py`

Best Result: Validation Loss: 0.0002, Bit Acc: 99.99%, Exact Match: 99.90%

6.2.2. RNN Model

Training Code: `train_lstm.py`

I chose a time step of 3 here.

Best Result: Validation Loss: 0.0001, Bit Acc: 100.00%, Exact Match: 100.00%

6.2.3. CNN Model

Training Code: `train_convnext.py`

I adopted the [5] ConvNeXt model.

Best Result: Epoch 6 | Training Loss: 0.00003849 | Validation Loss: 0.00001896 | Validation Bit Accuracy: 100.00000000% | Validation Exact Match Rate: 100.00000000%

6.3. Summary and Analysis

I used cellular automata experiments to prove symbolic rule deduction capability and Trapping Rain Water to prove algorithm fitting capability. This series of experiments proves that this paradigm is not exclusive to Transformers; many other models also have similar capabilities. This leads to an inevitable conclusion: this paradigm is an inherent capability of connectionist deep learning neural networks. Moreover, it is very likely that the Transformer is not necessarily the best model, or that the best model will vary for different problems in this paradigm. Here, due to time and energy constraints, I am personally unable to conduct further exploration.

Act VII: Bridging the Gap: The Continuum from Ideal Rules to the Real World

In this section, I mainly discuss two issues. In all the content discussed above, I focused my energy on datasets with precise transformation rules. So in the first part of this section, I will introduce random perturbations to the input and output of datasets with precise rules and observe the training progress and final loss convergence.

In the second part, I designed an experiment to prove that the rule learning capability and interpolation capability inherent in neural networks implied by this paradigm are not contradictory, but can exist simultaneously.

7.1. Perturbation of Precise Rules

I chose the one-dimensional cellular automata task, with evolution rule 110, evolution layer 2, and the number of bits of the one-dimensional cellular automata selected as 30. The specific perturbation practice is to randomly flip the input or output with a certain probability during the data generation process (i.e., 1->0 or 0->1). For simplicity, I only tested cases of flipping input alone or flipping output alone. The validation set uses the precise rule cellular automata dataset without random flipping.

Dataset Code: `generate_cellular_automata_1d_perturbed.py`
Training Code: `train_tiny_transformer.py`

7.1.1. Random Perturbation Test on Output

Random Flip Probability	Stable Eval Loss Value
0.1%	0.000692
1%	0.008490
10%	0.094094

7.1.2. Random Perturbation Test on Input

Random Flip Probability	Stable Eval Loss Value
0.1%	0.001895
1%	0.021377
10%	0.230316

7.1.3. Summary and Discussion

It can be seen that whether it is input or output perturbation, the difficulty of task learning increases with the increase of perturbation probability, which is also very natural. Additionally, input perturbation has a greater impact than output perturbation. This may be due to the inherent non-smooth nature of rule tasks themselves. It can be guessed that if the evolution layer is increased, input perturbation will make task learning even more difficult.

So, roughly speaking, there may be a continuum from noise-free precise transformation rule datasets to real-world datasets.

7.2. The Unity of Interpolation and Rule Learning

This section uses an experiment to prove that the rule learning capability and interpolation capability of neural networks are not mutually exclusive, but unified.

7.2.1. Experiment Description

To verify the ability of neural networks to fuse discrete rule learning and continuous value processing in a single task, I designed a logic-perception hybrid experiment. This experiment is based on the cellular automata image generation task. In the input and output images, the 1D 36-bit cellular automata state is arranged in row-major order into a 6*6 checkerboard grid image, where black represents 1 and white represents 0. But the following modifications were made to the input and output:

- **Input End:** The input image is no longer pure black and white blocks, but the grid points representing logical black and white are replaced by colored blocks taking random values within two different greyscale intervals, thereby introducing continuous perceptual information.
- **Output End:** The output image generated by the model is required to satisfy two conditions simultaneously:
  - **Logical Correctness:** The black and white patterns of the grid points must strictly follow the multi-step evolution rules of the cellular automata.
  - **Perceptual Relevance:** The final greyscale value of each grid point must be precisely calculated based on its corresponding input grid point's original greyscale value and a simple conditional function (preserve if white, invert if black). Specifically, assuming the input grid point's greyscale value is x, if the logical pattern of the output grid point is the same as the input grid point, the output grid point's greyscale value remains x; otherwise, it is $255 - x$.

This design ingeniously forces the model to simultaneously look through the continuous greyscale values of the input to execute discrete logical reasoning, and remember these greyscale values to complete the final continuous value mapping, thereby testing its two core capabilities of rule learning and interpolation simultaneously in an indivisible task.

Dataset Code: `generate_cellular_automata_1d_to_grid_image_interp.py`
Training Code: `train_image2image.py`
Loss: MSELoss

Some task parameters:

- Grid dimension: 6*6
- Grid size: 40*40 pixels
- Image resolution: 240*240

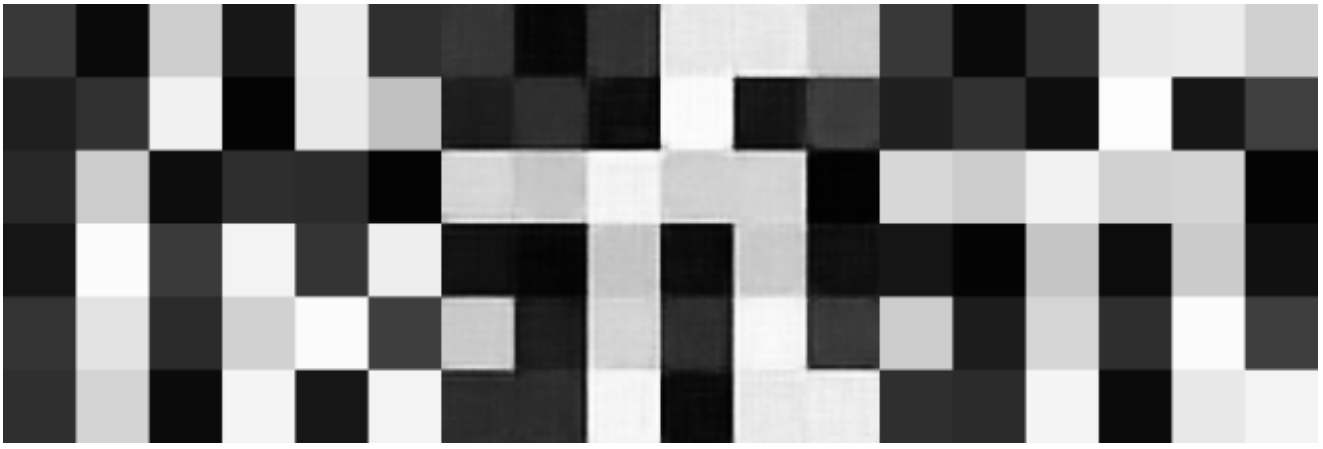
- Cellular automata rule: rule 110
- Rule evolution layers: 2

7.2.2. Experimental Results

After 3200 steps of training, the validation set mseloss had already reached 0.000899.

Here is the eval image at 3200 steps.

Left is input image, right is ground truth, middle is generated image.



7.2.3. Summary and Discussion

It can be seen that this experiment achieved great success. It fully proves that both interpolation and rule learning capabilities are inherent essential capabilities of neural networks. Moreover, they may not be parallel, but unified. For example, in this example, the model did not try to first understand the evolution rules of the cellular automata and then remember the greyscale value of each grid point; what it actually did should just be gradient descent. Still as mentioned earlier, connectionist gradient descent can approximate discrete symbolic rules.

7.3. The Unity of Pattern Recognition and Rule Learning

7.3.1.

This section uses an experiment to prove that the rule learning capability and pattern recognition capability of neural networks are not mutually exclusive, but unified.

7.3.1.1. Experiment Description

To verify the ability of neural networks to fuse discrete rule learning and pattern recognition in a single task, I designed an experiment that can prove both pattern recognition and symbolic reasoning are intrinsic capabilities of neural networks. This experiment is based on the cellular automata image generation task. In the input and output images, the 1D 36-bit cellular automata state is arranged in row-major order into a 6*6 checkerboard grid image, where black represents 1 and white represents 0. But the input was modified as follows: now I replace the input black grid points with the digit 1 from the MNIST dataset, and white grid

points with the digit 0 from the MNIST dataset. It can be anticipated that it should not converge perfectly, but I can compare the eval loss of the cellular automata representation of the input position (i.e., evolved 0 layers) (all eval loss here is due to pattern recognition) with the eval loss after evolving, say, 3 layers, to confirm how much extra eval loss the evolution of the cellular automata position brings. The output format remains unchanged, still black and white checkerboard grid points.

This design ingeniously forces the model to simultaneously manifest pattern recognition capability to understand the initial cellular automata position, and execute discrete logical reasoning through symbolic reasoning capability, thereby testing its two core capabilities of rule learning and pattern recognition simultaneously in an indivisible task.

Dataset Code: `generate_mnist_ca_110.py`
Training Code: `train_image2image.py`
Loss: MSELoss

Some task parameters:

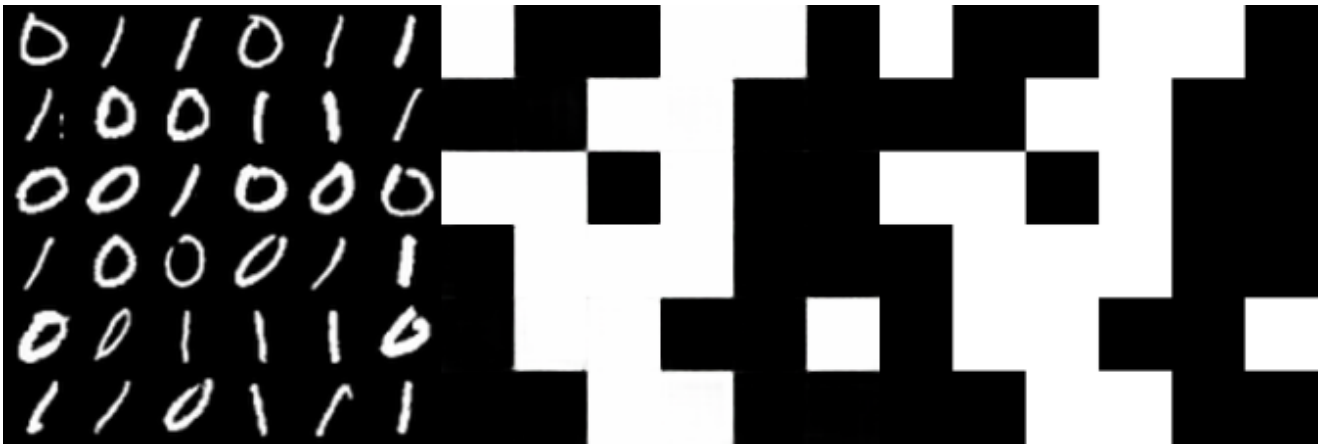
- Grid dimension: 6*6
- Grid size: 40*40 pixels
- Image resolution: 240*240
- Cellular automata rule: rule 110
- Rule evolution layers: 3

7.3.1.2. Experimental Results

- When the output is the cellular automata representation of the input position (i.e., evoluted 0 layers), after 2600 steps of training, the validation set mseloss had already reached 0.000393. This is the eval image at 2600 steps. It can be seen that the network has no problem recognizing MNIST digits. Left is input image, right is ground truth, middle is generated image.

- When the output is the cellular automata representation of the input position evolved by 3 layers according to rule 110, after 6800 steps of training, the validation set mseloss had

already reached 0.001320. This is the eval image at 6800 steps. Left is input image, right is ground truth, middle is generated image.



7.3.1.3. Summary and Discussion

It can be seen that this experiment achieved great success. It fully proves that both pattern recognition and rule learning capabilities are inherent essential capabilities of neural networks. Moreover, they may not be two separate capabilities, but parallel and unified, because these are two completely different capabilities exhibited by the same network structure. Furthermore, I have proven above that this network structure is not the root of symbolic rule learning capability, but rather an essential capability of neural networks. And, in this example, the model did not try to first understand and recognize the digit values of MNIST, convert them into the black and white checkerboard image representation of the initial position, and then evolve the position according to the cellular automata evolution rules. The learning processes of pattern recognition and symbolic rule learning capabilities are also coupled, with no sequential distinction. What it actually did should just be gradient descent. Still as mentioned earlier, connectionist gradient descent can approximate discrete symbolic rules. Moreover, since pattern recognition and symbolic rules are coupled capabilities, a more accurate description might be that connectionist gradient descent can simultaneously approximate discrete symbolic rules and fit pattern recognition; as for what capability the neural network manifests, it is mainly determined by the dataset.

7.3.2.

This section uses an experiment to prove that the rule learning capability and pattern recognition capability of neural networks are not mutually exclusive, but unified, and emphatically proves that neural network reasoning is coupled rather than having clear sequential steps.

7.3.2.1. Experiment Description

To verify the ability of neural networks to fuse discrete rule learning and pattern recognition in a single task, and integrating rule control aspects, I designed an experiment that can prove both pattern recognition and symbolic reasoning are intrinsic capabilities of neural networks. This experiment is based on the cellular automata image generation task. In the input and output images, the 1D 36-bit cellular automata state is arranged in row-major order

into a 66 *checkerboard grid image. But the composition of the input image is as follows: first randomly take the image of digit 0 or 1 from MNIST, stretch it to the full image of 240*240, then modify the initial image according to the initial cellular automata state. Where the cellular automata is 0, keep the grid point unchanged; otherwise, invert the color. For the output image, perform similar processing on the image of the same digit before modification of the input image, according to the evolved state of the cellular automata. The cellular automata evolution layer is still 3, but the evolution rule is determined by the MNIST digit 0 or 1; 0 represents rule 30, and 1 represents rule 110.

This design ingeniously forces the model to simultaneously manifest pattern recognition capability to understand the initial cellular automata position, recognize the digit of the initial position, and correspond to the evolution rule, then execute discrete logical reasoning through symbolic reasoning capability, and finally modify the initial MNIST digit image according to the final evolution state of the cellular automata to correspond to the final output image. The complexity of this task not only tests the neural network's two core capabilities of rule learning and pattern recognition simultaneously but also powerfully proves the coupling of time scale and spatial scale of neural network symbolic reasoning and pattern recognition through its long step chain.

Additionally, the reason why this experiment design is feasible is also because MNIST is white digits on black background; the edge is certainly black, giving the neural network an anchor to understand and correspond to the state of the cellular automata.

Dataset Code: `generate_rulemnist_ca.py`
Training Code: `train_image2image.py`
Loss: MSELoss

Some task parameters:

- Grid dimension: 6*6
- Grid size: 40*40 pixels
- Image resolution: 240*240
- Cellular automata rules: Digit 0 in the input image corresponds to rule 30, digit 1 in the input image corresponds to rule 110.
- Rule evolution layers: 3

7.3.2.2. Experimental Results

- After 10800 steps of training, the validation set mseloss had already reached 0.001011. This is the eval image at 10800 steps. Left is input image, right is ground truth, middle is generated image.

7.3.2.3. Summary and Discussion

It can be seen that this experiment achieved great success. It fully proves that both pattern recognition and rule learning capabilities are inherent essential capabilities of neural networks. Moreover, they may not be two separate capabilities, but parallel and unified, because these are two completely different capabilities exhibited by the same network structure. Furthermore, I have proven above that this network structure is not the root of symbolic rule learning capability, but rather an essential capability of neural networks. Also, because the work the neural network needs to do is too complex, including recognizing digits, corresponding to cellular automata evolution rules, recognizing initial cellular automata states, evolving cellular automata, and modifying the original MNIST digit image to the output according to the evolved cellular automata state. Such a long chain further implies that it is absolutely impossible for the neural network to execute the task in steps. And the discussion later will further verify this point directly.

Therefore, the learning processes of pattern recognition and symbolic rule learning capabilities are also coupled, with no sequential distinction. What it actually did should just be gradient descent. Still as mentioned earlier, connectionist gradient descent can approximate discrete symbolic rules. Moreover, since pattern recognition and symbolic rules are coupled capabilities, a more accurate description might be that connectionist gradient descent can simultaneously approximate discrete symbolic rules and fit pattern recognition; as for what capability the neural network manifests, it is mainly determined by the dataset.

Act VIII: Assisting Training with Decoupling — Rethinking the Relationship Between Symbolism and Connectionism

In this section, I will apply the decoupling idea from the Trapping Rain Water experiment mentioned above to accelerate the training of some tasks, and even make originally completely unconvergeable tasks converge—namely the binary number mod 3 task. Finally, I will demonstrate that not all decoupling formats accelerate training, and bad decoupling formats can even slow down training, and discuss the profound philosophical ideas contained therein.

The following details what decoupling is, that is, considering that direct prediction of the final output by the neural network may be difficult, one or several steps in the process are manually given as additional labels to assist the neural network in training. For example, in the Trapping Rain Water experiment mentioned above, I did not predict the binary representation of the total rainwater amount required by the original problem, but separately predicted the binary representation of the rainwater amount held by each bar. This method greatly reduced the time required for convergence. But this can be challenged as me changing the original goal. Below I introduce two decoupling methods that do not change the training goal.

The first method is similar to taking the output of the Trapping Rain Water decoupling experiment above—i.e., the amount of rainwater held by each bar—as input, and the original goal—total rainwater amount—as output, and training with another neural network (of course, the same neural network can be used without changing the network structure, just changing the number of outputs of the linear head layer). This method is equivalent to requiring different networks to predict different steps and then joining them together. Of course, a larger network can also be manually cut into several parts, with each part responsible for a segmented goal. The advantage of this method is that it is relatively consistent with human natural intuition: the front network completes the front task, and the back network completes the back task. But this method imposes an extra burden on people, making this method not the most natural one. That is, people need to manually specify which part of the network completes which stage of the task. Apart from intuition and experience, it is difficult to have better guidance. Moreover, this method is difficult to fully utilize network capacity. In addition, below, I will present a study on an already converged neural network, fully illustrating that a naturally converged neural network does not calculate step by step serially logic as conceived by this decoupling method, but is a computational logic implementation method where different stages and steps are quite coupled.

The second method is to directly concatenate the decoupled labels into the final result. If there are more detailed decoupling steps, they can also be added very conveniently because only the output labels need to be changed. Moreover, the advantage of this method is that there is no need to choose which part of the network performs the calculation of which stage, allowing for fuller utilization of network capacity.

Below I consistently choose the second decoupling method.

8.1. Binary Number Mod 3 Experiment

8.1.1. Experimental Part

I have already introduced the non-decoupled format experiment earlier; the neural network cannot learn any knowledge from the data. Below, the input and output formats of the decoupled experiment are introduced.

- **Input Format:** 30-bit binary number.
- **Output Format:** DFA trajectory, a total of 60 bits, where every two bits correspond to the mod 3 result of the preceding part of the input binary number. For example, bits 1-2 correspond to the mod 3 result of the 1st binary bit, bits 3-4 correspond to the mod 3 result of the 2-bit binary number composed of the 1st and 2nd bit, so clearly, bits 59-60 correspond to the mod 3 result of the entire 30-bit binary number.
- **Decoupling Format Discussion:** This decoupling format represents a state with every two bits, containing only three states:
  - **State S0:** Represents the current processed prefix value V satisfies $V \pmod 3 = 0$. If read $b = 0$, then $S_{next} = (0 * 2 + 0) \pmod 3 = 0$. State transition: S0 -> S0. If read $b = 1$, then $S_{next} = (0 * 2 + 1) \pmod 3 = 1$. State transition: S0 -> S1.
  - **State S1:** Represents the current processed prefix value V satisfies $V \pmod 3 = 1$. If read $b = 0$, then $S_{next} = (1 * 2 + 0) \pmod 3 = 2$. State transition: S1 -> S2. If read $b = 1$, then $S_{next} = (1 * 2 + 1) \pmod 3 = 0$. State transition: S1 -> S0.
  - **State S2:** Represents the current processed prefix value V satisfies $V \pmod 3 = 2$. If read $b = 0$, then $S_{next} = (2 * 2 + 0) \pmod 3 = 1$. State transition: S2 -> S1. If read $b = 1$, then $S_{next} = (2 * 2 + 1) \pmod 3 = 2$. State transition: S2 -> S2.

Training Set Script: `generate_binary_mod3_dfa_explain.py`
Training Script: `train_mlp.py`
Training Result: Using an MLP network, convergence was rapid, reaching 99.96% DFA sequence exact match in 25 epochs.

8.1.2. Discussion of Experimental Results

This is a very surprising result, not only because this decoupling format can assist training, but it can even make a problem that was originally completely unlearnable become perfectly learnable! This is clearly not a problem of network capacity. And the most puzzling thing is, if this problem is inherently learnable, why must the DFA auxiliary sequence be added to converge, while training alone yields no result? This is a question I currently cannot answer, but the only certainty is the undoubted effectiveness of this decoupling method. Although it seems to increase the burden on the neural network by making it predict more labels, it makes the entire training possible and faster, which is also a very counterintuitive interesting result.

8.2. Another Decoupling of the Trapping Rain Water Experiment

In this section, I decouple the Trapping Rain Water experiment, which has already been decoupled above, once again, exposing different algorithmic details.

8.2.1. Experimental Part

- **Input Format Introduction:** `N*BITS_PER_HEIGHT` bit binary number, where N is the number of bars, and `BITS_PER_HEIGHT` is the number of binary bits for the height of each bar.
- **Output Format Introduction:** Algorithm decoupling process part + Result part.
  - **Result Part:** `N*BITS_PER_HEIGHT` bit binary number, N is the number of bars, `BITS_PER_HEIGHT` is the number of binary bits for the amount of water held by each bar.
  - **Algorithm Decoupling Part**
    Three algorithm ideas introduction: 42. Trapping Rain Water - LeetCode
    - **DP Decoupling Format:** A vector of length `2*N*BITS_PER_HEIGHT`. It is directly concatenated from two parts:
      - First half: Binary representation of the `leftMax` array. The i-th element of this array represents the highest bar height from position 0 to i in the input sequence.
      - Second half: Binary representation of the `rightMax` array. The i-th element of this array represents the highest bar height from position i to N-1 in the input sequence.
      - Each height value is encoded as a `BITS_PER_HEIGHT` bit binary number.
    - **Monotonic Stack Decoupling Format:** A vector of length `N*(2*BITS_PER_INDEX+BITS_PER_HEIGHT)`. This vector is concatenated from N fixed-size "event slots." Each event slot records a triplet `(left_idx, right_idx, top_height)` information:
      - `left_idx`: The index of the left boundary wall forming the "concave groove," encoded as `BITS_PER_INDEX` bits.
      - `right_idx`: The index of the right boundary wall forming the "concave groove," encoded as `BITS_PER_INDEX` bits.
      - `top_height`: The height of the "groove bottom" bar where water is calculated, encoded as `BITS_PER_HEIGHT` bits.
      - Unused event slots are filled with 0.
    - **Two Pointers Decoupling Format:** A vector of length `N*BITS_PER_HEIGHT`. The i-th element block (length `BITS_PER_HEIGHT`) of this vector records the value of `leftMax` or `rightMax` relied upon when calculating the rain at position i.
      - If the rain at position i is determined by `leftMax` (i.e., when the left pointer moves), record the `leftMax` value at that time.
      - If it is determined by `rightMax` (i.e., when the right pointer moves), record the `rightMax` value at that time.
      - If position i holds no water or acts as a boundary wall, its corresponding value is 0.

Specific Parameters:
N=15, `BITS_PER_HEIGHT`=5, trained for 10 epochs on a training set of 500,000 entries.

Training Set Script: `generate_rain_water_final_showdown.py`
Training Script: `train_mlp.py`

Experimental Results (Validation Set):

	DP	Monotonic Stack	Two Pointers	No Decoupling Format
Decoupling part exact match	96.02%	17.36%	95.46%	
Result part exact match	75.60%	6.20%	85.16%	56.60%

If only caring about the exact match of the result part, it can be seen that the sorting of training effect from good to bad is Two Pointers > DP > No Decoupling Format > Monotonic Stack.

8.2.2. Discussion of Experimental Results

It can be seen that the DP algorithm and Two Pointers algorithm promoted the convergence of the task, but the Monotonic Stack algorithm actually slowed down the convergence of the task! First, the fact that a certain decoupling method can promote convergence once again proves that end-to-end training is not necessarily the best choice. Since the era of deep learning, there has been a belief: do not introduce too many manual biases, let it discover how to solve the problem. I personally believe that on one hand, this approach might be naturally effective for perceptual problems, and on the other hand, it is also because in the pattern recognition paradigm, it is difficult to give a clear auxiliary label without information loss and without introducing artificial bias. Now, in the paradigm I discovered, the latter difficulty no longer exists; for complex rule problems, different decoupling information can be given very simply to assist training. So, this experiment clearly illustrates that if some decoupling is given as guidance, the difficulty of learning the task can be completely reduced. Moreover, it should be noted that this is just a small algorithmic problem, and convergence can already be accelerated by this means. So it is easy to imagine that for a complex multi-stage large-scale rule/algorithm learning task, if decoupling step information is not added, simply extending the training time will be of almost no help.

Additionally, the Monotonic Stack decoupling method actually slowed down task convergence, which is a very interesting phenomenon. According to the description of this decoupling format, it is generally the longest of the three decoupling labels. Moreover, it describes not just information but the execution process of a serial algorithm, which is completely different from the decoupling formats of the DP algorithm and Two Pointers algorithm. It reducing the convergence speed of the task likely implies that the effort the neural network uses to fit the Monotonic Stack algorithm exceeds the benefits. So, the final convergence speed is actually even worse than training the task objective directly without decoupling auxiliary labels.

8.3. Discussion on Related Philosophical Meanings

If end-to-end connectionist learning is viewed as intuition, although many experiments above have shown that this intuition can cope with many tasks related to symbolic rules, experiments have also proven that this intuition is far from perfect. Whether in terms of convergence speed or learnability for certain problems, adding some decoupling—meaning information about intermediate process stages—might promote convergence. And I believe the significance of this point goes far beyond a mere technical experimental observation; just like humans cannot cope with the world relying solely on intuition, some logical thinking is absolutely necessary. The trust in end-to-end training of neural networks since the arrival of the deep learning era basically corresponds equivalently to the currently popular so-called "first principles thinking" or "deconstruction." Its root lies in the fact that many ideas are outdated and can no longer reflect facts, such as the aforementioned computer vision relying on handcrafted features (like [17] SIFT, [18] HOG), which can be seen as a "prior symbolic knowledge" of the visual world. Or the aforementioned string of logical chains without context leading to wrong conclusions; in this context, end-to-end training is akin to re-examining the problem without relying on past experience. As for the Monotonic Stack algorithm decoupling method in the Trapping Rain Water experiment mentioned above, if the neural network learned that Monotonic Stack algorithm step by step, could it solve the Trapping Rain Water problem? Yes! But, ultimately, it is found to be unnecessary, because adopting this decoupling method leads to even slower convergence than not using decoupling assistance. This is somewhat like: ideas were originally meant to help people solve problems and live better, but sometimes they completely detach from context and lose efficacy, becoming worse than a person dealing with it based on their own independent thinking—the neural network performed better based on its own native capabilities. But this does not deny that some decoupling methods accelerate training, such as the DP and Two Pointers methods mentioned above.

Maybe decoupling can be regarded as an operation that destroys the distributed representation of thought and makes it explicit, similar to making the subconscious conscious in psychoanalysis. This operation can clarify thoughts entangled together in some scenarios, making thinking easier, while in other scenarios, it might reduce the computational efficiency of distributed representation due to excessive symbolization.

This is absolutely not a simple contradiction. As the old Chinese saying goes: *Learning without thought is labor lost; thought without learning is perilous.* In MBTI personality types, there are P-types biased towards perception and sensing, and J-types biased towards planning and judging. In reinforcement learning, there is the contradiction between explore and exploit. Perhaps the formation of human concepts corresponds to the choice that maximizes cognitive economic utility, and with the passage of time and environmental changes, cognition is destined to be deconstructed and reconstructed time and again. Thinking involves making judgments; judgments will be wrong and might be corrected in the future, but not making judgments might be even worse. There is no absolutely correct choice; there only exist certain cognitive tendencies.

But this decoupling format returns to human manual selection again; it cannot emerge naturally, which seems to violate the philosophy of deep learning. So is it possible for neural networks to spontaneously form this decoupling format information or increase insight into the problem structure to accelerate training? I cannot answer this question, but if neural networks have the ability to do so, an even more incredible scenario will appear: neural networks having the ability to quickly fit any extremely complex problem. It's too good to be true; I personally believe intelligence itself is not that simple.

But regarding this problem, I think there is another roundabout way. If the same neural network is trained on multiple tasks simultaneously, when the neural network is forced to solve a large number of different but related symbolic tasks simultaneously within the same parameter space, the cognitive economic pressure of gradient descent will urge it to discover and construct those intermediate representations that are common and reusable across tasks—this might be related to my previously manually designed decoupling formats. For example, if images of cats from ImageNet and images of 1s from MNIST are taken out and merged together, asking the neural network to distinguish these two classes, it might find a shortcut based on texture or color; but a network that needs to simultaneously distinguish 1000 fine-grained categories in ImageNet is forced to form reusable, hierarchical abstract representations internally, from edges to parts to objects. So, the continuation of the philosophy of spontaneous emergence in deep learning might depend on multi-task simultaneous training. The birth of this insight into problem structure might not lie in deep drilling into a single task, but in the breadth of experience. Just as we observe in human cognition, there is hardly depth of cognition without breadth of experience. Of course, whether this hypothesis is correct still depends on future experiments and theoretical research.

Act IX: Research on Already Converged Neural Networks

The main purpose of this section is to study a neural network that has already converged for a specific problem, including its calculation method, information carried by each hidden layer, etc. The neural network used is MLP. The research method is mainly to freeze a certain hidden layer after the neural network converges, connect another MLP network behind it to try to decode some specific information, and then judge whether this information exists in this hidden layer and whether it is excessively entangled based on the convergence speed and final convergence accuracy. As for why one cannot judge solely based on final convergence accuracy, it is because in some cases, given sufficient time, convergence might be possible for many specific pieces of information. For example, in some extreme cases, exact input information can even be decoded in some hidden layers, so it is possible for the neural network to first decode the input information and then fit the designated specific information. However, in these cases, the speed of convergence for different specific information is often different, so one can only judge the existence mode of information in the hidden layer through the speed of convergence.

9.1. Studying the Last Hidden Layer via Trapping Rain Water Experiment

9.1.1. Experimental Part

I used the result part of the Trapping Rain Water from Section 8.2 as labels to train an MLP network. After it converged, I extracted the last hidden layer and connected a new MLP network behind it, with the output pointing to some designated information, to detect the existence and mode of existence of this information in this hidden layer.

These designated pieces of information are the decoupling information of the three algorithms introduced in Section 8.2.1—DP, Monotonic Stack, Two Pointers—which will not be repeated here.

Dataset Parameter Settings:

- Number of bars: 10
- Binary bits occupied by bar height: 4
  Training Script: `train_mlp_prefer.py`

Experimental Results:

Decoupling Algorithm	Best exact match
DP	26.90%
Monotonic Stack	42.34%
Two Pointers	92.74%

9.1.2. Discussion of Experimental Results

This experimental result indicates that after convergence, the neural network will have a large amount of task-related information in the last hidden layer. This information can be probed to try to understand how the neural network learns this task. Moreover, when learning a certain task, the neural network will form spontaneous cognitive style tendencies. For example, on this problem, the neural network's preference for these algorithms: Two Pointers > Monotonic Stack > DP. So, why is it like this? This is a very interesting question that will not be easily answered. Even considering that these decoupled formats cannot be completely equivalent to the corresponding algorithms, answering this question is still not that easy. Plus, considering that the Monotonic Stack decoupling format is actually longer than the DP decoupling format, this result is even more interesting.

9.2. Probing the Hidden Layers of a Converged MLP with Cellular Automata Task

9.2.1. Experimental Part

Experimental Setup: First, fit the state of a 1D 30-bit cellular automata after 8 layers of evolution under rule 110. Then sequentially extract each hidden layer and fit the states after 1 to 8 layers of evolution using another MLP network, training for the same number of epochs, and observe the exact match values.

Dataset Parameter Settings:

- Cellular automata bits: 30
- Evolution rule: 110
- Evolution times: 8
  Training Script: `train_mlp_ctscan.py`

Experimental Results: See the table below, `S_1 EM` represents the exact match for fitting the state of 1 layer of evolution.

Start fitting from which hidden layer	S_1 EM	S_2 EM	S_3 EM	S_4 EM	S_5 EM	S_6 EM	S_7 EM	S_8 EM
H_0	100.00%	100.00%	99.96%	59.20%	3.40%	0.04%	0.00%	0.00%
H_1	100.00%	100.00%	99.90%	90.12%	76.56%	33.72%	1.50%	0.24%
H_2	95.02%	86.64%	72.88%	65.08%	28.02%	11.42%	8.26%	99.88%
H_3	32.92%	29.18%	31.52%	35.40%	21.24%	21.90%	41.08%	100.00%

9.2.2. Discussion of Experimental Results

Based on the experimental result chart above, several qualitative conclusions can be drawn:

- Fitting directly from H_0/H_1, the more evolution layers, the harder to fit. This is easy to imagine and reasonable because H_0/H_1 are very close to the initial input, so as the problem difficulty increases, it naturally becomes harder to fit.
- The difficulty of fitting the final state decreases gradually from H_0 to H_3. This is also reasonable. If the neural network is viewed as a way of distributed computing, the sum of the entire neural network's calculations is fitting this task. The earlier neural network layers have already undertaken a lot of calculations, making the representations of later hidden layers closer to the final target to be fitted.
- From H_0 to H_3, the fitting ability for S_1/S_2/S_3 gradually decreases. For S_4/S_5/S_6, the fitting ability basically follows a rise then fall pattern (H_2 seems to be an exception for S_6, but it does not affect the final conclusion). For S_7/S_8, the fitting ability gradually increases. This is an interesting phenomenon, indicating that as the hidden layers move from the front to the back of the neural network, the information they contain becomes closer to the final task, gradually starting to lose the initial input information and shallow evolution state information, but not completely losing it, because these data are only obtained with very small epochs and small-scale MLPs. Below we will see that for a cellular automata with fewer evolution layers, it is even possible to

recover complete evolution information, including complete input, from the last hidden layer of a converged neural network.
- Another conclusion that can be drawn is that the neural network's calculation method does not follow the clear evolution logic of rule 110 layer by layer, because if it did, the exact match of H_3 or H_2 for S_7 would also be very high. The experimental results prove that H_3 fitting S_7 is far less easy than imagined; for example, actually, H_3's exact match for S_7 is less than 10% higher than H_3's exact match for S_1. So, the entire neural network is just pointing towards the fitting of S_8, and S_7 does not naturally occupy any special status.

9.3. Decoding the Complete Evolution Sequence of Cellular Automata from the Last Hidden Layer

9.3.1. Experimental Part

Experimental Setup: First, fit the state of a 1D 30-bit cellular automata after 4 layers of evolution under rule 110, training for 15 epochs. Then extract the last hidden layer, use another MLP network to fit the concatenated states of 1 to 4 layers of evolution, and observe the exact match values.

Dataset Parameter Settings:

- Cellular automata bits: 30
- Evolution rule: 110
- Evolution times: 4
  Training Script: `train_mlp_fulltrace.py`

Experimental Results: After sufficient training, an exact match of 100% can be achieved, i.e., 100% sequence accuracy for the evolution sequence of the cellular automata on the validation set.

9.3.2. Discussion of Experimental Results

This experimental result fully demonstrates that the last hidden layer contains a lot of information related to the task evolution sequence. For simple tasks, it can even be completely recovered. Moreover, this property may not be limited to the last hidden layer, but it is possible that the later the hidden layer, the more it contains high-level semantics corresponding to the task.

Act X: Some Further Explorations

This section will briefly describe some further explorations without excessive in-depth discussion and analysis.

10.1. Training Visualization

In those experiments above where the output is an image, it has been shown that the learning process can be visualized through periodic eval images during the eval process. So how to do this for general tasks? I found that as long as eval loss and accuracy statistics are performed separately for each bit during each eval, a relatively good visualization can be completed to study the learning process of the neural network.

When this method is used on binary addition, two phenomena were discovered. One is the easy-to-imagine serial learning, i.e., the accuracy gradually rises from low bits to high bits, because addition is difficult to predict high bit results without knowing accurate low bit results. The other phenomenon is that the highest bit of the result also converges relatively quickly; analysis suggests that some kind of heuristic simple pattern might have been discovered.

When this method is used on cellular automata experiments, it is found that almost every bit converges synchronously, and the convergence degree of each bit of the cellular automata is about the same in each eval.

Training Script: `train_mlp_visualize.py`

10.2. Autoregression

Surprisingly, using a decoder-only transformer can also fit the cellular automata dataset, only by completely changing its format to a string, for example:

`1011...1110 -> 1010...0001`

And then predicting the next token (a certain bit output of the cellular automata) from any position after the arrow; experiments prove that it can converge. But the problem is, this experiment might not actually depart from this paradigm. Imagine that predicting each bit of the output is actually a task type, and the neural network can distinguish which task it is through the input format and learn to predict. This is actually just a symbolic rule learning with branches. As for whether the current large model paradigm can precisely and hallucination-free evolve cellular automata to a relatively deep layer, I cannot give a definitive answer, but I tend to think it is unlikely to be done without the aid of tools, and it indeed made errors in actual tests, just as large models cannot perform reliable large number multiplication without tools, etc.

Dataset Script: `generate_ca_text_format_dataset.py`
Training Script: `train_ar_transformer.py`

Act XI: Theoretical Interpretation

This section attempts a rough phenomenological analysis of these experimental phenomena. If we momentarily set aside some tentative experiments conducted later, and also temporarily ignore experiments that only converge with added decoupling information, we can summarize it as such a fact: given a very small amount of data relative to the task space, for the vast majority of tasks, neural networks can universally fit to the precise rules

behind them. Here we might need to define "task" precisely. Considering that I did not have any restrictions when conceiving and selecting tasks, its real restriction is the input-to-output transformation written by python programs. So now considering this fact, neural networks can universally fit to the precise rules behind them with extremely small amounts of data, what does this mean? First, we can see that because the given training set data is extremely small, it is an extremely underdetermined problem, because corresponding to the given training set, there can actually correspond almost infinite rules. Moreover, considering such an experimental observation that the training set samples need a certain quantity to converge, we can view the process of increasing training set samples as a process of increasing constraints. Each sample is a constraint, and the training of the neural network is similar to solving some equation, or it can also be seen as a process of Bayesian belief update as samples increase. In this way, only when the number of samples is more than a certain amount, based on the constraints of samples and the form of the equation solved by the neural network, the solution starts to stabilize, and from the perspective of Bayesian belief update, the posterior also gradually narrows. So the question is, why can the neural network always choose the real rule behind our generated training set? We can now only draw this conclusion: the neural network has its internal preference or bias. The neural network solves equations according to some preference or bias, and so is Bayesian belief update. So what is this preference or bias? Considering our experimental results, the neural network consistently fits the real rule behind it. The common point among them is that this is the simplest way to describe the dataset, and because the best way to describe or compress the training set is to find this logical transformation rule. So we can conclude that under ideal training data, the neural network implicitly executes [25] Kolmogorov compression or searches for the shortest program description by finding flat minima through SGD, which is the [26] MDL principle.

This is a strong assumption, so we need to give a definition as precise as possible for the rule learning tasks focused on in this paper. Simplified, it can be defined as programmatically generated deterministic transformations.

The rules focused on in this study can be formalized as a special class of functions, whose core characteristics stem from their generation process:

- **Source Code Definition:** Defined not by data points, but by a deterministic, finite-length computer program. This means Kolmogorov complexity is far smaller than the size of its input space.
- **Zero-Entropy Determinism:** For any input, the output generated by the program is unique. This eliminates chance and uncertainty, making the theoretical minimal loss reach absolute zero, and thus possessing algorithmic compressibility.

Below we consider specific supporting evidence.

11.1. Theoretical Assumptions and Evidence

First, we observed in a large number of experiments that as task difficulty increases, the number of samples needed in the training set to support neural network fitting increases. This is exactly consistent with the MDL principle because complex rules need more samples to cover patterns. Moreover, it is observed that as task difficulty increases, the time required for training also increases. We can phenomenologically understand why neural networks learn rules: the neural network makes a choice between learning rules and memorizing samples. For symbolic rules: the complexity of the rule itself is constant and relatively low. However, the difficulty of memorization increases with the increase of sample size. Once the sample size exceeds a certain low threshold, fitting rules becomes "cheaper" than memorization, thereby forcing the model to choose generalization. Contrast with pattern recognition: in traditional tasks (like images), learning is from coarse-grained to fine-grained. In the late stage of training, the difficulty of mining extreme details exceeds the difficulty of memorization, leading the model to turn to memorization and overfitting. Regarding the rapid growth of memorization pressure, perhaps it can be explained this way: the input and output of symbolic data are discrete step jumps, lacking local smoothness, i.e., changing one bit of input may change the entire output. This rugged topological structure makes the cost of memorization based on interpolation extremely high. We might call it the endogenous regularization of data.

If we consider this problem with the above theory, when the number of samples is less than the number of samples needed to fit the rule, the neural network shows overfitting during training. But here comes a question: if we understand that the neural network chooses memorization when it finds memorization easier during training, and chooses fitting rules when it finds fitting rules easier, is this phase transition point consistent with the phase transition point mentioned above where complex rules need more samples to cover patterns according to the MDL principle? I think they are very likely consistent. In microscopic neural network dynamics, this manifests as gradient direction and magnitude; in phenomenological theory, it may manifest as a comparison of the difficulty of memorization and fitting rules; macroscopically, it may correspond to the description length principle of MDL: if training samples are few, the fitting mode of memorization or biased towards memorization may be smaller than the description length of the rule. And these three perspectives should correspond to the same thing; they should completely align.

Thinking from the perspective of [34] Implicit Regularization theory, the characteristics of stochastic gradient descent make it implicitly prefer flat regions: optimization trajectories continuously escape sharp minima because they are prone to failure due to perturbations, and finally settle in flat minima regions. In weight space, flat minima (broad regions with approximately constant error) allow weights to be stored with low precision, while sharp minima (narrow valley bottoms) require high precision. From the MDL perspective, the former corresponds to short program descriptions (only a few bits needed to locate "any point within the basin"), and the latter corresponds to long programs (need precise coordinates). Or thinking from another angle, the gradients of rule solutions are coherent across different training set samples and will reinforce each other, while the gradients of non-rule solutions or memorization solutions are incoherent across different training set samples

and will cancel each other out. This explains our observation. Therefore, SGD achieves implicit optimization for the shortest program description by finding flat minima. This is some inevitable phenomenon at the statistical level. The randomness of SGD is like a small ball perturbed in an energy landscape. Due to its free energy minimization tendency—pursuing both low energy (low loss) and high entropy (wide flat regions, allowing low precision storage)—it inevitably settles in flat basins. This is essentially the formation of dissipative structures in non-equilibrium thermodynamics: as an open system, the network spontaneously emerges low-entropy ordered rule structures from high-entropy chaos by continuously dissipating gradient energy. Its statistical essence follows the free energy minimization principle—the system automatically selects the path with the lowest complexity cost under constraints, making intelligence an inevitable emergence of energy-information conversion under non-equilibrium conditions.

Finally, linking to the concept of **compression involves intelligence**, combining [32] Ilya's speech in 2023 and [33] Jack Rae's online speech, their main idea is that optimal unsupervised learning is executing regular Kolmogorov compression (rather than compression conditional on data), and neural network + SGD is our current best practical method to approximate the uncomputable Kolmogorov compressor. Here, our experiments provide the purest test method for this theory. Since we use programmatically generated ideal data with zero noise, we eliminate the irreducible error inevitable in real-world data. Therefore, the training of the neural network is choosing what kind of way to compress the mapping from input to output in the training set. And our large number of experiments have also proven that, given enough training samples, the neural network will consistently fit to the true rules. This gives strong experimental evidence for the above proposition. This echoes the core assertion of "compression is intelligence": when a neural network selects a short program description for generating rules under zero-noise conditions, it is exactly performing lossless compression on the data—not superficial statistical pattern matching, but compression coding of the underlying causal mechanism. The limit of this compression, i.e., the minimum program length, is exactly the goal pursued by [24] Solomonoff Induction.

11.2. Relevant Experimental Evidence

Relevant scripts for this part can be found in the `research` subdirectory under the code repository root. For extreme over-parameterization experiments, refer to `train_mlp.py` in the root directory, changing hidden layers to 8192.

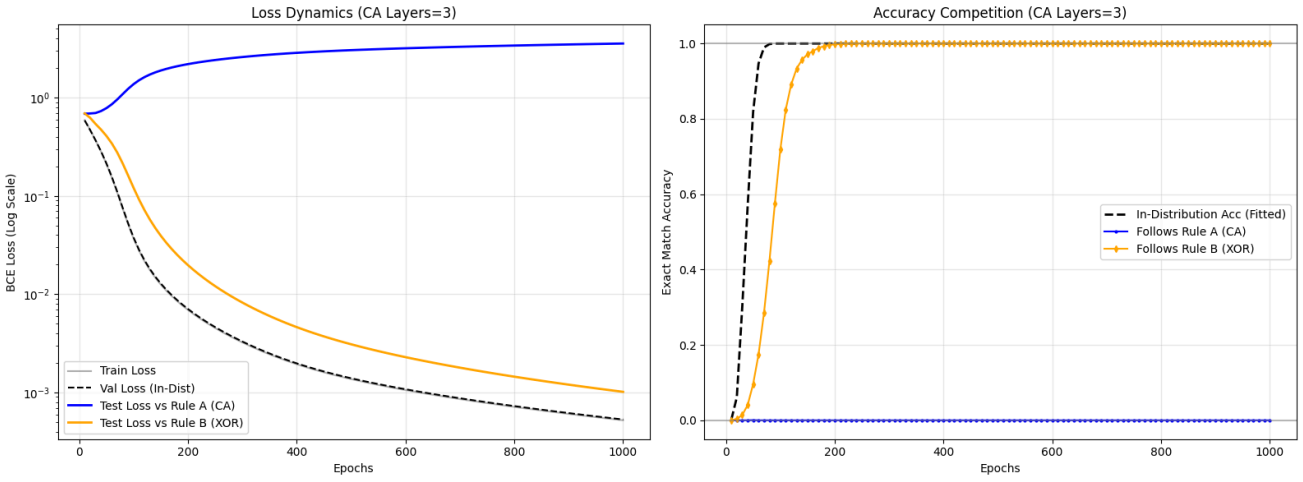
11.2.1. [27] NTK Related Experiments

I used an experimental script to roughly test whether my discovery only involves lazy learning in NTK theory. The answer is negative. For a task of a 1D 30-bit cellular automata evolving 3 layers of rule 110 with a training set of 30,000 samples, the analytical solution of NTK generalization only has 1.4% exact match accuracy on the test set, while with the same training set, real neural network training can reach 100% test set exact match accuracy. This is not explainable by error. Such contrast has also been confirmed in some other experiments.

11.2.2. Rule Preference Experiment

In this experiment, I designed two rules coexisting in the training set samples that can both precisely deduce output from input, and then tested which rule the neural network is more inclined towards internally on the test set. Experiments show that when the difficulty gap between the two rules is too large, after training for a period, the neural network will completely accept one rule and abandon the other. This clearly proves the neural network's tendency to choose simpler rules. Additionally, interestingly, the fitting speed of the in-distribution validation set is faster, reaching 100% accuracy in advance. Note that this is not memorization because it does not overlap with the training set; at this time, it should have learned some entangled state of simultaneously using both rules and corresponding data to predict. Some time later, the neural network completely turns to the simpler rule, completely turning from the entangled state of utilizing two rules and data to the simpler rule, and the accuracy on the test set of this rule reaches 100%. This is actually the [28] **Grokking** phenomenon! And this also explains that even after the neural network completely fits the training set, the implicit regularization of SGD is still driving the network weights to drift towards simpler rules. Therefore, SGD achieves implicit optimization for the shortest program description by finding flat minima.

As shown in the figure below:



When the difficulty gap between the two tasks is not large, i.e., when the cellular automata evolution layer is 1 or 2, both rules generalize poorly out-of-distribution, presenting an entangled state of rules.

11.2.3. Extreme Over-parameterization Experiment

Given a 2.69M naive MLP network, asking it to fit some training sets with extremely small samples, it still does not overfit. Including, for 1D 30-bit cellular automata, evolving 1 layer of rule 110 requires only 450 samples to ensure no overfitting; evolving 2 layers of rule 110 requires only 1800 samples to ensure no overfitting, and other experiments. Such extreme experimental configuration makes memorizing training set samples extremely easy, yet there is no overfitting, challenging the common sense of Large Model + Small Data = Overfitting, and supporting the hypothesis of deep neural networks as MDL solvers, i.e., neural networks are looking for a shortest program description.

11.3. Some Additional Thoughts

- Further thinking on the essence of MDL: Neural networks might be performing some kind of logical auto-simplification. Taking various experiments in this paper as examples, it is very likely that simplified logic is unintentionally introduced. For instance, creating a trivial example, if my program calculates `(x+1)+(x-1)`, then the data received by the neural network is just `2x`. It can only see data, so it will only fit data. As for how many equivalent logics describe the data, this is unknown and uncared for by the neural network. But the logic of the program is `(x+1)+(x-1)`, and maybe human understanding is also the same as the program. This is just an analogy; the actual situation is much more complicated. This situation is more likely to occur when superimposing many layers of logical steps; some simplifiable situations or logical shortcuts might appear, and at this time, the neural network will notice and perform simplification. A more practical example is the experiments I did; I did not specify (and of course impossible to specify) to the neural network which input corresponds to what meaning and which output corresponds to what meaning. When writing programs, these are mostly my own arbitrary designations and for the convenience of neural network training, but the neural network still fitted. Here, we have no reason to believe that the logic fitted by the neural network is consistent with human logic or the logic of dataset generation. So, if there is some logical shortcut, the neural network might utilize it. Note that the shortcut here refers to some logical simplification, not [35] shortcut learning different from the original logic. Moreover, I think the situation goes beyond this. Considering that the way neural networks carry logic is distributed representation and distributed computing, I think it might further simplify logic that appears unsimplifiable to humans in its own form and self-organizationally distribute it to different positions of neural network weights to satisfy its own MDL principle. Therefore, from this perspective, there exists such a possibility: looking for the interpretability of neural networks is inherently impossible because the calculation and representation of neural networks are not human-understandable formal logic. It is possible to preliminarily associate network weights with human-understandable concepts, but if precise correspondence is insisted upon, transforming into formal logic, the neural network will immediately lose its advantage and essence; it is no longer a neural network.
- My experiment on combining MNIST and cellular automata discussed in Section 7.3 implies that pattern recognition and rule learning might both be native capabilities of neural networks, or even the same capability. Then, a very natural inference is that our explanation of rule learning above, i.e., neural networks searching for the shortest program description, is very likely also applicable to the pattern recognition tasks we are familiar with, but this hypothesis needs further experimental verification and theoretical research.
- Regarding the problem of out-of-distribution generalization, first, the input and output formats and number of bits of my experimental network structure are constant, so it cannot be extended to more input or output bits, and thus does not have the characteristics of composability of true symbolism. Another case, if the network structure

is unchanged, then the performance in out-of-distribution generalization is mainly determined by the MDL principle, i.e., jointly determined by training data distribution and the MDL principle.

- A key question: Is the neural network's preference completely equivalent to the MDL principle? A more precise definition might be needed: Traditional "simple" usually refers to shorter description length and lower Kolmogorov complexity, but the "simple" for neural networks is more likely simple in the dynamical sense—that is, flat minima that are easy for SGD trajectories to reach and stable. If so, then how microscopic neural network dynamics correspond to the macroscopic MDL principle might be a question requiring serious discussion. The difficulty of tasks like mod 3 might reveal this difference: it is simple in description length, but might be located in a region difficult for SGD dynamics to reach. Currently, there seem to be few methods to study the properties of neural networks without going through actual experiments, so more experiments might be needed in the future to establish a relevant dynamic theoretical system and study how it corresponds to the macroscopic MDL principle.

11.4. Related Work

- **Connection with Grokking:** The previous rule preference experiment is highly isomorphic to Grokking experiments and also suggests a possible explanation for the Grokking phenomenon: after the neural network fits the training set, the implicit regularization of SGD is still driving network weights to drift towards simpler rules. Therefore, SGD achieves implicit optimization for the shortest program description by finding flat minima. And my extensive experiments also prove that for good datasets, including sufficient data volume and moderately difficult tasks, there is completely no need to go through a process of first overfitting then turning to generalization. The training process of the neural network will directly show the situation where train loss and val loss decrease synchronously, with val loss always slightly lower, until precise fitting is reached.
- **[29] Othello-GPT:** Their experiment proved that within a limited length, the autoregressive method can also achieve good precision. This experiment can be interpreted as the Transformer maintaining a world model, or it can be interpreted as performing multi-task learning, where different tasks correspond to different input lengths. If another similar game allows extension to longer variable inputs, whether it can still maintain high-precision prediction might still be a question temporarily. Additionally, my hidden layer probe experiment also has similarities with the probe experiment in this paper.
- **Amortized Planning with Large-Scale Transformers: A Case Study on Chess:** This paper inspired my research. They also reached an important conclusion that Transformer might be a general algorithm approximator. My current research proves that it might not only be Transformers that have such capabilities, and it is not limited to approximation; for some problems, it is a precise general algorithm fitter and executor. Moreover, the problem studied in this paper is also very interesting. Letting Transformer directly fit the algorithm for playing chess achieved significant success, indicating that as an algorithm

fitter, what neural networks can do goes far beyond the discrete rule transformations focused on in this paper.

- **[30] A Generalist Neural Algorithmic Learner:** My discovery shows that some algorithm fitting can be completed without specific architectures and intermediate signal supervision. For these tasks, the intermediate steps of the algorithm can in fact emerge spontaneously internally.
- **[31] What Can Neural Networks Reason About?:** This paper emphasizes the adaptability of different architectures to different problems. My discovery does not deny this point, but my paper mainly wants to use the fitting ability of general neural network architectures for different ideal data to trigger some kind of phase transition in neural networks, thereby exploring the properties of neural networks themselves, rather than focusing on the efficiency of specific neural network architectures in solving specific tasks.

Act XII: Conclusion — The Dawn of Reliable AI

The end of this study provides a clear and profound answer to a fundamental question that has plagued the field of artificial intelligence for decades: Can a neural network born for probability and statistics truly understand and execute precise, deterministic logical laws?

The experimental results strongly imply that this is affirmative. And the key to achieving this does not rely on larger, more complex model architectures, but stems from a fundamental paradigm shift: I abandoned the imitation of noisy real-world data and turned to using ideal data generated programmatically by pure rules; I abandoned the fragile autoregressive prediction chain and embraced a parallel, one-step holistic solving framework.

I did not stop at proving the superiority of specific advanced architectures but conducted a systematic test. I obtained an unexpected discovery that is incomparably consistent with first principles: a Multi-Layer Perceptron (MLP) purely composed of massive parameters without any structural bias exhibited significant generalization capabilities close to zero error on complex logical evolution tasks. The success of Recurrent Neural Networks pushes this discovery to a deeper philosophical level—it implies that neural networks can even "create" an iterative algorithm that I cannot intuitively understand but is functionally completely correct in order to satisfy the final mathematical constraints.

This series of experiments points to an unshakable conclusion: precise, algorithmic reasoning capability is not the "patent" of any specific architecture. It is a universal potential inherent in the connectionist system itself, deep-seated, and capable of being "awakened" by the correct paradigm. Under this paradigm, the superiority of architectures is no longer a distinction between "can and cannot," but a distinction determining the "efficiency" and "capability boundary" of awakening this potential.

This is not only a valuable exploration but provides a new research perspective. It means that we have acquired the ability to "sculpt" abstract, precise, human-understandable laws

into the "marble" of neural networks without worrying about it producing random "hallucinations."

This heralds the possibility of a new era: an era driven by truly reliable, controllable, and explainable neural networks. An era where we can not only converse with AI but also trust it to execute precision calculations, simulate the physical world, and even assist us in making the most rigorous scientific discoveries.

About the Nature of This Report

This research was completed independently by the author as an independent researcher. Due to the wide range of experiments covered and the massive workload, this initial version of the paper is released as a technical report. The author is aware that there is still room for improvement in presentation and format, welcomes constructive feedback, and plans to continuously improve it in subsequent versions.

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Finally, due to time constraints, some calculations and details in this text may contain omissions or errors. The author would be very grateful for criticisms and corrections from readers.

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Appendix:

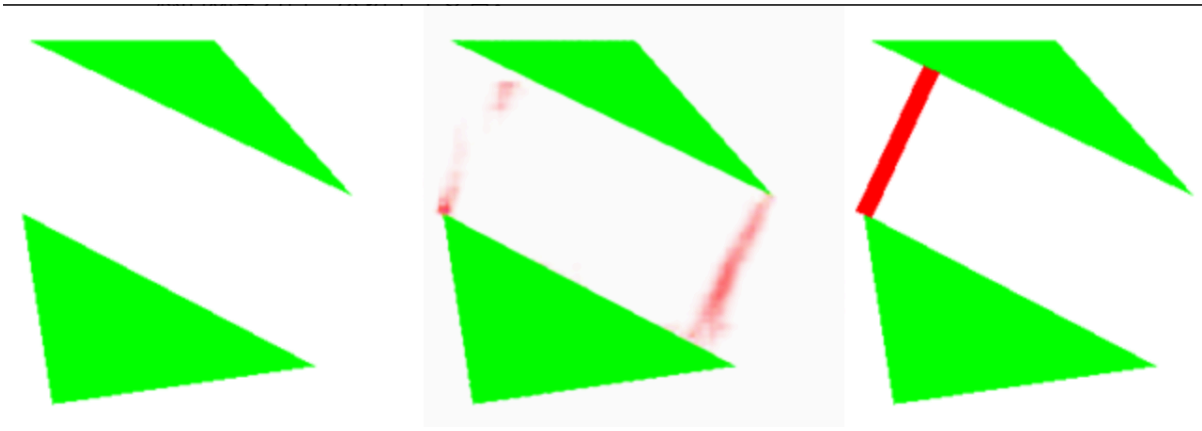
Some Interesting Experimental Images

- **Visualization of Learning Results**

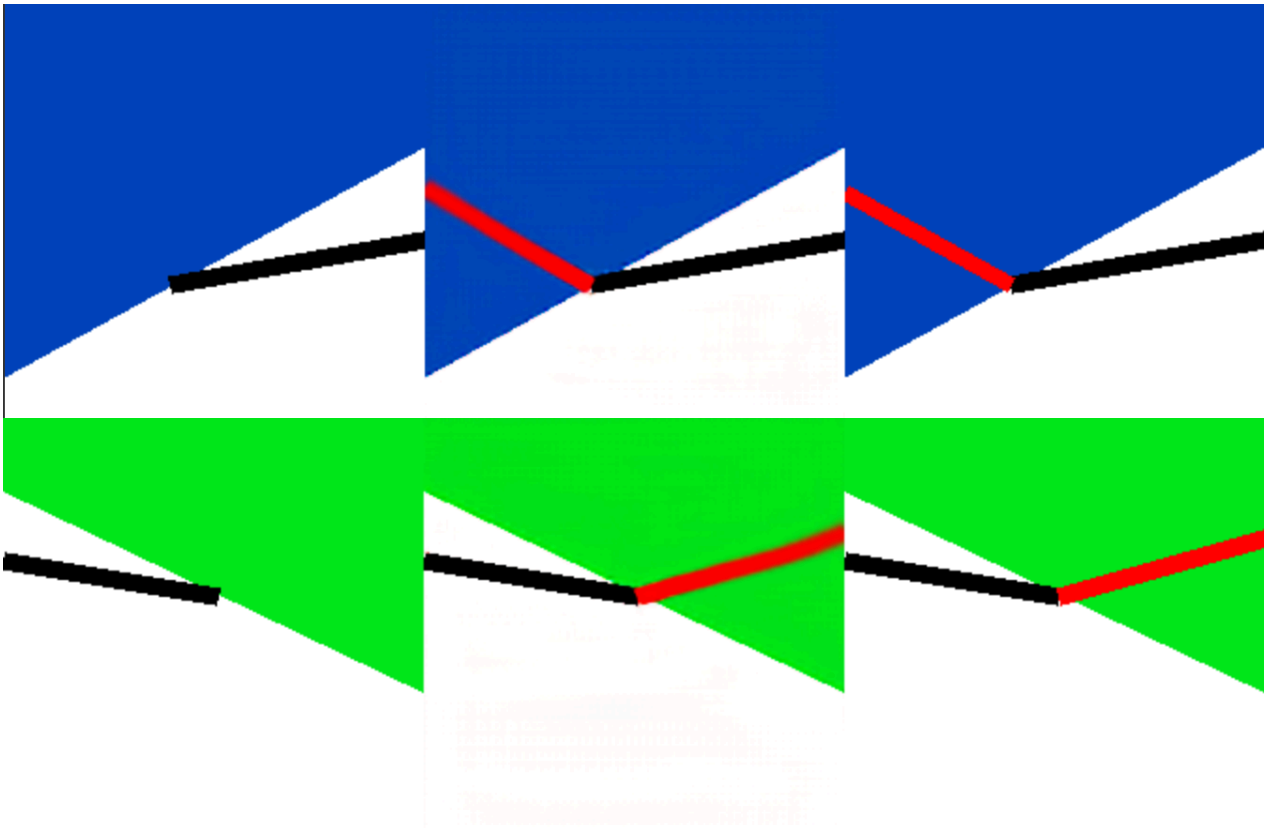
  Task Description: The input image has a white background with two arbitrary non-overlapping green solid triangles. The task is to draw a red line segment indicating the closest distance between the two triangles.

  Analysis: It seems the model learned that the closest distance line segment is either the perpendicular line from a vertex to the corresponding edge of the other triangle, or the connecting line between two vertices. However, here it is difficult for it to determine which distance appears closer, so it drew both perpendicular lines, but both are blurry.

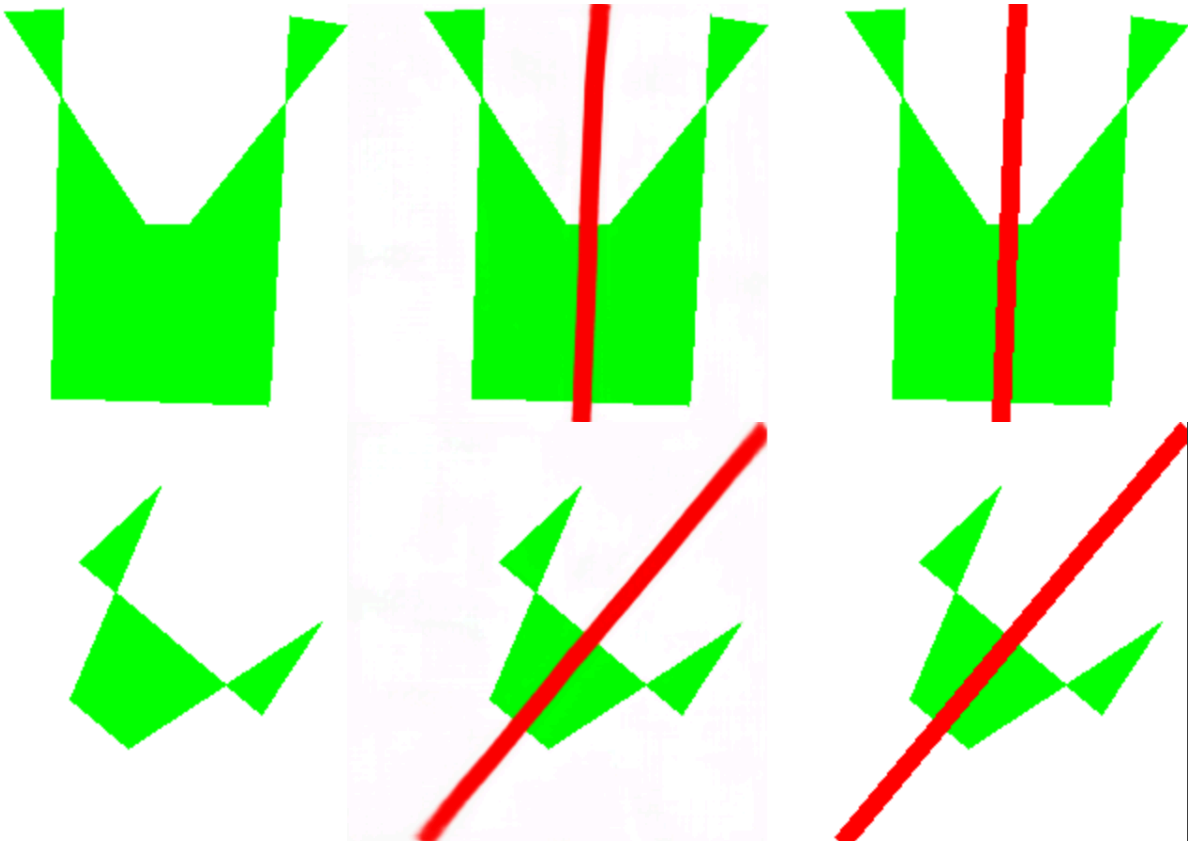
  Left is input image, middle is model generated image, right is ground truth.

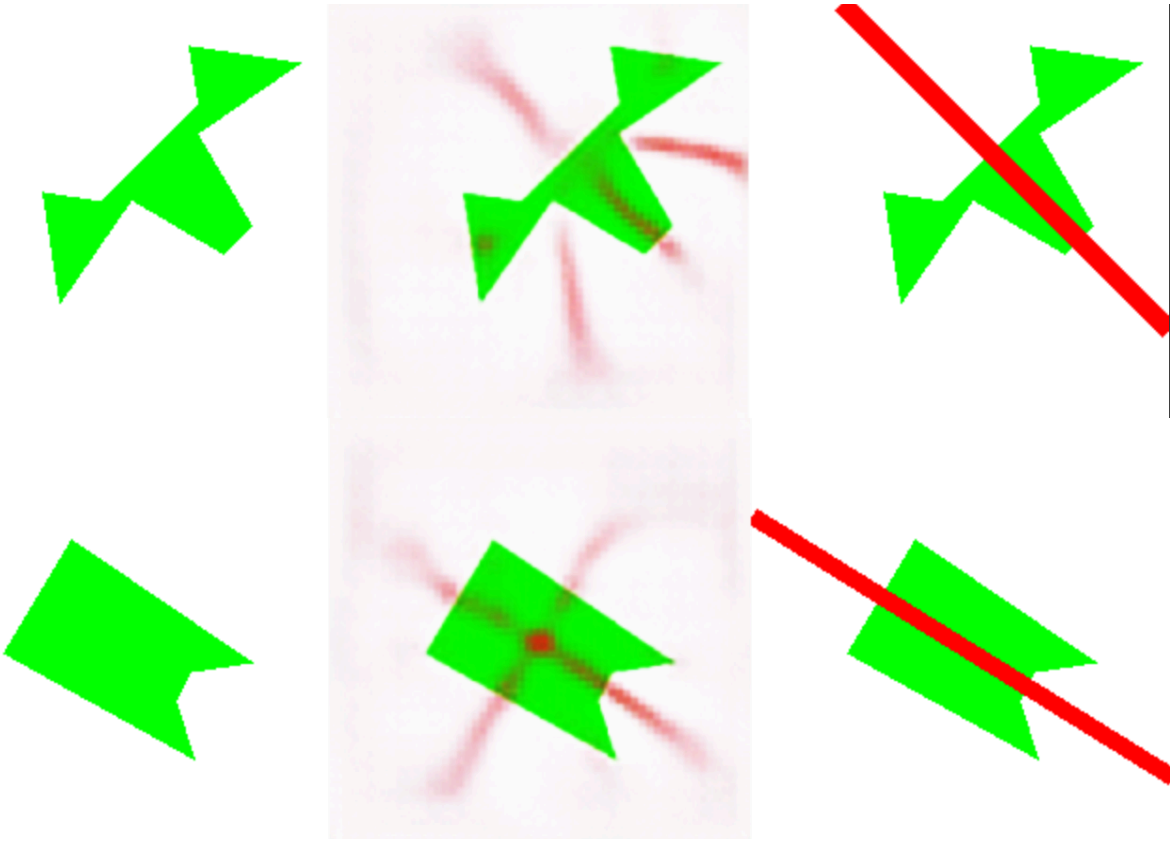
- **Training Results of Variable Refractive Index**

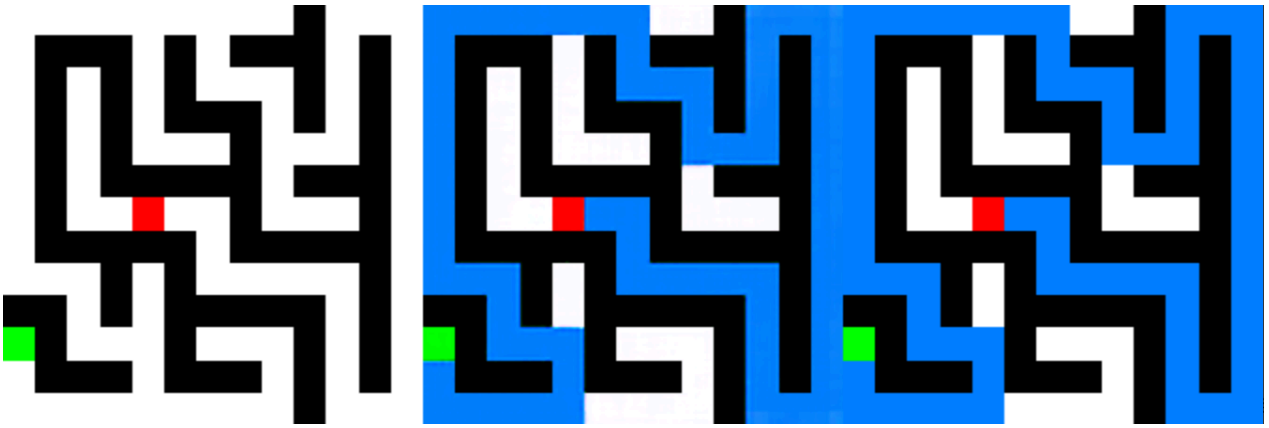
- **Symmetry Axis Learning**



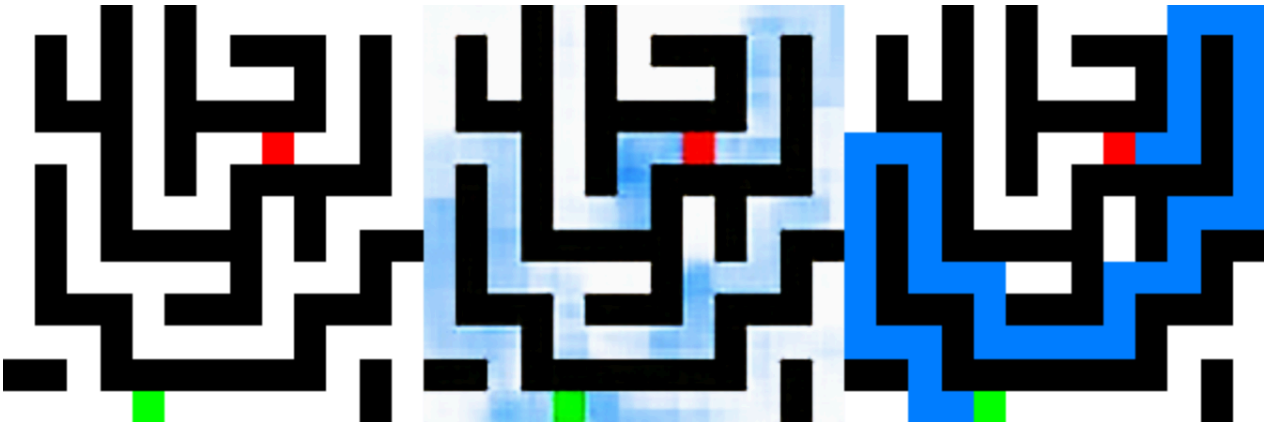
Learning Intermediate States



- **Maze Experiment**



Learning Intermediate States



List of All Experiments Conducted

A. Symbolic Math Logic

- `generate_conditional_add_subtract.py` : Generates addition or subtraction (absolute value) problems for two N-bit integers. Contains two modes.
- `generate_add_binary_modulo.py` : An early basic arithmetic experiment to test the model's ability to learn modular addition (or "truncated addition"), common in fixed-width integer arithmetic in computer hardware.
- `generate_multiply_binary.py` : A benchmark test for binary arithmetic capability, generating N-bit integer multiplication datasets.
- `generate_multiply_binary_no_carry_phase1.py` : Phase 1 of the multiplication "decoupling" experiment. Aims to test if the model can learn the first step of multiplication: bitwise multiplication and shifted addition without carries, decomposing a complex multiplication problem into a simpler counting problem.
- `generate_multiply_binary_from_counts_phase2.py` : Phase 2 of the multiplication "decoupling" experiment. Aims to verify if an independent model can learn to handle complex carry logic, calculating the final binary product from a "carry-free count vector."
- `generate_add_hexadecimal.py` : Compares the model's learning capability under different symbol systems. This script aims to verify if the model learns the abstract mathematical concept of addition or just patterns specific to binary symbols.
- `generate_multiply_decimal.py` : Tests the model's ability to process non-binary symbol inputs (0-9 characters) and perform arithmetic operations (multiplication).

- `generate_add_binary_with_position_shuffle.py` : The "position shuffle" part of the "semantic shuffling" series of experiments. Aims to verify if the model relies on the fixed spatial structure of imports or can learn abstract relationships independent of position.
- `generate_symbol_add_shuffle_dataset.py` : A key decisive experiment in my research, aiming to thoroughly separate the model's "superficial pattern matching" capability from its "abstract structure learning" capability.
- `generate_add_hidden_constant.py` : Tests the model's ability to infer hidden rules or parameters from large numbers of samples without any direct clues. This is similar to a simplified System Identification problem.
- `generate_multitask_alu.py` : Aims to build a multi-task learning scenario simulating an Arithmetic Logic Unit (ALU). Tests if the model can execute multiple different, well-defined computing tasks in parallel on the same input in a single forward pass.
- `generate_modulo_operation.py` : Explores the model's ability to learn Modulo Operation, a crucial but "cyclic" operation in number theory and computer science.
- `generate_rsa_encryption.py` : Tests the model's ability to learn highly non-linear deterministic rules considered "hard" computationally. RSA encryption is a typical example.
- `generate_deduction_chain_text.py` : Generates multi-step logical reasoning tasks, testing the model's ability to perform symbolic deduction, similar to a simplified theorem prover.
- `generate_deduction_multirule_text.py` : Tests whether the model can correctly "route" to the corresponding rule based on a Query and make judgments when facing multiple independent, unrelated rules.
- `generate_deduction_multirule_text_v2.py` : Tests whether the model can correctly "route" to the corresponding rule based on a Query and make judgments when facing multiple independent, unrelated rules.
- `generate_deduction_multirule_binary.py` : An optimized format version of the multi-rule reasoning task, aiming to test if compact binary coding is more conducive to model learning than sparse text formats.
- `generate_deduction_fixed_depth.py` : Tests the model's multi-step reasoning capability in symbolic deduction tasks with clear structure and fixed depth.
- `generate_function_composition.py` : Tests the model's capability to learn Function Composition. Requires the model to parse instructions sequentially and transform data like an interpreter.
- `generate_count_set_bits.py` : Tests the model's capability to execute global aggregation operations. Unlike local rules, counting requires the model to synthesize information from the entire input sequence.
- `generate_sum_pattern_positions.py` : Tests the model's capability to execute more complex, grouped parallel aggregation tasks. The model needs to segmentation the input, classify each segmented pattern, and finally accumulate position information for patterns belonging to the same class.

- `generate_sum_pattern_positions_v2.py` : Tests the model's capability to execute more complex, grouped parallel aggregation tasks. The model needs to segmentation the input, classify each segmented pattern, and finally accumulate position information for patterns belonging to the same class.
- `generate_sum_pairwise_hamming_distance.py` : Tests the model's capability to execute a complex task requiring two layers of nested aggregation operations. The model needs to perform global statistics on each bit first, and then accumulate results from all bits.
- `generate_circular_shift.py` : Tests the model's capability to learn bitwise shift operations, especially circular shift, a common operation in cryptography and low-level programming.
- `generate_multiply_matrix_3x3.py` : Tests the model's capability to learn structured algebraic operations (matrix multiplication), which requires more complex "data routing" and "multiply-accumulate" capabilities than simple scalar operations.
- `generate_evaluate_boolean_expression_text.py` : Tests the model's capability to parse a simple Domain Specific Language (DSL) and execute evaluation, going further than fixed-structure expression evaluation.
- `generate_evaluate_arithmetic_expression.py` : Trains the model to execute symbolic expression evaluation tasks, requiring the model to understand operator precedence (implicitly expressed through tree structures), variable substitution, and arithmetic operations.
- `generate_evaluate_arithmetic_expression_no_multiply.py` : A simplified version of `generate_evaluate_arithmetic_expression.py` , aiming to reduce learning difficulty by removing multiplication to test the model's capability on more basic arithmetic expression evaluation.
- `generate_evaluate_arithmetic_expression_no_multiply_small_range.py` : A further simplification based on the previous "no multiplication" version, further reducing learning difficulty by creating a smaller numerical range, used to precisely diagnose the model's performance bottleneck on the simplest expression evaluation tasks.
- `generate_check_boolean_equivalence.py` : Tests the model's judgment capability on Boolean algebra logical equivalence. This is an abstract symbolic reasoning task requiring the model to understand expression structure and Boolean operation laws.
- `generate_polynomial_shift_coefficients.py` : Tests the model's capability to learn an abstract algebraic transformation rule. This task requires the model to understand the intrinsic structure of polynomial expansion.
- `generate_convolution_2d.py` : Tests the model's capability to learn 2D convolution (Conv2D), a basic image processing operation, and explores whether it can infer hidden fixed rules (i.e., the convolution kernel itself) from input-output pairs.
- `generate_simple_block_cipher.py` : Tests the model's capability to "crack" or learn a simple but non-trivial custom encryption algorithm. This task represents a class of complex symbolic transformation rules with high chaos and avalanche effects.
- `generate_sin_function_float32.py` : Tests the model's capability to fit continuous, periodic, non-linear functions (sin(x)) using standard 32-bit floating-point format for input

and output.

- `generate_sin_function_float64_to_int12_deprecated.py` : Another encoding attempt for the sin function fitting task, aiming to explore the impact of using higher precision floating-point input and lower precision quantized binary output on learning effects.
- `generate_sin_function_float32_to_quantized_int.py` : Tests the model's capability to fit continuous, periodic, non-linear functions (sin(x)) and explores the impact of different input/output encoding schemes on learning effects.
- `generate_multiply_binary_modulo.py` : Part of the basic arithmetic experiments, tests the model's mastery of truncated multiplication (or modular multiplication).
- `generate_explainable_two_step_calculation.py` : Tests the model's capability to output calculation "intermediate steps" or "chain of thought," a direct verification of "functional interpretability."
- `generate_min_swaps_for_checkerboard.py` : Solves LeetCode 782 "Transform to Chessboard" - https://leetcode.cn/problems/transform-to-chessboard/ - Minimum swaps of rows and columns to transform a 0/1 matrix into a "chessboard" pattern.
- `generate_min_flips_for_alternating_binary.py` : Tests the model solving a string optimization problem based on bit flips, which can be ingeniously mapped to a sliding window problem.
- `generate_min_swaps_for_checkerboard_v2.py` : Solves LeetCode 1536 "Minimum Swaps to Arrange a Binary Grid" - https://leetcode.cn/problems/minimum-swaps-to-arrange-a-binary-grid/ - Minimum swaps of adjacent rows to transform a binary grid into upper triangular form.
- `generate_min_prefix_flips.py` : Tests the model's capability to learn a greedy algorithm that depends on historical states and processes sequentially.
- `generate_min_flips_for_chunked_binary.py` : Tests the model's capability to learn a string transformation optimization problem based on local chunks.
- `generate_largest_island_by_adding_one_cell.py` : Solves an algorithm problem involving graph traversal and global optimization (LeetCode 827. Making A Large Island). Model needs to evaluate all possible "land reclamation" positions.
- `generate_largest_island_by_adding_one_cell_v2.py` : Solves an algorithm problem involving graph traversal and global optimization (LeetCode 827. Making A Large Island). Model needs to evaluate all possible "land reclamation" positions.
- `generate_sat_solver_text.py` : Tests the model's capability to solve a landmark NP-complete problem — Boolean Satisfiability (SAT).
- `generate_sat_solver_compact_text.py` : A variant of `generate_sat_solver_text.py` using different input encoding formats to solve the same 3-SAT problem.
- `generate_binary_mod3_dfa_explain.py` : An interpretability experiment aiming to test if the model can learn and output the internal state transition process of a Deterministic Finite Automaton (DFA).
- `generate_add_binary_explainable.py` : Tests if the model can learn the internal mechanism of binary addition (carry propagation). Requires outputting not just results but the carry state of each bit.

- `generate_add_quaternary_float_encoding.py` : Tests if MLP has the capability to parse real numbers, map them to encoded binary, and complete calculations. Proves that real numbers spaced at certain intervals can be used as pure symbols when training MLPs.
- `generate_add_binary_masked_output.py` : Tests the model's capability on "conditional output" tasks. Requires the model to not only calculate addition but also dynamically decide which bits of the result to output based on input Mask. Similar to Attention mechanism.
- `generate_multitask_prefixed_ca110.py` : Generates a "Prefixed CA Evolution + 4 Post-tasks" multitask learning dataset to test the model's capability to execute shared representation extraction and multiple independent downstream tasks in a single forward pass. An important extension to `generate_multitask_alu.py` .
- `generate_convergent_abstraction.py` : Generates a "Convergent Abstraction" (Multi-input -> Single Intermediate Representation Hub) experiment dataset. Aims to verify the key hypothesis in Act VIII: Can different input tasks naturally converge to a shared abstract intermediate representation and complete subsequent reasoning tasks based on it?
- `generate_multiply_binary_showdown.py` : Three-algorithm showdown for binary multiplication. Extends the idea of `generate_rain_water_final_showdown.py` to multiplication tasks, comparing three decoupling methods.

B. Algorithm Learning

- `generate_sort_integers.py` : Tests the capability to execute basic sorting algorithms, a classic non-local algorithm task requiring comparison and rearrangement of input elements.
- `generate_edit_distance.py` : Tests the capability to learn Dynamic Programming problems. Edit distance is a typical DP problem requiring the conceptual construction of a 2D solution matrix. LeetCode 72. Edit Distance
- `generate_edit_distance_explainable.py` : A core experiment for "functional interpretability." Requires outputting the complete "chain of thought" (editing process) along with the final answer. LeetCode 72. Edit Distance (Explainable / Path Construction Version)
- `generate_maze_random_walls.py` : Tests basic wayfinding capability in randomly generated "porous" mazes.
- `generate_maze_dense.py` : Tests path planning capability in complex, "dense" mazes similar to human designs, more challenging than random wall mazes.
- `generate_blocks_world_arbitrary_goal.py` : Solves the classic "Blocks World" planning problem. As discussed in the Apple Research paper [15], this is a benchmark for evaluating reasoning capabilities in large language models. This script implements the general version allowing arbitrary initial and goal states.
- `generate_blocks_world_fixed_goal.py` : A simplified version of "Blocks World" with fixed goal states, aiming to test learning capability under clearer goals and more

structured state spaces.

- `generate_blocks_world_fixed_goal_multilabel.py` : Improves "Blocks World" by allowing multiple optimal solutions, testing the capability to handle multi-label classification, reflecting equivalent optimal paths in planning problems.
- `generate_blocks_world_fixed_goal_multilabel_fixed_format.py` : The final optimized version of "Blocks World," improving input representation with fixed-slot format to provide a clearer, more structured learning target.
- `generate_checkers_jump_1d.py` : Solves the 1D checkers jump planning problem. A core control experiment for verifying reasoning capability in complex planning problems, as discussed in [15].
- `generate_river_crossing_puzzle.py` : Solves the classic "River Crossing" constraint satisfaction and state space search problem. A benchmark for testing symbolic reasoning limitations in LLMs.
- `generate_trapping_rain_water_aggregate.py` : Initial attempt at "Trapping Rain Water" with aggregate output. Proved much harder to learn than the decoupled version, demonstrating the systematic impact of output format design. Corresponding LeetCode problem: 42. Trapping Rain Water.
- `generate_trapping_rain_water_decoupled.py` : Solves LeetCode Hard 42. Trapping Rain Water. Success demonstrates the capability to learn complex algorithms requiring global information via problem decoupling.
- `generate_trapping_rain_water_2d.py` : Extension to 2D Trapping Rain Water. Requires understanding "enclosure" and "boundary" in 2D space. LeetCode 407. Trapping Rain Water II.
- `generate_skyline_max_height_aggregate.py` : Initial attempt at the "Skyline" problem with aggregate output. Used to compare learning difficulty with decoupled output. LeetCode 1840. Maximum Building Height.
- `generate_skyline_all_heights_decoupled.py` : Tests capability to solve global optimization problems with 1D spatial constraints. LeetCode 1840. Maximum Building Height (Decoupled version).
- `generate_hanoi_tower_path_strategy_sep_format.py` : Control Group A for Tower of Hanoi, using separator + binary coding input format.
- `generate_hanoi_tower_path_strategy_fixed_format.py` : Control Group B for Tower of Hanoi, using structured fixed-slot input format to verify the impact of data representation.
- `generate_hanoi_tower_compare_formats.py` : Comparative experiment script generating two input formats for the same Hanoi problem.
- `generate_hanoi_tower_compare_formats_and_strategies.py` : Comprehensive comparative script generating two input formats and two strategy datasets (path strategy vs. global strategy).
- `generate_hanoi_tower_build_full_state_graph.py` : A comprehensive tool for "Tower of Hanoi" research, building a full state graph to deeply analyze recursive structure understanding.

- `generate_hanoi_tower_sample_from_state_graph.py` : Sampling script using the full state graph to extract specific training data subsets like "twisted paths" for fine-grained ablation studies.
- `generate_sokoban_planning_astar.py` : Solves classic "Sokoban" planning problem. Harder than simple path planning due to state changes.
- `generate_sokoban_planning_full.py` : Solves "Sokoban" planning. A high-difficulty AI task involving search in a huge state space.
- `generate_sokoban_planning_claude_deprecated.py` : (Deprecated) Early attempt.
- `generate_matrix_flip_strategy.py` : Solves matrix optimization (Score After Flipping Matrix). Tests capability to learn a "strategy" rather than final result. LeetCode 861. Score After Flipping Matrix.
- `generate_matrix_flip_max_score.py` : Tests capability to learn matrix optimization requiring a two-step greedy strategy, outputting aggregate score. LeetCode 861. Score After Flipping Matrix.
- `generate_min_k_bit_flips.py` : Tests capability to learn a greedy algorithm dependent on historical states and sequential processing using variable 'k'. LeetCode 995. Minimum Number of K Consecutive Bit Flips.
- `generate_min_k_bit_flips_fixed_k.py` : Same as above but with fixed/hidden 'k'. Control experiment. LeetCode 995. Minimum Number of K Consecutive Bit Flips.
- `generate_special_binary_string_recursion.py` : Tests capability to learn a recursively defined string transformation rule. LeetCode 761. Special Binary String.
- `generate_count_connected_components.py` : Tests basic understanding of graph structure, specifically connectivity. LeetCode 323. Number of Connected Components in an Undirected Graph (Note: Premium problem).
- `generate_check_graph_connectivity.py` : Tests judgment of path existence between two points in a graph. LeetCode 1971. Find if Path Exists in Graph.
- `generate_minimize_malware_spread.py` : Solves graph-based virus spread optimization. 924. Minimize Malware Spread. Requires understanding connectivity and impact of node removal.
- `generate_count_islands_1d.py` : Tests pattern recognition and counting on 1D sequences.
- `generate_find_articulation_points.py` : Tests identification of articulation points/bridges in graphs. LeetCode 1568. Minimum Number of Days to Disconnect Island.
- `generate_nim_game_zeckendorf.py` : Tests ability to learn a non-intuitive game theory problem based on complex number theory (Zeckendorf representation).
- `generate_longest_subsequence_constrained.py` : Tests processing of complex optimization combining sequence operations and numerical constraints. LeetCode 2311. Longest Binary Subsequence Less Than or Equal to K.
- `generate_treasure_hunt_tsp.py` : Solves complex state space search combining BFS and status compression DP (TSP variant).

- `generate_freedom_trail_dp.py` : LeetCode 514. Freedom Trail. Tests capability to solve complex optimization requiring DP and path backtracking.
- `generate_sum_of_subset_with_mask.py` : Tests aggregation (sum) of elements selected by a binary mask. Verifies impact of task structure itself.
- `generate_sudoku_6x6.py` : Tests capability on strong Constraint Satisfaction Problems (Sudoku).
- `generate_valid_parentheses_path_random_deprecated.py` : (Deprecated) Early attempt at valid parentheses path.
- `generate_valid_parentheses_path_balanced.py` : Pathfinding on 2D grid constrained by stack structure (parentheses matching). LeetCode Hard Check if There Is a Valid Parentheses String Path.
- `generate_point_in_polygon.py` : Tests learning of Ray Casting Algorithm in computational geometry.
- `generate_shortest_path_in_matrix_bfs.py` : Tests finding shortest path in 2D grid based on BFS.
- `generate_sudoku_4x4_stepwise_deprecated.py` : (Deprecated) Tests stepwise reasoning capability.
- `generate_tiling_problem_deprecated.py` : (Deprecated) Tests tiling cover optimization.
- `generate_hanoi_tower_twisted_path_deprecated.py` : (Deprecated) Intended to generate "twisted path" dataset for Hanoi.
- `generate_checkers_jump_1d_v2.py` : Sequence learning comparative experiment for checkers jump, generating multiple types of complete path datasets.
- `generate_maze_symbolic_to_image.py` : Converts symbolic maze datasets to image format for visual model testing.
- `generate_trapping_rain_water_visualizer.py` : Converts symbolic "Trapping Rain Water" dataset to image-to-image format for visual models.
- `generate_shortest_path_in_tree_deprecated.py` : (Deprecated) Early experiment on shortest path in trees from images.
- `generate_rain_water_final_showdown.py` : Algorithm comparison experiment providing DP, Monotonic Stack, and Two Pointers intermediate processes as auxiliary labels.
- `generate_bricks_falling.py` : LeetCode 803. Bricks Falling When Hit. Tests understanding of physical stability and connectivity.
- `generate_maximal_rectangle_coords.py` : LeetCode 85. Maximal Rectangle.
- `generate_median_island_median_finder.py` : Complex symbolic reasoning combining sequence segmentation, median calculation, and positioning.
- `generate_rain_water_summation.py` : Subtask B of "Trapping Rain Water." Tests pure summation capability given pre-calculated water amounts.
- `generate_rain_water_then_cellular_automata.py` : Cross-domain chain reasoning Task D. Tests executing "Trapping Rain Water" then using results as initial state for "Cellular Automata."

- `generate_maze_decoupled.py` : Decoupled version of dense maze, outputting full BFS distance map as explanation.
- `generate_edit_nextstep_mlp.py` : Decoupled experiment for Edit Distance focusing on "next optimal operation" with operation masks.
- `generate_edit_distance_dp_table_full.py` : Full DP table decoupling experiment for Edit Distance.
- `generate_median_one_finder.py` : "Median '1' Locator" dataset. Basic statistical positioning.
- `generate_mean_of_medians.py` : "Mean of Medians of Odd/Even Isolated Islands" dataset. Complex logic task known as "Ninefold Purgatory."
- `generate_exam_seats.py` : "Exam Room Arrangement" dataset. (Maximum Students Taking Exam - LeetCode). Tests status compression DP on 2D grids.
- `generate_parity_accumulator.py` : "Parity Instruction Accumulator" task. Complex logic with sequential dependency and conditional arithmetic.
- `generate_neural_turing_machine.py` : Simulates conditional write operations on a cyclic tape like a simplified NTM. Direct test of algorithmic operation learning.
- `generate_lights_out_forward.py` : "Lights Out" game forward simulation. Tests local rule propagation.
- `generate_lights_out_gaussian_elimination.py` : "Lights Out" inverse problem modeled as linear equation solving with Gaussian elimination decoupling. Proves capability to learn complex inverse problems with appropriate decoupling.

C. Visual Reasoning

- `generate_checkerboard_to_binary.py` : Basic visual-to-symbol conversion test.
- `generate_line_angle_to_vector.py` : Tests extracting precise geometric info (angles) from images.
- `generate_count_shapes_from_image.py` : Tests multi-visual task capability: object recognition (shape), attribute recognition (color), counting.
- `generate_sokoban_symbolic_to_image_no_labels.py` : Converts symbolic Sokoban to image-only format.
- `generate_sokoban_symbolic_to_image_with_labels.py` : Converts symbolic Sokoban to image classification dataset for CV models.
- `generate_triangle_to_incircle.py` : Landmark experiment for "sculpting precise rules with gradient descent." Triangle Incircle.
- `generate_polygon_to_symmetry_axis.py` : Tests inferring implied symmetry axis from symmetric figures.
- `generate_triangle_to_centroid.py` : Tests learning centroids.
- `generate_triangle_to_tessellation.py` : Landmark display of paradigm capability. Tests infinite lattice-based generation rules (Tessellation) to rule out interpolation/memorization.

- `generate_shortest_distance_between_triangles.py` : Tests global geometric relationship reasoning (shortest distance).
- `generate_coords_to_triangle.py` : Basic symbol-to-geometry rendering (coordinates to pixels).
- `generate_triangle_coords_to_tessellation.py` : Advanced reasoning mixing symbolic instructions and geometric generation rules.
- `generate_trapping_rain_water_image_to_symbol.py` : Image-to-symbol conversion testing if CV models can extract precise physical quantities.

D. Cellular Automata

- `generate_cellular_automata_1d.py` : 1D CA evolution dataset logic learning test.
- `generate_game_of_life_2d.py` : 2D CA (Conway's Game of Life) dataset. More complex spatial relations.
- `generate_cellular_automata_1d_multistate.py` : 1D CA extension to non-binary states.
- `generate_cellular_automata_programmable.py` : Tests "programmability" or "meta-learning." Execution based on rules given in input.
- `generate_cellular_automata_inverse_rule90.py` : Tests "Inverse Problem" solving. Inferring input from output under constraints.
- `generate_game_of_life_image_to_image.py` : 2D CA image-to-image version.
- `generate_cellular_automata_spatial_conditional.py` : Tests partitioning and parsing "instruction" and "data" within a single image modality.
- `generate_cellular_automata_multimodal_deprecated.py` : (Deprecated) True multimodal (image+text) dataset.
- `generate_cellular_automata_1d_to_grid_image_interp.py` : Logic/Perception hybrid task testing unified rule learning and interpolation.
- `generate_cellular_automata_1d_to_grid_image.py` : Tests "rendering" 1D symbolic results to 2D structured images.
- `generate_cellular_automata_inverse_rule.py` : First attempt at Inverse Reasoning (inferring rule from input-output).
- `generate_cellular_automata_inverse_rule_and_steps.py` : Early version inferring rule and steps.
- `generate_cellular_automata_inverse_rule_and_steps_unique.py` : Major upgrade to unique solution inverse reasoning. Inferring applied rule and number of applications.
- `generate_cellular_automata_1d_perturbed.py` : Systematically tests robustness against imperfect data (input/output noise).
- `generate_ca110_full_trace.py` : Deep analysis experiment generating full evolution traces.
- `generate_ca_text_format_dataset.py` : Autoregressive experiment testing Decoder-only Transformers (e.g., GPT) on CA rules via text prompts.

- `generate_cellular_automata_image_and_label.py` : General dataset generator for image format and symbol format.
- `generate_mnist_ca_110.py` : Perception and Reasoning fusion. MNIST digits as CA cells.
- `generate_rulemnist_ca.py` : Advanced Perception and Reasoning fusion with rule control.
- `generate_ca_continuous_transform.py` : Continuous-Discrete-Continuous hybrid CA experiment.
- `generate_ca_nested_masked.py` : "Matryoshka" nested mask CA dataset testing multi-stage implicit reasoning under dynamic masks.
- `generate_ca_masked_output.py` : "Partially Observable" CA dataset.
- `generate_ca_to_abstract_real.py` : Tests encoding discrete evolution results into arbitrary real number symbol representations.
- `generate_ca_rule110_minimalist_trace.py` : Minimalist text explanation format for autoregressive models (CoT style).
- `generate_ca_multirule_cot_trace.py` : Extension with multi-rule support for CoT trace.
- `generate_ca_ood_hallucination_split.py` : CA OOD generalization test dataset (split by rules).
- `generate_hierarchical_ca_tree.py` : Hierarchical abstract CA tree dataset generator. Verifies emergence of intermediate representations.

E. Physics Simulation

- `generate_projectile_motion_simulation.py` : Tests learning simple dynamic physical processes (projectile motion).
- `generate_snell_refraction_simulation.py` : Tests learning Snell's Law.
- `generate_snell_refraction_with_contextual_index.py` : Tests learning Snell's Law with inferred physical parameters (refractive index) from context.
- `generate_reaction_diffusion_deprecated.py` : (Deprecated) Gray-Scott reaction-diffusion simulation. Continuous field simulation.
- `generate_cube_rotation_matplotlib_deprecated.py` : (Early version) 3D object rotation from abstract pose parameters.
- `generate_cube_rotation_pillow_v1.py` : (Technical upgrade) 3D rotation with better rendering.
- `generate_cube_rotation_pillow_with_anchor.py` : (Final version) 3D rotation with visual anchors.
- `generate_cube_rotation_pillow_wireframe.py` : (Variant) 3D rotation with wireframes and sparse input.
- `generate_catenary_curve_simulation_deprecated.py` : Early exploration of Catenary (non-linear curve from physical laws).
- `generate_catenary_curve_from_points.py` : Tests learning Catenary curve uniquely determined by minimum potential energy.

- `generate_orbital_path_from_initial_state.py` : Tests learning Kepler's laws/Law of Universal Gravitation.

F. ARC-AGI Exploration

- `generate_arc_contextual_color_swap.py` : Tests learning rules from local context/examples applied globally. Imitates ARC-AGI core concept.
- `generate_arc_find_cross_pattern.py` : Visual pattern recognition (object detection) in noise.
- `generate_arc_find_odd_one_out.py` : "Find the Odd One Out" meta-reasoning task.
- `generate_arc_connect_colored_pairs.py` : Identifies multiple "connection tasks" and implicit layer/priority rules.
- `generate_arc_conditional_perpendicular_lines.py` : Geometric operations conditional on object attributes and global references.
- `generate_arc_column_projection.py` : Recognizes complex context relations ("below... and within range...") for conditional column operations.
- `generate_arc_procedural_spiral.py` : Executes iterative, procedural generation algorithms (spirals).
- `generate_arc_fractal_stamping.py` : Recursive/fractal generation rules using input as "brush."
- `generate_arc_flood_fill.py` : Classic Flood Fill algorithm. ARC Prize - Puzzle #36a08778
- `generate_arc_layered_fill.py` : Complex filling algorithm dependent on topological distance and conditional judgment.
- `generate_arc_fluid_simulation.py` : Simulates fluid dynamic process with specific rules in image space.
- `generate_arc_periodic_conditional_fill.py` : Complex conditional formatting rule with periodicity and special cases.
- `generate_arc_fill_square_holes.py` : Multi-step visual reasoning: identifying "foreground in background" (holes) -> geometric attribute judgment -> coloring.
- `generate_arc_conditional_recoloring.py` : Understands visual layers and conditional object attribute modification.
- `generate_arc_sort_by_length_remap_position.py` : Executes "Attribute-Position Decoupling and Remapping" complex sorting task.
- `generate_arc_jigsaw_puzzle_simple.py` : Visual matching and transformation (Simple). ARC Prize - Puzzle #898e7135
- `generate_arc_jigsaw_puzzle_advanced.py` : Visual matching and transformation (Advanced). ARC Prize - Puzzle #898e7135
- `generate_arc_connect_path_by_sequence.py` : Parses external instruction sequences to execute multi-step, stateful path connection in images.

- `generate_arc_reflection_simulation_deprecated.py` : (Deprecated) Tests understanding of complex physical optics rules (reflection, collision, color change).